

ALGEBRAIC KUGA–SATAKE CORRESPONDENCES AND HODGE CONJECTURES ON A K3 QUADRATIC LOCUS

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ABSTRACT. We prove the rational Hodge and generalized Hodge conjectures in every cohomological degree of every self-power of a projective complex K3 surface whose transcendental quadratic space, with its cup-product form, admits a rational isometric embedding in $\mathbb{U}_{\mathbb{Q}}^{\oplus 2} \perp \langle -1 \rangle^4$. This includes every ample P -polarized K3 surface, including all Picard jumps, for $P = \mathbb{U} \oplus D_8(-1) \oplus D_4(-1)$. On this locus we construct algebraic correspondences inducing every prescribed standard even-Clifford Kuga–Satake tensor. More generally, for any projective complex K3 surface, algebraicity of one exact standard even-Clifford Kuga–Satake tensor implies both conjectures for every self-power.

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1. INTRODUCTION

The Hodge conjecture asks which cohomology classes on a smooth projective complex variety come from algebraic subvarieties. In its rational form it asserts surjectivity of

$$\mathrm{cl} : \mathrm{CH}^r(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^r(X) := H^{2r}(X, \mathbb{Q}) \cap H^{r,r}(X)$$

for every r . The generalized Hodge conjecture asks the corresponding support question for whole Hodge substructures. Write

$$N^p H^j(X, \mathbb{Q}) = \sum_{\substack{B \subset X \text{ closed} \\ \mathrm{codim}_X B \geq p}} \ker(H^j(X, \mathbb{Q}) \longrightarrow H^j(X \setminus B, \mathbb{Q})).$$

The generalized conjecture says that this equals the largest rational sub-Hodge structure whose types (a, b) satisfy $a, b \geq p$. We use rational coefficients for both conjectures throughout, and abbreviate them as HC and GHC.

For a projective K3 surface S , let

$$T_S = \mathrm{NS}(S)_{\mathbb{Q}}^{\perp} \subset H^2(S, \mathbb{Q})$$

with its cup-product form q . Its extreme Hodge spaces are $T_S^{2,0} = H^{2,0}(S)$ and its conjugate, both one-dimensional. The Hodge index theorem gives signature $(2, 20 - \rho(S))$. This small Hodge level makes K3 surfaces a natural setting for the generalized conjecture, but their powers contain tensor Hodge classes and substructures that need not be generated by divisor classes alone.

Kuga and Satake attached abelian varieties to polarized K3 Hodge structures through Clifford algebras [24], and Deligne developed the construction in families [12]. The distinguished correspondence is absolutely Hodge and motivated [14, Proposition 6.26(d) and Corollary 6.27], [1, Section 7.1]. Algebraicity is the additional assertion needed here: we require a cycle inducing the prescribed Clifford map, not merely some nonzero Hodge embedding into abelian cohomology.

1.1. One exact Kuga–Satake tensor. We specify the Kuga–Satake tensor used throughout the paper. For a quadratic space (T, q) of K3 type we use the Clifford relation $v^2 = q(v, v)$. Its full even algebra $W = C^+(T, q)$ carries the standard left Kuga–Satake Hodge structure. If e_1, e_2 are an oriented orthonormal basis of the positive period plane with $T^{2,0} = \mathbb{C}(e_1 + ie_2)$, its complex structure is left multiplication by $J = e_1 e_2$. Let A be an abelian variety with a specified rational Hodge identification $H^1(A, \mathbb{Q}) = W$. Such a realization is unique up to the usual isogeny choices. For a nonisotropic $w \in T$ the standard map is

$$(1.1) \quad j_w : T(1) \longrightarrow \text{End}(W), \quad j_w(v)(a) = vaw.$$

Here $\mathbb{Q}(1)$ has weight -2 . The right factor need not be a Hodge vector: conjugation by left spin multiplication makes (1.1) a Hodge map at every period.

Let E be a rational polarization of W and set $E^\flat : W \rightarrow W^\vee(-1)$, $y \mapsto E(y, -)$. In the ordering $\text{End}(W) = W \otimes W^\vee$, apply $(E^\flat)^{-1}$ to the second factor after twisting by $\mathbb{Q}(-1)$. This gives the specified tensor form

$$(1.2) \quad \iota_{w,E} : T \longrightarrow W \otimes W \subset H^2(A \times A, \mathbb{Q}).$$

The adjective “standard” below refers to this actual map, not to an unspecified nonzero embedding into the cohomology of an abelian variety; see [37, Proposition 6.3] for the full even-Clifford construction.

Theorem 1.1. *Let S be a projective K3 surface. Suppose that, for one rational nonisotropic $w \in T_S$, one rational polarization, and one full even-Clifford realization, the tensor of $j_w(v)(a) = vaw$ is induced by a cycle in $\text{CH}^2(S \times A^2)_{\mathbb{Q}}$. Then rational HC and rational GHC hold in every degree of every self-power of S .*

Section 2 proves this criterion independently of the concrete K3 family constructed later.

1.2. The quadratic locus. Let $\mathbb{U}$ be the hyperbolic plane with Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and put

$$P = \mathbb{U} \oplus D_8(-1) \oplus D_4(-1), \quad V_P = (\mathbb{U} \oplus \mathbb{U}(2) \oplus D_4(-1))_{\mathbb{Q}} \simeq \mathbb{U}_{\mathbb{Q}}^{\oplus 2} \perp \langle -1 \rangle^4.$$

The form V_P has dimension eight and signature $(2, 6)$. Lemma 1.3 relates it to primitive copies of P in the K3 lattice. An *ample P -polarization* on S means a primitive embedding $P \hookrightarrow \text{NS}(S)$ whose image contains an ample class. At a Picard jump its fixed perpendicular still has dimension eight, although it is then larger than T_S .

Theorem 1.2. *Let S be a smooth projective complex K3 surface. Suppose that there is a rational isometric embedding*

$$(T_S, q) \hookrightarrow V_P.$$

For any full even-Clifford realization $H^1(A, \mathbb{Q}) = C^+(T_S, q)$, any rational nonisotropic $w \in T_S$, and any rational polarization E of this weight-one Hodge structure, there is a cycle

$$\Gamma_S \in \text{CH}^2(S \times A \times A)_{\mathbb{Q}}$$

whose action on every $x \in T_S$ is exactly $\iota_{w,E}(x)$ in $H^2(A \times A, \mathbb{Q})$. The same assertion holds on a transported isogenous model with its specified tensor.

Moreover, the rational Hodge conjecture and the rational generalized Hodge conjecture hold in every cohomological degree of every self-power S^m , for every integer $m \geq 1$.

In particular the theorem applies to every ample P -polarized K3 surface, including all Picard jumps. Its hypothesis concerns the actual transcendental quadratic space with its cup-product form. Standard motivic decompositions also extend these HC and GHC conclusions to all self-powers of Hilbert schemes of points and to the smooth projective moduli spaces specified in Corollary 9.5.

1.3. Context and proof strategy. Geometric constructions for special K3 families include Paranjape’s curve-product and Prym construction for six-line double planes [29, Sections 2–4], and Bolognesi–Laterveer’s treatment of order-three non-symplectic automorphisms whose fixed locus contains a curve [5, Theorem 4.4 and Section 6]. Floccari and Fu explicitly distinguish an algebraic embedding from the prescribed Kuga–Satake map [18, Conjectures 2.4 and 2.6]; they prove algebraicity of the latter for the K3 surfaces associated with projective singular OG6-varieties [18, Corollary 4.7].

For the lattice P used here, Poon computes the Clifford algebra and constructs a modular map to quaternionic abelian eightfolds [31, Theorem 3.1.4 and Section 4]. Van Geemen and Verra study related quaternionic Prym eightfolds and algebraic quaternionic classes [39, Sections 3–4]. These modular and quaternionic-class constructions do not by themselves give the prescribed K3-to-abelian correspondence. The Clifford and half-spin descriptions are classical [37, Proposition 6.3 and Example 6.6]; our recovery argument keeps the full even module and both tensor orientations.

For powers of a K3 surface, Zarhin’s description of its Hodge group reduces tensor Hodge classes to classical invariant theory [43]. When the endomorphism field of the transcendental Hodge structure is CM, ordinary HC is already known for every self-power [41, Corollary 2.3]. The totally real case also requires relative determinant tensors; Varesco identifies these generators and explains why classes from the square alone need not suffice [41, Theorem 2.16]. For specified four-dimensional families with generic Picard number 16, he also derives ordinary HC for all powers from algebraicity of the Kuga–Satake correspondences [41, Theorem 0.2].

From the exact tensor to HC and GHC. Section 2 proves the general criterion from its one exact standard tensor. Alternated compositions of j_w inject $(\wedge^k T_S)(k)$ into the same endomorphism algebra: Clifford PBW and invertibility of the right factor prove injectivity. Algebraic adjoints and a polynomial inverse give return correspondences from $H^2(A^2)$ to $\wedge^k T_S$ with target support of codimension $k - 1$. The returns first algebraize relative volume tensors in the totally real case; together with the endomorphism-field actions and classical invariant theory, these give ordinary HC. A separate highest-weight argument bounds the number of exterior factors needed to carry a rational Hodge substructure in terms of its Hodge coniveau. Their returns, followed by an HC-realized map, then supply the required geometric support. This proves GHC for whole substructures, not just for Hodge classes.

A curve source and its K3 comparison. Sections 3 and 4 construct double covers X of abelian surfaces. Their resolved inversion quotients Y carry a rank-eight variation U with quadratic form V_P and an open period image. Resolved pullback identifies it with $U_X \subset H^2(X)$, multiplying the cup form by two. On a nonempty geometric open, a Gauss-pencil construction gives a product of curves dominating X ; when U_X has no nonzero rational $(1, 1)$ -class, it supplies an algebraic injection

$$U_X \hookrightarrow H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q}).$$

Section 5 realizes U , with its Y -form, in an auxiliary K3 family S^0 over the original pencil line. The larger integral anti-invariant lattice is retained to control vanishing roots and descent. The

remaining task is to make the specified comparison $U \rightarrow U_X$ algebraic; the curve domination alone does not do so.

Section 6 accomplishes this by two quadratic substitutions. The first, $t = s + u^2$, gives uniruled threefold models on both sides, whose third cohomology has algebraic splittings through curve cohomology with a Tate twist. The required degree-three Hodge comparison is therefore algebraic. The second, $s = b + v^2$, produces a conic with two components over $t = b$. Their difference defines a cylinder that returns this comparison to the surface fibers. Its class lies in the Leray term controlled by degree-three cohomology. Fixing the first models and correspondence before b varies makes the discarded boundary's Hodge structures fixed, so nonconstant simple periods eliminate the boundary error. Composition with the curve injection now gives an algebraic curve source for $T_{S_b^0} = U_b$.

The prescribed tensor and all periods. Section 7 upgrades a curve injection at a full special-orthogonal period to every rational Hodge map into the full even-Clifford tensor square. Among the nontrivial constituents, weight-one effectivity leaves the two half-spin representations; a multiplicity-space calculation and rational Hodge-Hom descent recover the prescribed tensor. The auxiliary K3 need not be P -polarized. In Section 8, a rational change of period coordinates puts the construction into any local ample- P target ball without altering its algebraic pencils. Successive choices give a target seed with $\text{NS} = P$; Buskin's theorem transports the auxiliary curve source there, and the half-spin argument realizes the fixed target tensor. Proper cycle parameters then extend that same action on the whole rank-eight perpendicular of P , in the full degree-two target, throughout the ball. Only afterward does Section 9 compress to the actual T_S at a Picard jump. A separate rational period and Buskin transfer reaches the theorem's full rational quadratic-embedding locus.

1.4. Conventions and the lattice. Cohomology means rational Betti cohomology, except when an integral lattice is specified; integral cohomology is taken modulo torsion. For smooth projective varieties M, N , a cycle $Z \in \text{CH}^r(M \times N)_{\mathbb{Q}}$ acts by

$$Z_*x = (p_N)_*(\text{cl}(Z) \smile p_M^*x), \quad Z_* : H^k(M) \longrightarrow H^{k+2r-2\dim M}(N).$$

Degree-changing maps carry the corresponding Tate twist as Hodge maps. We use only cohomological projectors represented by cycles, not idempotence modulo rational equivalence. The geometric U below is distinct from the hyperbolic plane $\mathbb{U}$; the auxiliary K3 is S^0 , the double-cover surface is X , and the final K3 is S .

Lemma 1.3. *There is a primitive embedding of P in $\Lambda_{K3} = \mathbb{U}^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$ with orthogonal complement $T_P = \mathbb{U} \oplus \mathbb{U}(2) \oplus D_4(-1)$. For every primitive embedding of P in the K3 lattice, the rational orthogonal complement is isometric to V_P .*

Proof. For an even lattice L , write $A_L = L^*/L$ and $q_L : A_L \rightarrow \mathbb{Q}/2\mathbb{Z}$ for its discriminant form. The forms of $D_8(-1)$ and $D_4(-1)$ are respectively the split and anisotropic forms u, v on $(\mathbb{Z}/2)^2$. Their values are integral modulo $2\mathbb{Z}$, so each is isomorphic to its negative. Also $q_{\mathbb{U}(2)} = u$. Hence

$$(A_P, q_P) \simeq u \oplus v \simeq (A_{T_P}, -q_{T_P}).$$

An anti-isometry glues $P \oplus T_P$ to an even unimodular lattice of signature $(3, 19)$, hence to Λ_{K3} , with both summands primitive. This gives the asserted embedding. For another embedding, Witt's extension theorem [30, Corollary 7.2.8] applied to the two copies of the nondegenerate rational space $P_{\mathbb{Q}}$ gives an isometry of $\Lambda_{K3, \mathbb{Q}}$ carrying one copy to the other. It carries their rational perpendiculars to each other. Finally rescaling one isotropic basis vector identifies $\mathbb{U}(2)_{\mathbb{Q}}$ with $\mathbb{U}_{\mathbb{Q}}$, and the coordinate description of D_4 gives $D_4(-1)_{\mathbb{Q}} \simeq \langle -1 \rangle^4$. $\square$

The lattice P occurs as an exact Néron–Severi lattice. Choose a period in the domain of T_P outside the countably many hyperplanes defined by integral vectors not in P . Period surjectivity

gives a K3 surface with Néron–Severi group exactly P ; it is projective because P contains a positive class. Its ample cone contains a primitive integral class of P . These are standard consequences of the K3 period and chamber theorems [22].

2. HODGE CLASSES AND GEOMETRIC SUPPORTS FROM ONE EXACT TENSOR

We prove Theorem 1.1. The argument is independent of the concrete K3 family constructed later; its only Clifford premise is one algebraic realization of the exact standard tensor in that theorem.

2.1. Algebraic summands and coefficient extensions. Let S be a projective K3 surface, and write

$$N = \text{NS}(S)_{\mathbb{Q}}, \quad T = N^{\perp} \subset H^2(S, \mathbb{Q}), \quad V = T(1).$$

We denote the intersection form on T , and its weight-zero version on V , by q . Choose a point $o \in S$, a divisor basis $D_1, \dots, D_{\rho}$ of N , and let (b^{ij}) be the inverse of $(D_i \cdot D_j)$. The following cycles in $\text{CH}^2(S \times S)_{\mathbb{Q}}$ act as the indicated cohomological projectors:

$$(2.1) \quad \pi_0 = [o] \times S, \quad \pi_4 = S \times [o], \quad \pi_N = \sum_{i,j} b^{ij} D_i \times D_j, \quad \pi_T = \Delta_S - \pi_0 - \pi_4 - \pi_N.$$

The form on N is nondegenerate by the Hodge index theorem. In particular, these projectors and the pairing and inverse pairing on T are algebraic. The latter are given by intersection and the transcendental component of the diagonal. Throughout this section, “projector” means a projector on cohomology represented by a cycle; no assertion of idempotence modulo rational equivalence is needed.

The product of k copies of π_T , followed by the alternating projector

$$(2.2) \quad \text{Alt}_k = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} \text{sgn}(\sigma) \sigma,$$

is an algebraic projector $\pi_{\wedge, k}$ on S^k with image $\wedge^k T \subset H^{2k}(S^k, \mathbb{Q})$. Here permutations act by their graphs, with the convention giving the ordinary permutation of the tensor factors. All these factors have even degree. The projector is self-adjoint for the middle intersection pairing: π_T is self-transpose, and transposition replaces a permutation by its inverse. The induced pairing on $\wedge^k T$ is a nonzero rational multiple of the exterior pairing associated with q , hence is nondegenerate.

We will also use algebraic degree-one projectors on abelian varieties. For example, if A has dimension g and $a \geq 2$ is an integer, the polynomial

$$\pi_1^A = \prod_{\substack{0 \leq j \leq 2g \\ j \neq 1}} \frac{[a]^* - a^j \text{id}}{a - a^j}$$

in the transpose graph of multiplication by a acts as the projector onto $H^1(A, \mathbb{Q})$.

An algebraic class with coefficients in a field $K \supset \mathbb{Q}$ will mean an element of the scalar extension of the rational image of the cycle-class map. This convention makes rational descent transparent. If $A_{\mathbb{Q}} \subset H_{\mathbb{Q}}$ is that rational image in a fixed cohomology group, then

$$(2.3) \quad H_{\mathbb{Q}} \cap (A_{\mathbb{Q}} \otimes_{\mathbb{Q}} K) = A_{\mathbb{Q}} \quad \text{inside } H_{\mathbb{Q}} \otimes_{\mathbb{Q}} K.$$

Indeed tensoring $H_{\mathbb{Q}}/A_{\mathbb{Q}}$ with a field is faithful. Products, permutations and contractions by algebraic pairings preserve this notion of algebraicity. All varieties remain over $\mathbb{C}$; only the coefficients are extended.

2.2. Algebraic returns from exterior powers.

Proposition 2.1. *Let S be a projective K3 surface with transcendental space (T, q) of rational dimension t , put $V = T(1)$, and let $W = C^+(T, q) = H^1(A, \mathbb{Q})$ be an even-Clifford Kuga–Satake realization. Suppose that for one rational nonisotropic $w \in T$ the tensor form of*

$$(2.4) \quad j_w : V \longrightarrow \text{End}(W), \quad j_w(v)(a) = vaw,$$

is induced by a cycle in $\text{CH}^2(S \times A^2)_{\mathbb{Q}}$, using a rational polarization of W . Then, for every $1 \leq k \leq t$, there is an injective algebraic Hodge map

$$F_k : \bigwedge^k V \longrightarrow \text{End}(W)$$

with an algebraic left inverse on cohomology. More precisely, put $Z = A^2$ and $n = \dim Z$. There are correspondences

$$\Gamma_k \in \text{CH}^{k+1}(S^k \times Z)_{\mathbb{Q}}, \quad R_k \in \text{CH}^{n+k-1}(Z \times S^k)_{\mathbb{Q}}.$$

Viewing $\bigwedge^k V$ in $H^{2k}(S^k, \mathbb{Q})(k)$ and identifying $\text{End}(W) \simeq (H^1(A, \mathbb{Q}) \otimes H^1(A, \mathbb{Q}))(1) \subset H^2(Z, \mathbb{Q})(1)$ by a rational polarization, their weight-zero actions satisfy $(\Gamma_k)_|_{\bigwedge^k V} = F_k$ and $(R_k)_*F_k = \text{id}$. Every rational Hodge class in $\bigwedge^k V$ is the class of a rational algebraic cycle of codimension k on S^k .*

Proof. Choose an ample divisor class ℓ on A , and use the polarization

$$e(x, y) = \int_A x \cup y \cup \ell^{g-1}, \quad g = \dim A.$$

Changing the rational polarization in the hypothesis preserves algebraicity: the change on the identified dual factor is a rational Hodge automorphism of $H^1(A, \mathbb{Q})$, and hence comes from an abelian homomorphism with rational coefficients. The elementary argument for this fact is given in the proof of Lemma 9.1. Here $n = 2g$; put

$$B_W = H^1(A, \mathbb{Q}) \otimes H^1(A, \mathbb{Q}) \subset H^2(Z, \mathbb{Q}).$$

The polarization identifies $\text{End}(W)$ with $B_W(1)$.

The Clifford calculation. Define F_k by alternating the composition of k copies of j_w . Associativity gives

$$j_w(v_1) \cdots j_w(v_k)(a) = v_1 \cdots v_k a w^k.$$

Thus, if $c_k : \bigwedge^k T \rightarrow C(T, q)$ is the antisymmetrized Clifford multiplication, then

$$(2.5) \quad F_k(\alpha)(1) = c_k(\alpha)w^k.$$

The Clifford PBW decomposition identifies c_k with the degree- k exterior summand, so it is injective. Moreover $w^{-1} = w/q(w, w)$, and right multiplication by w^k is invertible. Equation (2.5) proves injectivity in every degree under consideration. For odd k , the left and right Clifford multiplications are both odd and their composite still preserves $C^+(T, q)$.

The map j_w is a Hodge morphism for the left spin action; tensor products, composition and alternation preserve this property. Consequently

$$(2.6) \quad F_k : \bigwedge^k T \longrightarrow B_W(1 - k)$$

is the corresponding untwisted Hodge morphism. In particular, the construction uses the exact multiplication in (2.4); an unspecified algebraic copy of T in $H^2(A^2, \mathbb{Q})$ would not imply (2.5).

Algebraic realization and degree. Represent the tensor form of j_w by the cycle in the hypothesis, projected by π_T and $\pi_1^A \boxtimes \pi_1^A$. Take its k -fold exterior product and project the input by $\pi_{\wedge, k}$.

Composition in $\text{End}(W)$ contracts $k - 1$ pairs of W -slots by e , in the appropriate tensor order. Each such contraction is algebraic: the cycle

$$(\Delta_A)_*(\ell^{g-1}) \in \text{CH}^{2g-1}(A^2)_{\mathbb{Q}},$$

regarded as a correspondence to a point, realizes $W \otimes W \rightarrow \mathbb{Q}(-1)$. It lowers cohomological degree by two. The two uncontracted slots give Z ; any signs arising from permuting degree-one cohomology factors are included in the cycle coefficients. Hence the resulting correspondence, denoted by Γ_k , has

$$(2.7) \quad \Gamma_k \in \text{CH}^{k+1}(S^k \times Z)_{\mathbb{Q}}, \quad (\Gamma_k)_*|_{\wedge^k T} = F_k.$$

Indeed its degree shift is $2 - 2k$, which agrees with $2((k+1) - \dim S^k)$. Its output twist is $1 - k$, as in (2.6).

An algebraic left inverse. Let $L = \text{pr}_1^* \ell + \text{pr}_2^* \ell$ on Z . The summand B_W is primitive in degree two for L . To see this, expand L^{n-1} : every term acting on B_W has odd degree in each abelian factor, whereas the only possible top bidegree on A^2 is $(2g, 2g)$. Thus $L^{n-1}B_W = 0$. The Hodge–Riemann bilinear relations imply that

$$Q_Z(x, y) = \int_Z x \cup y \cup L^{n-2}$$

is a polarization on B_W , up to a fixed sign. Retwisting (2.6) gives a morphism of weight-two Hodge structures $(\wedge^k T)(k-1) \rightarrow B_W$. Thus its image is a Hodge substructure of B_W . The restriction of Q_Z to this image is nondegenerate: the associated definite Hermitian form restricts to a definite form there.

Multiplication by L^{n-2} , followed by the transpose of Γ_k and by $\pi_{\wedge, k}$, defines an algebraic map

$$(2.8) \quad G_k : H^2(Z, \mathbb{Q}) \longrightarrow \bigwedge^k T, \quad G_k \in \text{CH}^{n+k-1}(Z \times S^k)_{\mathbb{Q}}.$$

Here the cohomological degrees are

$$H^2(Z) \xrightarrow{L^{n-2}} H^{2n-2}(Z) \xrightarrow{t\Gamma_k} H^{2k}(S^k).$$

If Q_k denotes middle intersection on $\wedge^k T$, self-adjointness of $\pi_{\wedge, k}$ gives

$$(2.9) \quad Q_k(G_k F_k x, y) = Q_Z(F_k x, F_k y).$$

The pairing on the right is nondegenerate, and Q_k is nondegenerate by the discussion of (2.2). Therefore $D_k = G_k F_k$ is invertible on $\wedge^k T$.

Its characteristic polynomial has nonzero constant term, so Cayley–Hamilton supplies a polynomial $P_k \in \mathbb{Q}[X]$ such that $P_k(D_k)D_k = \text{id}$ on that summand. Composition of cycles realizes $P_k(D_k)$ in codimension $2k$ on $S^k \times S^k$, using $\pi_{\wedge, k}$ as the identity of the summand. The correspondence

$$(2.10) \quad R_k = P_k(D_k)G_k \in \text{CH}^{n+k-1}(Z \times S^k)_{\mathbb{Q}}$$

satisfies $R_k F_k = \text{id}$; in weight-zero notation it maps $H^2(Z, \mathbb{Q})(1)$ to $(\wedge^k T)(k)$. This construction uses multiplication by an ample class, transposition and a polynomial inverse on a finite-dimensional summand. It requires no algebraic inverse to a Lefschetz operator. The composite $\Gamma_k \circ R_k$, of codimension n on $Z \times Z$, is the corresponding cohomological projector onto $F_k(\wedge^k T) \subset H^2(Z, \mathbb{Q})$.

Finally, let $\alpha \in (\wedge^k V)_{\mathbb{Q}}^{0,0}$. Its image is a rational $(1, 1)$ class in B_W , so Lefschetz’s $(1, 1)$ theorem gives a divisor class on Z representing it. Applying R_k yields α . The resulting cycle has codimension $1 + (n + k - 1) - n = k$ on S^k , as asserted. $\square$

2.3. Geometric supports for exterior powers. The same return that algebraizes exterior Hodge classes also places each whole exterior power on a fixed geometric support. We use geometric coniveau N^p as defined in the introduction and set $N^p H^j(X, \mathbb{Q}) = H^j(X, \mathbb{Q})$ for $p \leq 0$. Recall that a sub-Hodge structure has Hodge coniveau at least p when all its types (a, b) satisfy $a, b \geq p$.

Lemma 2.2 (Support of a correspondence). *Let X, Y be smooth projective varieties, with $\dim X = N$ and $\dim Y = D$, and let $C \in \text{CH}^{N+p}(X \times Y)_{\mathbb{Q}}$, where $p \geq 0$. There is a fixed closed subset $B \subset Y$ of codimension at least p supporting the image of C_* on cohomology. In particular,*

$$C_* H^j(X, \mathbb{Q}) \subset N^p H^{j+2p}(Y, \mathbb{Q}).$$

Proof. Choose a finite cycle representative for C . Each component has dimension $D - p$, so the union B of its proper images in Y has dimension at most $D - p$. The restriction of the correspondence to $X \times (Y \setminus B)$ is zero. Compatibility of its action with restriction proves the assertion, including for rational linear combinations of components. $\square$

Proposition 2.3 (Exterior-power coniveau). *Let S satisfy the exact even-Clifford algebraicity hypothesis of Proposition 2.1, and put $t = \dim_{\mathbb{Q}} T$. For every $1 \leq k \leq t$,*

$$\bigwedge^k T \subset N^{k-1} H^{2k}(S^k, \mathbb{Q}).$$

Moreover every rational sub-Hodge structure of this exterior power satisfies the generalized Hodge conjecture in its ambient $H^{2k}(S^k, \mathbb{Q})$. In particular, the top exterior line is supported in codimension t .

Proof. Retain $Z = A^2$ and $n = \dim Z$ from Proposition 2.1. Equation (2.10) gives

$$R_k \in \text{CH}^{n+k-1}(Z \times S^k)_{\mathbb{Q}}, \quad R_k F_k = \text{id} \quad \text{on} \quad \bigwedge^k T.$$

The support of this cycle has dimension $n + 2k - (n + k - 1) = k + 1$. Lemma 2.2 therefore puts its entire image from $H^2(Z, \mathbb{Q})$ in coniveau $k - 1$, and that image contains $\bigwedge^k T$.

The only possible types of $\bigwedge^k T$ are $(k + 1, k - 1)$, (k, k) and $(k - 1, k + 1)$, since the extreme Hodge spaces of T are one-dimensional. Thus a sub-Hodge structure requiring more than coniveau $k - 1$ is necessarily of type (k, k) . Every rational vector in such a substructure is algebraic by Proposition 2.1. Choosing a finite rational basis and taking the union of the supports of its codimension- k cycles proves coniveau k . This also applies to $\bigwedge^t T$, whose only type is (t, t) . These are all possible coniveau requirements. $\square$

The exterior returns now have both properties needed for the criterion. We first use their algebraicity on Hodge classes to prove ordinary HC for arbitrary powers of S . Then Section 2.6 combines that result with the support property to treat arbitrary rational Hodge substructures.

2.4. The endomorphism field at an individual surface. The Hodge structure T is simple. Indeed, semisimplicity of polarizable Hodge structures splits any sub-Hodge structure, and at most one summand can contain its one-dimensional $(2, 0)$ part. Any other summand would be of type $(1, 1)$, contrary to the definition of T and Lefschetz's $(1, 1)$ theorem. Thus $E = \text{End}_{\text{Hdg}}(T)$ is a division algebra; its faithful action on $H^{2,0}(S)$ embeds it into $\mathbb{C}$, so it is a number field.

We use Zarhin's Hodge-group theorem [43] in the precise forms restated in [38, Section 2.1 and Theorem 2.8] for the field, adjoint and totally real group, and in [41, Section 2.1] for the CM group. The field E is either totally real or CM. The involution induced by the q -adjoint is respectively the identity or CM conjugation. If E is totally real, there is a nondegenerate symmetric E -bilinear form ϕ on T whose trace is q , and

$$(2.11) \quad \text{Hg}(V) = \text{Res}_{E/\mathbb{Q}} \text{SO}(T, \phi).$$

If E is CM with real subfield E^+ , the analogous form is Hermitian and

$$(2.12) \quad \mathrm{Hg}(V) = \mathrm{Res}_{E^+/\mathbb{Q}} \mathrm{U}(T, \phi).$$

Here $\mathrm{Hg}(V)$ is the connected rational group generated by the Hodge circle. Its invariant tensors are precisely the rational Hodge tensors of the weight-zero structure V .

Put $d = \dim_E T$, and choose a finite Galois field $K/\mathbb{Q}$ containing the images of all embeddings of E . Write

$$(2.13) \quad V_K = \bigoplus_{\eta: E \hookrightarrow K} V_\eta.$$

In the real case, the summands are mutually orthogonal and nondegenerate for q , and (2.11) becomes the product of their full special orthogonal groups after extending scalars to an algebraic closure. In the CM case, q pairs V_η perfectly with $V_{\bar{\eta}}$ and pairs it trivially with all other summands. Equation (2.12) becomes a product of full general linear groups, one per conjugate pair, acting on V_η and $V_{\bar{\eta}}$ as standard and dual representations.

2.5. The ordinary Hodge assertion. We first obtain the invariant tensors needed for ordinary HC. The following constructions are used only inside this proof of Theorem 1.1.

Volume tensors and real multiplication. Suppose first that E is totally real. Each line

$$\bigwedge^d V_\eta \subset \left(\bigwedge^d V \right)_K$$

is fixed by the group in (2.11). Formation of invariants of a rational algebraic group commutes with extension of scalars. Hence these lines belong to the scalar extension of the rational Hodge-class space of $\bigwedge^d V$. Proposition 2.1 proves their algebraicity. These relative volume tensors are the additional special-orthogonal generators identified in [41, Theorem 2.16].

Choose a nonzero volume tensor α_η on V_η . Contracting $\alpha_\eta \otimes \alpha_\eta$ by q in $d-1$ pairs of slots gives

$$(2.14) \quad c_\eta q_\eta^{-1} \in V_\eta \otimes V_\eta, \quad c_\eta \in K^*,$$

where q_η is the restricted form. For example, in an orthogonal basis $u_1, \dots, u_d$ with $q(u_i, u_i) = a_i \neq 0$, the remaining tensor is a nonzero scalar multiple of $\sum_i a_i^{-1} u_i \otimes u_i$; its scalar contains the nonzero product $\prod_i a_i$. This verifies in particular that the contraction cannot vanish.

These contractions are algebraic. They are obtained by intersecting along the relevant diagonals and pushing forward on powers of S . Two codimension- d tensors initially give codimension $2d$ on S^{2d} ; contracting $d-1$ pairs of K3 slots leaves codimension $2d - 2(d-1) = 2$ on S^2 . Pairing q_η^{-1} with q in one slot gives the orthogonal projector p_η onto V_η in V_K , so that projector is algebraic as well. For $a \in E$, its action is

$$(2.15) \quad a = \sum_\eta \eta(a) p_\eta.$$

This action is rational; (2.3), applied to its cohomological correspondence class on S^2 , proves rational algebraicity.

CM endomorphisms and pairings. Now suppose E is CM. Its norm-one elements span E over $\mathbb{Q}$. Here is an elementary verification. Their rational span F is a unital subalgebra of the finite field extension E , closed under multiplication, and is therefore a field. If $\bar{s} = -s$, then

$$u = \frac{1+s}{1-s}, \quad u\bar{u} = 1, \quad s = \frac{u-1}{u+1} \in F.$$

The denominators are nonzero, and $u+1 = 2/(1-s)$. Fixing a nonzero such s , every $a \in E^+$ satisfies $as \in F$, hence $a = (as)/s \in F$. Thus $F = E$.

Each norm-one element is a Hodge isometry of T . Extending it by the identity on N gives a rational Hodge isometry of $H^2(S, \mathbb{Q})$, which is algebraic by [9, Theorem 1.1]. Rational linear

combinations and projection by π_T give every endomorphism in E . Choose a primitive element b of $E/\mathbb{Q}$. The formulas

$$p_\eta = \prod_{\tau \neq \eta} \frac{b - \tau(b) \text{ id}}{\eta(b) - \tau(b)}$$

then realize the individual eigenprojectors with coefficients in K . Applying $p_\eta \otimes p_{\bar{\eta}}$ to the inverse metric gives the separate standard–dual pairing tensors. This argument requires that the *endomorphism field* be CM; it does not require that T be one-dimensional over that field.

Invariant generation and Künneth assembly. First work over an algebraic closure of K . The first fundamental theorem for a special orthogonal group on a standard space with nondegenerate symmetric form says that its invariant tensors are generated by the metric and a volume tensor. The first fundamental theorem for a general linear group on copies of its standard space and dual says that invariant tensors are generated by the standard–dual pairings. In the multilinear formulation the number of standard and dual slots must agree, and complete pair contractions span the invariant space. These statements apply to arbitrary numbers of slots; there is no stable range restriction on the spanning assertion [23, Sections 4.2 and 10.2]. A tensor formulation of the orthogonal statement is also given in [25, Theorem 2.1, Remark 2.2 and Corollary 4.2]: the top exterior tensor is harmonic and has nonzero norm, and adjoining it to the ordinary orthogonal contractions gives all special orthogonal invariants.

For each tensor power of V , use the algebraic projectors of the preceding construction to decompose every slot according to (2.13). Invariants of the product of the independent groups in (2.11) or (2.12) are then obtained factor by factor. In the real case all metric tensors and each relative volume tensor are algebraic by the preceding construction. In the CM case the full groups are general linear, so the algebraic pairings already suffice. Products, permutations and the algebraic duality on T therefore realize every invariant tensor. Scalar extension of invariant spaces and (2.3) show that every rational Hodge tensor in $V^{\otimes r}$ is algebraic over $\mathbb{Q}$, as a codimension- r class on S^r .

Finally apply the algebraic projectors (2.1) to every factor of S^m . The only non-Tate summand of $H^*(S, \mathbb{Q})$ is T in degree two. Every Hodge class in a Künneth component is therefore a sum of Hodge tensors in $T^{\otimes r}$ tensored with point, divisor and unit classes. If a component has r transcendental slots, a divisor slots and b top-degree slots, then its degree is $2r + 2a + 4b$ and the cycles constructed above have codimension $r + a + 2b$. They thus give all classes in the required codimension on S^m .

This proves the ordinary Hodge assertion of Theorem 1.1. The only Clifford input in this deduction was the single map in its hypothesis, through Proposition 2.1.

2.6. Generalized Hodge conjecture for K3 powers. To pass from exterior powers to arbitrary Hodge substructures, we combine their return maps with ordinary HC on products of powers. The latter algebraizes maps between actual rational summands, so no rational splitting of a complex middle-exterior representation is required. The Hodge level of a nonzero pure Hodge structure is $\max\{|a - b| : H^{a,b} \neq 0\}$.

Let $H \subset H^{2d}(S^m, \mathbb{Q})$ be an actual simple sub-Hodge structure of Hodge coniveau at least p , with $0 \leq p \leq d$. Then $H(d)$ has Hodge level at most $2(d - p)$. We will realize each nontrivial $H(d)$ as a quotient of a tensor product of $c \leq d - p$ exterior powers of $V = T(1)$. The number c , rather than the sum of their exterior degrees, is what controls support. We will show that the product of the c return correspondences, followed by an ordinary-HC realization of the quotient map, has target support of codimension at least $d - c$. The next two lemmas construct the required quotient with this bound on c ; ordinary HC handles trivial $H(d)$. The assembled proof below gives the correspondence calculation.

Lemma 2.4 (Orthogonal exterior carriers). *Let $V = T(1)$ for a projective K3 surface with $\mathrm{Hg}(V) = \mathrm{SO}(V, q)$ and $t = \dim V \geq 3$. Let H be a nontrivial rational simple representation occurring in a direct sum of tensor powers of V , endowed with its induced weight-zero Hodge structure. If its Hodge level is at most $2b$, there are integers $c, s \geq 1$ and $k_1, \dots, k_c \geq 1$ such that*

$$c \leq b, \quad s = \sum_i k_i, \quad k_i \leq t, \quad \bigotimes_{i=1}^c \bigwedge^{k_i} V \longrightarrow H$$

is a surjective rational Hodge map.

Proof. Over $\mathbb{C}$, choose orthogonal torus coordinates so that the Hodge cocharacter has vector weights $1, 0, \dots, 0, -1$. The integral highest weights of tensor representations of SO_t have the form

$$\begin{cases} \lambda_1 \geq \dots \geq \lambda_l \geq 0, & t = 2l + 1, \\ \lambda_1 \geq \dots \geq \lambda_{l-1} \geq |\lambda_l|, & t = 2l. \end{cases}$$

Only integral tensor weights occur here; half-integral spin weights are not representations in tensor powers of the standard module. For an irreducible constituent of highest weight λ , the largest value of the Hodge cocharacter on its weights is λ_1 : the cocharacter is dominant, and all weights lie in the convex hull of the Weyl orbit of λ . The largest value is attained at λ . The orthogonal Weyl group also gives the value $-\lambda_1$. Hence this constituent has Hodge level exactly $2\lambda_1$.

Put $c = \lambda_1$. In odd dimension take the columns of the partition $(\lambda_1, \dots, \lambda_l)$. Their number is c . The tensor product of their exterior highest-weight vectors is a highest-weight vector of weight λ . Complete reducibility therefore places the desired irreducible representation in the tensor product of these c exterior powers. In even dimension use the columns of $(\lambda_1, \dots, \lambda_{l-1}, |\lambda_l|)$. For columns of height l , use the highest-weight vector of the appropriate sign in $\bigwedge^l V_{\mathbb{C}}$; the two available weights are $(1, \dots, 1, 1)$ and $(1, \dots, 1, -1)$. Their sign is chosen to agree with λ_l . Shorter columns are unchanged. The same highest-weight argument applies. We keep the full exterior powers, including both middle-exterior constituents, as the carriers.

Choose a complex irreducible constituent H_{λ} of $H_{\mathbb{C}}$. It has $c = \lambda_1 \leq b$. The preceding construction defines a rational representation, for $\mathbf{k} = (k_1, \dots, k_c)$, $C_{\mathbf{k}} = \bigotimes_i \bigwedge^{k_i} V$ whose complexification shares this constituent with $H_{\mathbb{C}}$. Consequently

$$\mathrm{Hom}_{\mathrm{SO}(V, q)}(C_{\mathbf{k}}, H) \otimes_{\mathbb{Q}} \mathbb{C} = \mathrm{Hom}_{\mathrm{SO}(V_{\mathbb{C}}, q)}((C_{\mathbf{k}})_{\mathbb{C}}, H_{\mathbb{C}}) \neq 0.$$

This scalar-extension identity follows by taking invariants in $(C_{\mathbf{k}})^* \otimes H$. Any nonzero rational map in this space is surjective by rational simplicity of H , and it is a Hodge map because the Hodge cocharacter lies in $\mathrm{SO}(V, q)$. This proves the required rational assertion without descending an individually chosen complex projector.

For $t = 3$ the argument is the single-coordinate case, with columns of height one. For $t = 4$ the coordinates are $\lambda_1 \geq |\lambda_2|$, and columns have heights one and two; the two signs in the latter give the same construction. Thus the small orthogonal ranks cause no exception. $\square$

Lemma 2.5 (Exterior carriers with endomorphism fields). *Let $V = T(1)$ for a projective K3 surface and let $G = \mathrm{Hg}(V)$. If a nontrivial rational simple representation H occurring in a direct sum of tensor powers of V has induced Hodge level at most $2b$, then there are $1 \leq c \leq b$ and $1 \leq k_1, \dots, k_c \leq \dim_{\mathbb{Q}} V$ with a surjective rational Hodge map*

$$\bigotimes_{j=1}^c \bigwedge^{k_j} V \longrightarrow H.$$

Proof. Put $E = \mathrm{End}_{\mathrm{Hdg}}(T)$ and $a = \dim_E T$. We use the full groups in (2.11) and (2.12). Choose an algebraic closure of $\mathbb{Q}$ inside $\mathbb{C}$ over which these groups split. Highest-weight constituents can be described over that closure and then extended to $\mathbb{C}$. The multiset of constituents of H

is invariant under the absolute Galois group because H is rational. We shall use this rationality before applying any Hodge-level bound.

The possible orthogonal ranks. In the totally real case, $a \geq 3$. Indeed $\text{End}_G(V) = \text{End}_{\text{Hdg}}(V) = E$, because commuting with the Hodge circle is equivalent to commuting with its rational Zariski closure. This equality remains true after scalar extension, by scalar extension of invariants. Put $e = [E : \mathbb{Q}]$. If $a = 1$, the split orthogonal group is trivial, so its commutant on V has dimension $e^2 > e$; here $e = \dim_{\mathbb{Q}} V \geq 2$ because the two extreme K3 Hodge lines are nonzero. If $a = 2$, each split standard orthogonal module is the sum of two distinct characters of SO_2 , so the commutant has dimension at least $2e > e$. Both contradict the dimension e of E . Thus the orthogonal factors below have the ranks covered by Lemma 2.4.

Totally real widths and columns. Suppose first that E is totally real. Over $\mathbb{C}$ the group is $\prod_{\sigma: E \hookrightarrow \mathbb{C}} \text{SO}(V_{\sigma}, q_{\sigma})$. There is one distinguished embedding σ_0 for which V_{σ_0} has the types $(1, -1)$ and $(-1, 1)$; the other summands have type $(0, 0)$. Accordingly the Hodge cocharacter is trivial on the other factors and has vector weights $1, 0, \dots, 0, -1$ on the distinguished factor.

Fix one complex irreducible constituent of $H_{\mathbb{C}}$, with tuple of orthogonal highest weights λ_{σ} , and write $w_{\sigma} = \lambda_{\sigma,1}$ in the coordinates of Lemma 2.4. Galois acts transitively on the embeddings of E and transports the actual standard eigenspace of each orthogonal factor to the actual standard eigenspace at the conjugate embedding. The conjugacy class of cocharacters with weights $1, -1, 0, \dots, 0$ on that standard module is therefore preserved. The maximum of this class on a constituent is its first highest-weight coordinate, so Galois transports w_{σ} to the corresponding width at σ_0 . This argument also covers type D_4 : a triality automorphism exchanging the vector and a half-spin module does not preserve the actual standard module and cannot arise from this rational eigenspace transport. That conjugate constituent still occurs in $H_{\mathbb{C}}$. The Hodge-level calculation in Lemma 2.4 consequently gives $w_{\sigma} \leq b$ for every σ .

Set $c = \max_{\sigma} w_{\sigma}$. For each factor form its signed orthogonal columns as in that lemma, with heights $h_{\sigma j}$ for $1 \leq j \leq c$, padding with zeros. In the even case the sign of a middle column is retained. Set $k_j = \sum_{\sigma} h_{\sigma j}$. The exterior decomposition of a direct sum includes

$$\bigotimes_{\sigma} \bigwedge^{h_{\sigma j}} V_{\sigma} \subset \bigwedge^{k_j} V_{\mathbb{C}}.$$

Taking the Cartan components over all columns shows that $\bigotimes_j \bigwedge^{k_j} V_{\mathbb{C}}$ contains the chosen tuple of irreducibles. Columns from different embeddings thus occupy the same exterior carrier; their widths are maximized, not added. The ranks three and four are included by Lemma 2.4.

CM widths and signed columns. Now suppose that E is CM. Choose one embedding η from each conjugate pair and write

$$G_{\mathbb{C}} = \prod_{\eta} \text{GL}(W_{\eta}), \quad V_{\mathbb{C}} = \bigoplus_{\eta} (W_{\eta} \oplus W_{\eta}^*), \quad \dim W_{\eta} = a.$$

Choose the representatives so that the distinguished embedding containing $V^{1,-1}$ is represented by a standard module. On this factor the Hodge cocharacter is conjugate to $\text{diag}(z, 1, \dots, 1)$; on the other factors it is trivial. Its action on the distinguished dual is the inverse. A complex irreducible constituent of $H_{\mathbb{C}}$ has, on each factor, an integral highest weight

$$\lambda_{\eta,1} \geq \dots \geq \lambda_{\eta,a}.$$

These are highest weights for the full general linear group: negative entries and central determinant characters are retained. The extrema of the distinguished cocharacter on a factor of highest weight λ_{η} are $\lambda_{\eta,1}$ and $\lambda_{\eta,a}$, by the weight-polytope description and the permutation Weyl group.

Galois acts transitively on the conjugate embedding pairs. Since the decomposition of V comes from its rational E -action, this action transports standard modules to standard modules or

their duals. On highest weights the latter possibility replaces $(\lambda_1, \dots, \lambda_a)$ by $(-\lambda_a, \dots, -\lambda_1)$; it introduces no additional central twist. Transporting each factor to the distinguished one, rationality of H and its level bound therefore imply

$$w_\eta := \max\{\lambda_{\eta,1}, -\lambda_{\eta,a}, 0\} \leq b.$$

In particular this bound controls signed central weights as well as differences of adjacent weights.

Set $c = \max_\eta w_\eta$. For $1 \leq j \leq c$ put

$$h_{\eta j}^+ = \#\{i : \lambda_{\eta,i} \geq j\}, \quad h_{\eta j}^- = \#\{i : \lambda_{\eta,i} \leq -j\}, \quad k_j = \sum_\eta (h_{\eta j}^+ + h_{\eta j}^-).$$

The positive and negative columns can be placed in the same exterior power, because its direct-sum decomposition contains

$$\bigotimes_\eta \left(\bigwedge^{h_{\eta j}^+} W_\eta \otimes \bigwedge^{h_{\eta j}^-} W_\eta^* \right) \subset \bigwedge^{k_j} V_{\mathbb{C}}.$$

The highest weight of $\bigwedge^h W$ has h initial entries equal to 1, and that of $\bigwedge^h W^*$ has h final entries equal to -1 . Summing these weights over j gives exactly λ_η . Their tensor highest-weight vector therefore produces the desired Cartan component, including its central character. Thus the product of these c full exterior carriers contains the chosen constituent. The cost is the maximum of the positive and negative widths, not their sum.

Rational descent of the carrier. In both cases $c = 0$ would make the chosen constituent trivial. A rational simple representation with a trivial complex constituent is trivial, by scalar extension of invariants, so our nontriviality assumption gives $c \geq 1$. A factor attaining the maximum width has a nonempty column for each $1 \leq j \leq c$; therefore $k_j \geq 1$. The displayed direct-sum exterior decompositions also give $k_j \leq \dim_{\mathbb{Q}} V$.

Finally, for $\mathbf{k} = (k_1, \dots, k_c)$, keep the rational carrier $C_{\mathbf{k}} = \bigotimes_j \bigwedge^{k_j} V$, rather than selecting any of its complex summands. Sharing a complex constituent yields

$$\mathrm{Hom}_G(C_{\mathbf{k}}, H) \otimes_{\mathbb{Q}} \mathbb{C} = \mathrm{Hom}_{G_{\mathbb{C}}}((C_{\mathbf{k}})_{\mathbb{C}}, H_{\mathbb{C}}) \neq 0.$$

The equality follows by scalar extension of invariants in $(C_{\mathbf{k}})^* \otimes H$. A nonzero rational equivariant map is surjective because H is rationally simple, and it is a Hodge map because G contains the Hodge cocharacter. This handles nonsplit forms, Galois-conjugate constituents and their multiplicities without assuming any complex projector rational. $\square$

Proof of the generalized Hodge assertion in Theorem 1.1. Put $V = T(1)$ and $G = \mathrm{Hg}(V)$, with the group descriptions (2.11) and (2.12). The ordinary Hodge assertion just proved gives rational HC for every power of S under the same exact-map hypothesis. We use that conditional conclusion in both endomorphism-field cases.

Fix a rational sub-Hodge structure $H \subset H^{2d}(S^m, \mathbb{Q})$ of Hodge coniveau at least p , where $0 \leq p \leq d$. By semisimplicity it suffices to treat H simple. The K3 Künneth decomposition shows that $H^{2d}(S^m, \mathbb{Q})(d)$ is a direct sum of tensor powers of V : its other factors, coming from H^0 , the Néron–Severi part of H^2 , and H^4 , are Tate and become trivial after the total normalization. Thus $H(d)$ is a rational simple representation of G , of Hodge level at most $2(d-p)$.

If $H(d)$ is trivial, H has type (d, d) . Ordinary HC and a finite basis of cycle classes put H in $N^d H^{2d}(S^m, \mathbb{Q}) \subset N^p H^{2d}(S^m, \mathbb{Q})$. Otherwise Lemma 2.5 supplies a tuple $\mathbf{k} = (k_1, \dots, k_c)$ and a surjective rational Hodge map

$$f : C_{\mathbf{k}}(s) \longrightarrow H(d), \quad C_{\mathbf{k}} = \bigotimes_{i=1}^c \bigwedge^{k_i} T \subset H^{2s}(S^s, \mathbb{Q}), \quad s = \sum_i k_i, \quad c \leq d-p.$$

Extend f by zero on a rational Hodge complement of $C_{\mathbf{k}}(s)$ in $H^{2s}(S^s, \mathbb{Q})(s)$, which exists by polarizability. Poincaré duality identifies this extended map to $H(d)$ with a rational Hodge class of codimension $s + d$ on $S^s \times S^m$. Ordinary HC for S^{s+m} therefore gives

$$C_f \in \mathrm{CH}^{s+d}(S^s \times S^m)_{\mathbb{Q}}$$

realizing it.

Retain $Z = A^2$ and $n = \dim Z$ from Proposition 2.1. For each column take the return R_{k_i} of (2.10). Their exterior product, with factors reordered, gives

$$R_{\mathbf{k}} \in \mathrm{CH}^{nc+s-c}(Z^c \times S^s)_{\mathbb{Q}}, \quad (R_{\mathbf{k}})_* : H^{2c}(Z^c, \mathbb{Q}) \longrightarrow H^{2s}(S^s, \mathbb{Q}),$$

whose image contains $C_{\mathbf{k}}$. Indeed the tensor product of the F_{k_i} is a right inverse on $C_{\mathbf{k}}$. Composition gives

$$C_f \circ R_{\mathbf{k}} \in \mathrm{CH}^{nc+d-c}(Z^c \times S^m)_{\mathbb{Q}},$$

since $(nc + s - c) + (s + d) - 2s = nc + d - c$. Its image on $H^{2c}(Z^c, \mathbb{Q})$ contains H , and its support has dimension

$$nc + 2m - (nc + d - c) = 2m - d + c.$$

Lemma 2.2 yields

$$H \subset N^{d-c}H^{2d}(S^m, \mathbb{Q}) \subset N^pH^{2d}(S^m, \mathbb{Q}).$$

For a nonsimple substructure, take the finite union of the supports obtained for its simple summands. This treats every actual occurrence, including multiplicities and substructures embedded diagonally across isomorphic Künneth summands.

For the converse, put $X = S^m$ and let $B \subset X$ be any closed support of codimension at least p , including a singular or non-equidimensional support. For each irreducible component B_i choose a smooth projective resolution and let $q_i : \tilde{B}_i \rightarrow X$ be its proper map; put $c_i = \mathrm{codim}_X B_i \geq p$. Deligne's exactness theorem for a proper smooth variety and a closed support, using the disjoint union of these resolutions as a proper smooth surjective source [13, Proposition 8.2.7 and Corollary 8.2.8], gives

$$\ker(H^j(X, \mathbb{Q}) \longrightarrow H^j(X \setminus B, \mathbb{Q})) = \sum_i (q_i)_* H^{j-2c_i}(\tilde{B}_i, \mathbb{Q}).$$

Each proper Gysin map is a Hodge morphism $H^{j-2c_i}(\tilde{B}_i, \mathbb{Q})(-c_i) \rightarrow H^j(X, \mathbb{Q})$. The source before twisting has nonnegative Hodge indices, since $\tilde{B}_i$ is smooth projective; after twisting both indices are at least $c_i \geq p$. Taking sums over B therefore proves that geometric coniveau p has Hodge coniveau at least p .

Thus the two filtrations agree. If $p > d$, the same reverse inclusion forces $N^pH^{2d}(S^m, \mathbb{Q}) = 0$, since a pure structure of weight $2d$ cannot have both indices at least p . Cases $p \leq 0$ are immediate. Odd cohomology of S^m vanishes. The result therefore holds in every degree, not only in middle cohomology or the pure transcendental tensor summand. $\square$

3. ABELIAN DOUBLE COVERS AND THEIR PERIODS

We construct a rank-eight variation of K3 type with quadratic space V_P and show that its period map has rank six. The variation occurs in the cohomology of a double cover of an abelian surface. A resolved inversion quotient supplies the quadratic form that will be used in the K3 realization.

3.1. The double cover. Work on a smooth finite-level chart in the moduli space of principally polarized abelian surfaces. Let B be an indecomposable principally polarized abelian surface, let $M = \mathcal{O}_B(H)$ be a symmetric line bundle defining the principal polarization, and choose $\chi \in \text{Pic}^0(B)[4]$ of exact order four. The symmetric representative M and the torsion point χ add only finite choices. Products of line bundles will be written without tensor symbols when convenient.

We recall the theta facts needed below. Ample vanishing and Riemann–Roch on B give

$$h^0(B, M\chi) = h^0(B, M\chi^{-1}) = 1, \quad h^0(B, M^2) = 4,$$

with no higher cohomology for these bundles. The linear system $|M^2|$ is basepoint-free: the sums of opposite translates of H belong to it, and for any point of B one can choose both translates to avoid that point. The normalized symmetric linearization of M^2 fixes every section. For example, in period coordinates its four second-order theta sections are $\theta[\epsilon; 0](2z, 2\tau)$, with $\epsilon \in (\frac{1}{2}\mathbb{Z}/\mathbb{Z})^2$; reindexing the defining series by $n \mapsto -n - 2\epsilon$ proves that they are even. Their distinct Fourier congruence classes give a basis. Changing the symmetric theta representative tensors M by two-torsion and leaves its square, with this normalized linearization, unchanged. These descriptions agree with the standard theta constructions in [28, 4].

Choose $[d] \in \mathbf{P}H^0(B, M^2)$, and write $D = (d = 0)$. We restrict to the nonempty open where D is smooth and avoids $B[2]$; nonemptiness follows from Bertini and basepoint-freeness, by avoiding the finitely many evaluation hyperplanes at $B[2]$. Every such D is invariant under inversion. The parameter space is thus an open in a projective 3-space bundle over the three-dimensional abelian-data chart, and has dimension six.

All families below are formed on algebraic affine charts of this projective bundle. After trivializing its rank-four section bundle, choose a nonzero relative functional ℓ . On the chart $\ell(d) \neq 0$ use the actual normalized universal section $\ell(d) = 1$. On a simply connected product polydisc, two such normalizations differ by a nonvanishing scalar with a holomorphic square root. Rescaling the cover coordinate by that square root identifies the resulting covers, their involutions, and their rational eigenspaces.

Let

$$\pi : B' \longrightarrow B, \quad \pi_* \mathcal{O}_{B'} = \mathcal{O}_B \oplus \chi^2$$

be the degree-two isogeny attached to χ^2 . Choose a trivialization of $\pi^* \chi^2$ and form

$$(3.1) \quad L = \pi^*(M\chi), \quad f : X = \text{Spec}_{B'}(\mathcal{O}_{B'} \oplus L^{-1}) \longrightarrow B', \quad z^2 = \pi^*d.$$

The chosen trivialization identifies L^2 with π^*M^2 , so π^*d defines the multiplication in this algebra. The surface X is smooth. Write δ for its sheet involution.

The line bundle $\pi^*\chi$ has order two, so L is symmetric. Normalize its symmetric linearization at the origin. Its square and the pullback of the normalized linearization of M^2 agree at the origin and hence everywhere, since an automorphism of a line bundle on an abelian variety is scalar. Inversion therefore lifts to an involution j of X commuting with δ . Put

$$U_X = H^2(X, \mathbb{Q})^{j=1, \delta=-1}.$$

3.2. The eigenspace and its quadratic form.

Proposition 3.1. *There is a smooth double cover $Y \rightarrow K = \text{Km}(B')$, with K the resolved Kummer surface, satisfying the following assertions.*

(i) *The anti-invariant lattice*

$$V = H^2(Y, \mathbb{Z})^{\delta=-1}$$

(with torsion removed) is even, has signature $(2, 14)$, and has 2-elementary discriminant group. For fixed (B, M, χ) , monodromy acts trivially on this discriminant group.

- (ii) *There are eight mutually orthogonal algebraic anti-invariant classes $a_1, \dots, a_8$, each of square -4 . Their perpendicular*

$$U = \langle a_1, \dots, a_8 \rangle^\perp \subset V_{\mathbb{Q}}$$

is a Hodge structure of signature $(2, 6)$ and, with the intersection form inherited from Y , is rationally isometric to

$$V_P = (\mathbb{U} \oplus \mathbb{U}(2) \oplus D_4(-1))_{\mathbb{Q}}.$$

- (iii) *Resolved pullback gives a flat Hodge isomorphism $\beta : U \xrightarrow{\sim} U_X$ with*

$$(\beta x, \beta y)_X = 2(x, y)_Y.$$

The algebraic projector

$$(3.2) \quad e = \frac{1}{4}(\Delta_X + \Gamma_j - \Gamma_\delta - \Gamma_{j\delta}) \in \text{CH}^2(X \times X)_{\mathbb{Q}}$$

has image U_X in degree two and is zero in every other degree.

The constructions are compatible with variation on the indicated smooth parameter opens.

Proof. We first determine the fibre signs of the normalized involution on L . Write $B = \mathbb{C}^2/\Lambda$ and represent χ by a character $c : \Lambda \rightarrow \mu_4$ of exact order four. Then $B' = \mathbb{C}^2/\Lambda'$, where $\Lambda' = \ker(c^2)$. Choose $\lambda \in \Lambda \setminus \Lambda'$ and put $k = \lambda \bmod \Lambda' \in \ker \pi$. Translation by k pairs the sixteen points of $B'[2]$. The fibre sign of π^*M is the same at x and $x + k$, since these points have the same image in B . For the flat factor, representing x by v with $2v \in \Lambda'$ shows that its sign changes by

$$\frac{c(2v + 2\lambda)}{c(2v)} = c(2\lambda) = -1.$$

Thus the signs of L in each pair are opposite: eight are positive and eight negative.

The branch curve is absent near these sixteen points. At a positive-sign point, j preserves each local sheet of X and acts by $(u, v) \mapsto (-u, -v)$. The quotient has two ordinary double points, whose resolutions give two disjoint (-2) -curves over the corresponding exceptional curve of K . At a negative-sign point, j exchanges the local sheets. The quotient is smooth, and its map to $B'/\{\pm 1\}$ is locally $\mathbb{C}^2 \rightarrow \mathbb{C}^2/\{\pm 1\}$. Blowing up this smooth source point resolves the map to K and gives a double cover branched along the exceptional curve of K . These local models construct the smooth double cover $Y \rightarrow K$.

Set $D' = \pi^{-1}D$. It is smooth and connected, since it is ample, and $(D')^2 = 16$. Adjunction gives $g(D') = 9$. Inversion acts freely on D' , so its quotient has genus five. The branch divisor of $Y \rightarrow K$ is the disjoint union of that quotient and the eight exceptional curves belonging to the negative signs. Therefore

$$e(Y) = 2e(K) - ((2 - 2 \cdot 5) + 8 \cdot 2) = 48 - 8 = 40.$$

The cover algebra, Serre duality, and ample vanishing give $H^1(B', L^{-1}) = 0$ and hence $H^1(X, \mathbb{Q}) = f^*H^1(B', \mathbb{Q})$. Inversion is -1 on this space. Taking quotient invariants, and using invariance of H^1 under the intervening blowups, gives $q(Y) = 0$. For holomorphic two-forms,

$$H^{2,0}(X) = H^{2,0}(B') \oplus H^0(B', L), \quad H^0(B', L) = H^0(B, M_\chi) \oplus H^0(B, M_\chi^{-1}).$$

The first summand is invariant under both involutions. Inversion replaces χ by χ^{-1} ; the trivialization of $\pi^*\chi^2$ used to identify the pullbacks has the nontrivial deck character, so it exchanges the two one-dimensional summands in $H^0(B', L)$. They are sheet-anti-invariant. Thus $p_g(Y) = 2$. Consequently $b_2(Y) = 38$, and Hodge index gives signature $(5, 33)$ on $H^2(Y)$.

The sheet-invariant rational cohomology is the pullback of $H^2(K, \mathbb{Q})$, with pairing multiplied by two, and has signature $(3, 19)$. Its perpendicular $V_{\mathbb{Q}}$ has signature $(2, 14)$. The canonical

class K_Y is invariant. For $v \in V$ we have $v \cdot K_Y = 0$, and Wu's characteristic parity formula [27, Theorem 11.14 and Problem 14-B]

$$v^2 \equiv v \cdot K_Y \pmod{2}$$

shows that V is even.

For the integral assertions, let $\mathcal{H} = H^2(Y, \mathbb{Z})/\text{torsion}$ and let $\mathcal{H}^+$ be its invariant lattice. Integral Poincaré duality makes $\mathcal{H}$ unimodular. Both eigensublattices are primitive, and

$$2x = (x + \delta x) + (x - \delta x) \quad \text{for every } x \in \mathcal{H}.$$

The quotient $\mathcal{H}/(\mathcal{H}^+ \oplus V)$ is therefore killed by two. Unimodular gluing identifies this quotient with each of $A_{\mathcal{H}^+}$ and A_V , proving that their discriminant groups are 2-elementary. For fixed B' , pullback from the fixed Kummer surface makes the invariant rational cohomology pointwise constant. Its integral sublattice and its discriminant group are consequently fixed as well. The gluing identification is equivariant, so monodromy acts trivially on A_V .

At a positive-sign point, denote the two exceptional curves on Y by E_i^+ and E_i^- . They are exchanged by δ , and

$$a_i = [E_i^+] - [E_i^-], \quad a_i^2 = -4.$$

The eight differences are mutually orthogonal. Parallel transport can permute their indices or exchange the two curves, but preserves their span, so its perpendicular U is flat. Near a degeneration at which the branch passes through a positive-sign point, the two labels are governed by a scalar square root; the local action on a_i changes its sign.

The same local models identify U with U_X . Resolving the two quotient nodes at a positive-sign point adds precisely one sheet-anti-invariant exceptional direction, namely a_i ; the blowup at a negative-sign point adds only a sheet-invariant direction. On a common resolution of $X \dashrightarrow Y$, pullback of a class in U is orthogonal to all exceptional classes over X , and hence comes from $H^2(X, \mathbb{Q})$. Its image is exactly the $(j, \delta) = (1, -1)$ eigenspace. The degree-two projection formula gives

$$(\beta x, \beta y)_X = 2(x, y)_Y.$$

In particular, U has signature $(2, 6)$. We retain its Y -pairing throughout the rational quadratic-form and K3 arguments.

The correspondence (3.2) is the product of the commuting projectors $(1 + j)/2$ and $(1 - \delta)/2$. The sheet involution fixes H^0 and H^4 , and the calculation of $H^1(X, \mathbb{Q})$ shows that it fixes H^1 ; by duality it fixes H^3 as well. Thus e vanishes outside degree two and has image exactly U_X there.

We now determine the rational form on U with its Y -pairing. The invariant lattice $\mathcal{H}^+$ contains the pullback lattice $H^2(K, \mathbb{Z})(2)$ with finite index, necessarily a power of two. The latter has absolute determinant 2^{22} , so $|\det \mathcal{H}^+|$ is a square in $\mathbb{Q}^*/\mathbb{Q}^{*2}$. Unimodular gluing gives the same absolute determinant for V . Since the negative index of V is fourteen, its determinant is positive. Splitting off the eight orthogonal lines of square -4 therefore gives

$$\det(U) = 1 \quad \text{in } \mathbb{Q}^*/\mathbb{Q}^{*2}.$$

For an odd prime p , the projectors $(1 \pm \delta)/2$ split $\mathcal{H} \otimes \mathbb{Z}_p$ orthogonally, so $V \otimes \mathbb{Z}_p$ is unimodular. The a_i span a unimodular $\mathbb{Z}_p$ -lattice, since their Gram determinant is $(-4)^8$. Orthogonal projection using its inverse Gram matrix is integral over $\mathbb{Z}_p$. Its perpendicular thus gives a unimodular $\mathbb{Z}_p$ -lattice in $U \otimes \mathbb{Q}_p$.

The lattice T_P has rank eight, signature $(2, 6)$, and determinant $(-1)(-4)(4) = 16$; it too is unimodular at every odd prime. A unimodular quadratic form over $\mathbb{Z}_p$, for odd p , diagonalizes with unit entries, and the Hilbert symbol of two units is 1. Hence both rational forms have Hasse invariant 1 at every odd prime. Their real Hasse invariants are $(-1)^{\binom{6}{2}} = -1$. Hilbert reciprocity then gives the same invariant, namely -1 , at 2. The local classification and the rational local-to-global isometry criterion now give $U \simeq V_P$. For a symmetric pairing q we use the quadratic

polynomial $Q(x) = q(x, x)$, whose diagonal coefficients are those of q itself in the conventions of [30, Equations (3.1)–(3.2)]. The precise Hilbert-symbol, reciprocity, and classification results are [30, Theorems A.4.4, A.4.6, 3.3.1, and 3.3.5].

The covers, blowups, quotients, and graph correspondences are all relative after the finite choices and restriction to the stated smooth opens. $\square$

3.3. The character eigenspace and the period differential.

Lemma 3.2. *Put $E = T_B(-\log D)$. There is a flat complex identification of U_X with a character eigenspace having Hodge components*

$$(3.3) \quad U_X^{p,q} \simeq H^q(B, \Omega_B^p(\log D) \otimes M^{-1}\chi), \quad p + q = 2.$$

In particular, $U_X^{2,0} = H^0(B, M_\chi)$ is one-dimensional. Fix a nonzero translation-invariant volume form Ω on B , and let $\theta \in H^0(B, M_\chi)$ represent this line. Under (3.3), the period differential is the map on H^1 induced by

$$(3.4) \quad E \xrightarrow{\theta} E \otimes M_\chi \simeq \Omega_B^1(\log D) \otimes M^{-1}\chi.$$

The Kodaira–Spencer map identifies the tangent space of the full six-dimensional family $(B, M, \chi, [d])$ with $H^1(B, E)$.

Proof. Let $\alpha : \tilde{B} \rightarrow B$ be the cyclic degree-four isogeny trivializing χ , with generator g , and let $Z \rightarrow B$ be the untwisted double cover $w^2 = d$ defined using M . On $\tilde{X} = Z \times_B \tilde{B}$, let δ change the sign of w , and let g act through the isogeny. A trivialization of $\alpha^*\chi$ identifies

$$X = \tilde{X} / \langle \delta g^2 \rangle.$$

On sheet-anti-invariant cohomology, invariance under δg^2 selects exactly the g -eigenvalues i and $-i$. Inversion conjugates g to g^{-1} and exchanges these eigenspaces. Projection to either one therefore identifies their j -invariant part with that eigenspace over $\mathbb{C}$. Choose the generator so that the selected character sheaf is χ ; the other convention replaces χ by χ^{-1} . These finite-group projectors commute with parallel transport.

The eigensheaves can be read directly from the local double-cover equation. If $D = (x = 0)$ and $w^2 = x$, the sheet-odd holomorphic forms are generated over the base by

$$w; \quad w \, dy, \quad w \frac{dx}{x}; \quad w \frac{dx}{x} \wedge dy$$

in degrees zero, one, and two. The odd generator has the transition rule of M^{-1} . Thus the sheet-odd direct image of holomorphic p -forms is $\Omega_B^p(\log D)M^{-1}$; selecting the character tensors this by χ . Its logarithmic flat differential has local term $\frac{1}{2}d \log x$, together with the flat connection on χ . The Hodge decomposition of the smooth projective finite cover, with the character projector applied, gives (3.3).

Infinitesimal changes of local coordinates for the pair (B, D) give a Čech cocycle of logarithmic vector fields in E . Locally, a logarithmic field v satisfying $v(f) = af$ lifts on $w^2 = f$ to $v + (a/2)w\partial_w$. This lift is unique: an infinitesimal vertical change preserving $w^2 = f$ has coefficient b with $2b = 0$. Contraction with the sheet-odd two-form computes the Kodaira–Spencer action on the Hodge line. The local generators just displayed identify this contraction with multiplication by θ in (3.4). The bundle isomorphism there is the rank-two determinant identity

$$E \otimes \mathcal{O}_B(D) \simeq \Omega_B^1(\log D), \quad \mathcal{O}_B(D) = M^2.$$

The chosen finite-order line bundles have unique infinitesimal extensions with their torsion data, so they add no tangent directions.

It remains to identify all tangent directions of our family. Adjunction and the logarithmic normal sequence give

$$0 \longrightarrow E \longrightarrow T_B \longrightarrow \mathcal{O}_D(D) = K_D \longrightarrow 0.$$

An infinitesimal translation preserving D is zero: its flow would lie in the stabilizer of $\mathcal{O}_B(D)$, the finite kernel of the polarization isogeny. Thus $H^0(E) = 0$, and the cohomology sequence yields

$$0 \longrightarrow \frac{H^0(K_D)}{H^0(T_B)} \longrightarrow H^1(E) \longrightarrow \ker(H^1(T_B) \longrightarrow H^1(K_D)) \longrightarrow 0.$$

Here $h^0(K_D) = 5$, $h^0(T_B) = 2$, and $h^1(T_B) = 4$. The sequence

$$0 \longrightarrow \mathcal{O}_B \xrightarrow{d} M^2 \longrightarrow K_D \longrightarrow 0$$

identifies $H^1(K_D)$ with $H^2(\mathcal{O}_B)$. Under this identification both normal-sequence boundary maps are contraction with $c_1(D)$. Indeed, choose local frames $e_j = g_{ij}e_i$ of M^2 and write $d = f_i e_i$, so $f_i = g_{ij}f_j$. Pulling the displayed extension back along $T_B \rightarrow K_D$, the local lift of a vector field v is $v(f_i)e_i$. Differences of these lifts, divided by d , are $v(\log g_{ij})$. The pulled-back extension therefore has class $c_1(D) \in H^1(\Omega_B^1)$, and its connecting maps in degrees zero and one are the asserted contractions. In degree one the target is the one-dimensional $H^2(\mathcal{O}_B)$. This contraction is nonzero and its three-dimensional kernel is exactly the tangent space to polarized deformations of B . Hence $h^1(E) = 3 + 3 = 6$.

Extending M imposes the same condition, since $c_1(M) = \frac{1}{2}c_1(M^2)$. Among its infinitesimal extensions the symmetric representative is unique: differences form an $H^1(\mathcal{O}_B)$ -torsor on which inversion acts by -1 . The order-four point χ and the associated isogenies extend uniquely because torsion of the relative Picard variety is finite étale. Thus the root and torsion data create no additional infinitesimal condition.

For fixed B , the section directions are $H^0(M^2)/\langle d \rangle$. Their images in $H^0(K_D)$ have zero boundary in $H^1(\mathcal{O}_B)$. On a translation vector v , that same boundary is $v \lrcorner c_1(D)$, and the map

$$H^0(T_B) \longrightarrow H^1(\mathcal{O}_B), \quad v \longmapsto v \lrcorner c_1(D)$$

is an isomorphism because D is ample. Section directions therefore identify with the entire quotient $H^0(K_D)/H^0(T_B)$. Every polarized first-order deformation lifts to our projective bundle of sections, by ample vanishing and base change. The induced map on the right-hand term is the polarized Kodaira–Spencer isomorphism. The two identifications in the exact sequence prove that the full Kodaira–Spencer map is an isomorphism. $\square$

3.4. Full rank of the period map.

Proposition 3.3. *The period map of U , equivalently of U_X , has rank six on a nonempty Zariski-open subset of the six-dimensional parameter space. For a very general parameter, its rational Hodge structure is simple, has no nonzero rational $(1,1)$ -class, and has Hodge group $\mathrm{SO}(U)$.*

Proof. Let $F = (\theta = 0)$. It is a translate of the theta curve of B , hence is a smooth genus-two curve with $\mathrm{Jac}(F) \simeq B$. For clarity, these facts use indecomposability as follows. An integral singular theta curve would have normalization of genus at most one and could not generate the abelian surface. If the theta divisor is not integral, ampleness and square two force two reduced elliptic components meeting once, and hence the product polarization: each component has positive integral intersection with H and nonnegative even self-intersection. These possibilities are excluded. Adjunction gives genus two, and the restriction pairing on $H^1(B, \mathbb{Z})$, obtained by cup product with the principal theta class, is unimodular. The induced map on Jacobians, of equal dimension, is therefore an isomorphism.

Set $K = K_F$ and $\eta = \chi^{-2}|_F$. Restriction $\text{Pic}^0(B) \rightarrow \text{Pic}^0(F)$ is an isomorphism, so η is nontrivial two-torsion. Adjunction gives

$$(M_\chi)|_F = K, \quad M^2|_F = K^{\otimes 2} \otimes \eta.$$

For a transverse D , multiplication by θ yields

$$(3.5) \quad 0 \longrightarrow E \longrightarrow E \otimes M_\chi \longrightarrow (E \otimes M_\chi)|_F \longrightarrow 0.$$

We shall prove that the last sheaf has no global sections for general d . By the cohomology sequence and Lemma 3.2, this implies injectivity of the period differential on all six tangent directions.

Put $Z = D \cap F$, a reduced divisor of degree four. Restricting the logarithmic normal sequence to F is exact because the intersection is transverse. After tensoring by K , its map on global sections is

$$(3.6) \quad A_d : H^0(F, K)^{\oplus 2} \longrightarrow H^0(Z, (K^{\otimes 3} \otimes \eta)|_Z), \quad v \longmapsto (v \lrcorner dd)|_Z.$$

Its kernel is $H^0(F, (E \otimes M_\chi)|_F)$, and source and target both have dimension four. Differentiating a section at its zero scheme is intrinsic, so A_d is independent of local bundle frames.

The restriction sequence for F , together with ample vanishing, gives a surjection

$$H^0(B, M^2) \longrightarrow H^0(F, K^{\otimes 2} \otimes \eta)$$

whose kernel is generated by $\theta\theta'$, where $0 \neq \theta' \in H^0(B, M_\chi^{-1})$. Its restriction $t = \theta'|_F$ is a nonzero section of $K \otimes \eta$: a section vanishing along F would give a section of the nontrivial flat bundle χ^{-2} . The bundle $K^{\otimes 2} \otimes \eta$ is basepoint-free by Riemann–Roch on the genus-two curve. We may therefore fix a general restriction $s = d|_F$ whose four zeros are simple and avoid the zeros of t .

Replacing d by $d + \lambda\theta\theta'$ fixes s and Z and replaces A_d by $A_d + \lambda A'$. Choose a basis v_1, v_2 of translation-invariant vector fields on B . The normal derivatives

$$\theta_i = (v_i \lrcorner d\theta)|_F, \quad i = 1, 2,$$

form a basis of $H^0(F, K)$: the map to this space is injective because the theta divisor has finite translation stabilizer, and both spaces have dimension two. In these coordinates,

$$A'(a, b) = t(a\theta_1 + b\theta_2)|_Z.$$

The canonical pencil of a genus-two curve gives an isomorphism $\text{Sym}^2 H^0(K) \simeq H^0(K^{\otimes 2})$. Moreover, $K^{\otimes 2}(-Z) \simeq \eta$ has no section, so restriction to Z is injective. Since t does not vanish on Z , we obtain

$$\text{im } A' = t H^0(K^{\otimes 2})|_Z, \quad \text{rank } A' = 3.$$

The kernel is spanned by the skew gradient $k = (\theta_2, -\theta_1)$. The canonical pencil has no base point, so this is a nowhere-zero tangent section of $T_F \otimes K \simeq \mathcal{O}_F$.

Under $T_F \otimes K \simeq \mathcal{O}_F$, the vector k is a constant $c \neq 0$. With the adjunction convention $\text{Res}(\Omega/h)$, where $\theta = he$, this constant is $c = -\Omega(v_1, v_2)$: writing $\Omega = J dh \wedge du$ identifies k with $-\Omega(v_1, v_2)\partial_u \otimes du$. Consequently $A_d(k) = c d_F s|_Z$. Here $d_F s|_Z$ is an intrinsic section of $(K^{\otimes 3} \otimes \eta)|_Z$. To show that this vector is outside $\text{im } A'$, consider the sequence

$$0 \longrightarrow \eta \xrightarrow{\cdot s} K^{\otimes 2} \longrightarrow K^{\otimes 2}|_Z \longrightarrow 0,$$

where $\eta^{\otimes 2} \simeq \mathcal{O}_F$ is understood. The connecting class of $(d_F s)/t|_Z$ pairs by Serre duality with $t \in H^0(K \otimes \eta)$. This pairing is computed by local principal parts. Near $z \in Z$, choose a coordinate u and an η -frame ϵ with $\epsilon^2 = 1$, and write $s = f(u)(du)^2\epsilon$ and $t = g(u)du\epsilon$. Here $f(z) = 0$, $f'(z) \neq 0$, and $g(z) \neq 0$. A local lift of $(d_F s)/t|_Z$ is $(f'/g)(du)^2$. Dividing by s gives the η -valued principal part $[f'/(gf)]\epsilon$; pairing it with t gives $(f'/f)du$. Thus

$$\langle \partial((d_F s)/t|_Z), t \rangle = \sum_{z \in Z} \text{Res}_z \frac{df}{f} = 4,$$

up to the fixed nonzero normalization of Serre duality. Each zero is simple. Changing a frame adds a regular differential and leaves the residues unchanged. The formula concerns the four local principal parts; it does not require a global meromorphic differential with those four poles. The connecting class is nonzero, and hence $A_d(\ker A')$ is not contained in $\operatorname{im} A'$.

Choose source and target bases in which A' has an invertible 3×3 block and is zero in the remaining row and column. The coefficient of λ^3 in $\det(A_d + \lambda A')$ is the determinant of that block multiplied by the induced map $\ker A' \rightarrow \operatorname{coker} A'$. It is nonzero by the preceding pairing. Thus $A_{d+\lambda\theta\theta'}$ is invertible for general λ . This construction shows that invertibility is a nonempty open condition in the space of sections. Intersecting with the nonempty open of smooth divisors avoiding $B[2]$ gives the required general d .

It follows from (3.5) that $H^1(E) \rightarrow H^1(E \otimes M\chi)$ is injective. By Lemma 3.2, its source is the entire six-dimensional tangent space. The type-IV period domain of a quadratic space of signature $(2, 6)$ has complex dimension six; the period map is therefore a local isomorphism at these points. The rank condition is Zariski open, since it is the rank condition on the algebraic Kodaira–Spencer map between Hodge bundles.

Finally, on a type-IV period domain the Hodge circles rotate its oriented positive two-planes. These rotations generate the identity component of $\operatorname{SO}(U_{\mathbb{R}})$, and rotations from any nonempty open have the same Zariski closure. Hence the Hodge circles in our open period chart are not all contained in any proper rational algebraic subgroup of $\operatorname{SO}(U)$.

We recall why the exceptional conditions may be taken complex analytic. A proper rational algebraic subgroup stabilizes a rational line in a tensor representation that is not stabilized by $\operatorname{SO}(U)$. Indeed, take a finite-degree representation of regular functions containing generators of the subgroup's defining ideal, intersect it with that ideal, and take the top exterior power. The resulting line has precisely the required stabilizer. Polynomial representations and semisimplicity place this representation in a sum of tensor powers and duals of $U(1)$. A Hodge circle stabilizing a rational line fixes it pointwise, because it is connected and compact. Thus it forces a specified rational tensor to have type $(0, 0)$. Each such condition is a closed complex-analytic Hodge locus, and it is proper unless that tensor is fixed by all the Hodge circles in the domain. There are only countably many rational tensors. Outside their proper loci, $\operatorname{Hg}(U) = \operatorname{SO}(U)$. The standard representation of $\operatorname{SO}(U)$ is absolutely irreducible and has no invariants. Simplicity of U and the absence of nonzero rational $(1, 1)$ -classes follow. $\square$

4. COHERENT HALF SUMS AND DOMINATION BY A PRODUCT OF CURVES

We retain the data (B, M, χ, D) of Section 3, with $M = \mathcal{O}_B(H)$ and $D \in |2H|$ smooth and disjoint from $B[2]$. The regular Gauss fibers of D consist of four opposite pairs. Coherently taking half the sum of one point from each pair produces two curves mapping to B . We prove that their addition map has total degree eight and pulls D back to an even divisor. Separate covers of the two curves then give a morphism to the surface X of Equation (3.1).

4.1. The generic Gauss pencil.

Lemma 4.1. *For (B, D) in a nonempty Zariski-open subset of the parameter space, D has no nontrivial translation stabilizer. Its quotient $C = D/\{\pm 1\}$ is a smooth genus-three curve, and the connected étale cover $D \rightarrow C$ is defined by a nontrivial line bundle $\epsilon \in \operatorname{Pic}^0(C)[2]$. The Gauss map is the composite*

$$D \longrightarrow C \xrightarrow{|K_C \otimes \epsilon|} \mathbf{P}^1,$$

of degrees two and four. No fiber of the degree-four map has all its multiplicities even. The good open remains nonempty on every connected component obtained by adjoining the finite theta, torsion, and level choices used to define X .

Proof. The smooth ample divisor D is connected: the divisor sequence and $H^1(B, \mathcal{O}_B(-D)) = 0$ give $H^0(D, \mathcal{O}_D) = \mathbb{C}$. Thus it is irreducible. Adjunction gives $2g(D) - 2 = D^2 = 8$. Inversion acts freely because D avoids $B[2]$, so Riemann–Hurwitz gives $g(C) = 3$. Connectedness of the double cover makes ϵ nontrivial.

The curve D generates B : containment in a translate of a proper abelian subvariety would make D an elliptic curve with square zero. Hence $J(D) \rightarrow B$ is surjective and invariant one-forms on B restrict injectively to D . They are anti-invariant under inversion and span the two-dimensional space $H^0(C, K_C \otimes \epsilon)$, whose dimension follows from Riemann–Roch. They have no common zero on D , since its tangent line is a nonzero subspace of T_0B . They therefore define the basepoint-free pencil $|K_C \otimes \epsilon|$, of degree four. Identifying its target with the projective line of tangent directions identifies the composite with the Gauss map.

We justify that these pairs (C, ϵ) are general. The open projective bundle of smooth curves in $|2H|$ over the three-dimensional level moduli of principally polarized abelian surfaces has dimension six and maps algebraically to the moduli space $\mathcal{R}_3$ of genus-three curves with nonzero two-torsion. Its fibers are countable. In fact, fixing (C, ϵ) fixes the double cover D ; any map of that curve to an abelian surface is a homomorphism from $J(D)$ followed by a translation. Surjective homomorphisms, up to isomorphism of their targets, form a countable set: the connected kernel is determined by a rational subspace of $H_1(J(D), \mathbb{Q})$, and the further finite isogeny choices are countable. For a fixed homomorphism, requiring inversion on the image to be the given involution on D imposes an equation $2b = \text{constant}$ on the translation b , giving finitely many choices. Polarizations are discrete, and $[D] = 2[H]$ fixes their numerical class; the remaining symmetric theta and level choices are finite. A countable fiber of a finite-type morphism over $\mathbb{C}$ has dimension zero. Since $\mathcal{R}_3$ is irreducible of dimension six, the map is dominant on each component of its six-dimensional source. Irreducibility of $\mathcal{R}_3$ follows from the transitive symplectic monodromy on nonzero two-torsion. The Torelli theorem for curves, with their integral principal polarization [22, Chapter 3, Theorem 2.2], likewise gives zero-dimensional fibers for the Jacobian map from $\mathcal{R}_3$ to the moduli space $\mathcal{A}_3[\epsilon \neq 0]$ of principally polarized abelian threefolds with nonzero two-torsion. This target is irreducible of dimension six: it is a quotient of connected Siegel space, and the symplectic group is transitive on nonzero vectors modulo two. Hence the Jacobian map is dominant.

Choose a theta characteristic κ on a genus-three curve and set

$$\Theta_\kappa = \{\alpha \in J(C) : h^0(C, \kappa \otimes \alpha) > 0\}.$$

The Riemann theta theorem identifies Θ_κ with a symmetric theta divisor; see [28, Chapter II, Section 3, Corollary 3.6 and Theorem 3.10]. An all-even member of $|K_C \otimes \epsilon|$ exists precisely when some α with $2\alpha = \epsilon$ lies on Θ_κ : an effective divisor E with $2E \in |K_C \otimes \epsilon|$ corresponds to $\alpha = \mathcal{O}_C(E)\kappa^{-1}$. This condition is independent of κ : translating a symmetric theta divisor by two-torsion permutes the set of all sixty-four halves. The condition is closed on $\mathcal{A}_3[\epsilon \neq 0]$: after a finite theta-level cover, the theta divisor and the finitely many half sections are defined, and incidence with each section is closed. Their finite union descends to a closed subset.

It is a proper condition. On a product of three elliptic curves, choose ϵ nonzero on every factor. Every half then has coordinates of order four, whereas a symmetric product theta divisor is a union of coordinate divisors at two-torsion points. No half lies on that divisor. The resulting nonempty open in the moduli of principally polarized threefolds meets the dominant Jacobian locus, and hence pulls back to a nonempty open in our parameter space. This proves the assertion about fibers. The order-four choice χ on B is independent of D , C , and ϵ . Every connected component of its finite étale torsion/level cover surjects onto the connected base, so the same open remains nonempty in each component.

Finally, a translation preserving D belongs to the finite kernel $K(M^2) = B[2]$. Each nonzero element of that kernel acts nontrivially on $\mathbf{P}H^0(B, M^2)$: otherwise its commutator with all theta-group elements would be trivial, contrary to the nondegenerate theta-group pairing [4]. Its fixed

locus is proper. Avoiding the union of these finitely many loci gives a nonempty open on each section fiber. Its intersection with the preceding open proves the lemma. $\square$

4.2. Coherence, parity, and signed relations. Fix data in the open of Lemma 4.1. Let $R \subset \mathbf{P}^1$ be the regular-value open of the Gauss pencil. Locally on R its fibers on D can be written $\{\pm x_1, \dots, \pm x_4\}$. A *selection* means an unordered subset containing exactly one point of each opposite pair. The selections form a finite étale cover $\mathcal{S} \rightarrow R$ of degree sixteen. If s is a selection containing p , write $s[p]$ for the selection obtained by replacing p by $-p$.

This selection cover and its splitting by the parity of sign changes form the combinatorial part of Donagi's tetragonal construction [17, Sections 2.1–2.2 and 2.5]. The argument below constructs compatible half sums in the given abelian surface B and lifts their addition map through separate curve covers to X .

Lemma 4.2. *There is an algebraic map $h : \mathcal{S} \rightarrow B$ such that*

$$(4.1) \quad 2h(s) = \sum_{p \in s} p, \quad h(s[p]) = h(s) - p.$$

The selections split globally into two degree-eight covers $\mathcal{S}_+ \sqcup \mathcal{S}_-$ according to the parity of the number of sign changes. Their smooth projective completions $C_\pm$, which may be disconnected, carry extensions $a_\pm : C_\pm \rightarrow B$ of h . Wherever its differential is nonzero, either map is tangent to the direction recorded by its parameter in $\mathbf{P}^1$.

Proof. For a fixed regular fiber, a coherent assignment is a function h on its sixteen selections satisfying Equation (4.1). Choosing one selection s_0 and one half b of its sum determines it uniquely by

$$h(s) = b - \sum_{p \in s_0 \setminus s} p.$$

Thus coherent assignments form a torsor under the constant group $B[2]$. This torsor is algebraic and finite étale over R : start with the cover of pairs (s, b) satisfying $2b = \sum_{p \in s} p$, and identify (s, b) with $(s[p], b - p)$. After étale-locally labeling the four pairs, the flips commute, are involutions, and their equivalence classes have sixteen elements; the resulting finite étale quotient is precisely the assignment torsor. Parallel transport commutes with the constant $B[2]$ -action, so its monodromy acts by translations, despite possible permutations of the selections.

Consider a small disc about any omitted value, including infinity if it is omitted. At the central fiber choose one point from each pair of opposite points of D , and near each chosen point take the entire cluster specializing to it. Their union is a monodromy-invariant selection on the punctured disc. Its sum extends holomorphically to the disc. Because multiplication by two on B is étale, that sum has a holomorphic half on the simply connected disc. The corresponding coherent assignment is fixed by local monodromy. A translation of a torsor with a fixed point is the identity. Hence the assignment torsor has trivial monodromy about every puncture, extends étale over $\mathbf{P}^1$, and is trivial. Any of its global sections gives the algebraic map h .

Two selections have the same parity if they differ by an even number of flips. The two parity classes define a degree-two étale cover of R . The invariant selection over each punctured disc fixes its parity class, hence both classes. This cover also extends étale over $\mathbf{P}^1$ and is trivial. Labeling its two sections gives the global covers $\mathcal{S}_\pm$. Properness of B extends their maps uniquely to the smooth projective completions. Finally, differentiation of Equation (4.1), using the invariant tangent space of B , gives locally

$$(4.2) \quad da_\pm = \frac{1}{2} \sum_{i=1}^4 \sigma_i dx_i, \quad \sigma_i \in \{1, -1\}.$$

All the dx_i lie in the same Gauss direction. Their sum, whenever nonzero, lies in that direction as well. Continuity gives the same conclusion at the omitted values wherever the extended map has nonzero differential. $\square$

The common assignment on both parities gives an exact identity for any two selections over the same regular value:

$$(4.3) \quad h(s) + h(t) = 2h(s) - \sum_{p \in s \setminus t} p = \sum_{p \in s \cap t} p.$$

Here we used Equation (4.1). In particular, $h(s^c) = -h(s)$ for the opposite selection s^c . Changing the common assignment by $b_0 \in B[2]$ changes the left side of Equation (4.3) by $2b_0 = 0$, so the identity is independent of that choice.

Lemma 4.3. *On a regular-value disc, no signed sum of a nonempty subset of the four local branches x_i is constant. Every connected component of C_+ and C_- maps nonconstantly to B . No image component from C_+ is an opposite translate of an image component from C_- ; that is, their image curves cannot satisfy $Z_+ = b - Z_-$ for any $b \in B$.*

Proof. A one-term relation is impossible because a branch of D is nonconstant. A two-term relation identifies open subsets of D and a translate of D or $-D$. Since $D = -D$, triviality of its translation stabilizer forces the translate to be zero; this would identify two of the distinct opposite pairs, also impossible.

Suppose there is a three-term relation. The monodromy on the four unsigned pairs is transitive, since C is connected. Analytic continuation therefore supplies four constant relations, one omitting each index. Their coefficient matrix has zero diagonal and entries ± 1 off the diagonal. Let J denote the all-ones matrix. Modulo two the coefficient matrix is $J - I$ in dimension four, whose square is the identity over $\mathbb{F}_2$. Its determinant is therefore odd, in particular nonzero. The corresponding homomorphism $B^4 \rightarrow B^4$ is an isogeny. The four moving branches lie in a fixed finite fiber of this isogeny and must all be constant, a contradiction.

Two four-term relations whose sign vectors are neither equal nor opposite differ in between one and three positions. Subtracting them makes twice a nonempty proper signed partial sum constant. That partial sum is itself locally constant, since $[2]$ has finite fibers, contrary to the cases just proved. Thus a four-term relation, if it exists, determines its sign selection uniquely up to complement. Monodromy must preserve that selection or exchange it with its complement. Every local monodromy has a fixed signed point: a partition of four which is not all-even is one of $1 + 1 + 1 + 1$, $2 + 1 + 1$, or $3 + 1$, and there is a singleton in each case. The étale cover $D \rightarrow C$ gives two fixed signed points above that singleton. The selection therefore cannot be exchanged with its opposite. The local monodromies generate the full monodromy, so the four chosen signed points form a global invariant subset of the eight sheets of $D \rightarrow \mathbf{P}^1$. This contradicts connectedness of D .

A constant component of $a_{\pm}$ would give a constant four-term relation by Equation (4.1). Suppose image components from opposite parities satisfy $Z_+ = b - Z_-$. Choose a general smooth point $z_+ \in Z_+$ and its partner $z_- = b - z_+ \in Z_-$, and lift them along local inverse branches of a_+ and a_- away from ramification and the exceptional Gauss parameters. The tangent directions at these paired image points agree. Lemma 4.2 identifies both directions with their Gauss parameters, so the corresponding local selections s_+, s_- lie over the same parameter and satisfy $h(s_+) + h(s_-) = b$. Opposite parities have either one or three agreed signed points. Equation (4.3) then gives one of the excluded partial relations. $\square$

4.3. The degree-eight addition map.

Proposition 4.4. *For every parameter in a nonempty Zariski-open subset of the parameter space, there exist smooth projective connected curves C_1, C_2 and a dominant generically finite morphism*

$$g : C_1 \times C_2 \longrightarrow X.$$

Whenever additionally U_X has no nonzero rational $(1, 1)$ -class, an algebraic correspondence induces an injection

$$U_X \hookrightarrow H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q}).$$

In particular, this injection exists at a very general parameter.

Proof. Take the open of Lemma 4.1 and the common coherent assignment of Lemma 4.2.

Image classes and finiteness. For either parity, its eight sign vectors satisfy

$$\sum_{\sigma} \sigma_i \sigma_j = 8\delta_{ij}.$$

If ω is an invariant real two-form on B , Equation (4.2) therefore gives, over a regular-value disc,

$$\sum_{s \in \mathcal{S}_{\pm}} h(s)^* \omega = 2 \sum_{i=1}^4 x_i^* \omega.$$

The right side is exactly the sum over the eight branches $\pm x_i$ of D , since inversion fixes invariant two-forms. Integrating over the regular-value open, whose omitted points have measure zero, proves

$$(4.4) \quad (a_+)_*[C_+] = (a_-)_*[C_-] = [D] \quad \text{in } H_2(B, \mathbb{R}).$$

Invariant forms represent all of $H^2(B, \mathbb{R})$, so this equality of cycle classes includes the degrees of all image maps and every connected component of both sources.

Consider addition

$$F : C_+ \times C_- \longrightarrow B, \quad (p, q) \longmapsto a_+(p) + a_-(q).$$

It is finite on every product of connected components. Otherwise a positive-dimensional fiber would contain a curve. Neither projection of that curve could be constant, since each factor map is nonconstant and hence finite onto its image. Both projections would therefore be surjective, forcing the two image curves to be opposite translates, contrary to Lemma 4.3. Properness now gives finiteness. Each component product is two-dimensional and its finite image is all of B . Its total degree, summed over the component products, is, by Equation (4.4),

$$(4.5) \quad \deg F = [D]^2 = (2H)^2 = 8.$$

One can obtain this formula directly by intersecting one image cycle with a general translate of the negative of the other; the intersection multiplicities count the fibers of addition with the image-map degrees.

The pullback divisor. Fix a signed point x of a regular Gauss fiber. For each selection $s \in \mathcal{S}_+$ containing x , let $\tau = s^c[-x]$. It differs from s in three signs, so $\tau \in \mathcal{S}_-$, and Equation (4.3) gives

$$(4.6) \quad h(s) + h(\tau) = x.$$

There are exactly four such selections s . The resulting four points of $C_+ \times C_-$ are distinct, since their first selections differ.

As x varies locally on D , each construction defines a smooth local curve in the source, parametrized by the common regular Gauss parameter and mapped isomorphically to an open subset of D . At its points both factor differentials have image in the tangent line of D at x , so dF has rank at most one. Its restriction to the constructed local curve is nonzero. Choose target

coordinates (r, s) with $D = (s = 0)$ and source coordinates $u = r \circ F, v$, with the local curve given by $v = 0$. Then

$$F(u, v) = (u, G(u, v)), \quad G(u, 0) = 0, \quad G_v(u, 0) = 0.$$

Thus the multiplicity of F^*D along each constructed local curve is at least two.

Write $F^*D = \sum_{\alpha} e_{\alpha} E_{\alpha}$, with E_{α} running through the prime divisors on all component products. Every E_{α} dominates D , since F is finite. Put $f_{\alpha} = [\mathbb{C}(E_{\alpha}) : \mathbb{C}(D)]$. The valuation degree formula and Equation (4.5) give

$$\sum_{\alpha} e_{\alpha} f_{\alpha} = 8.$$

Above a general complex point of D , the normalization of E_{α} has f_{α} distinct local branches. The four branches constructed above already contribute at least $4 \cdot 2$ to this sum. They exhaust it, and each has multiplicity exactly two. Several of these local branches may belong to the same prime divisor. This proves, on every component product, the Cartier divisor identity

$$(4.7) \quad F^*D = 2E_0$$

for an effective integral divisor E_0 ; the smooth source makes E_0 Cartier.

Lifting to the double cover. Choose connected components $C_1^0 \subset C_+$ and $C_2^0 \subset C_-$. Pull back $\pi : B' \rightarrow B$ separately along their maps and choose a connected component of each pullback. This gives finite étale surjective maps $C_i' \rightarrow C_i^0$ and lifts $a_i' : C_i' \rightarrow B'$. Their sum

$$F' = a_1' + a_2' : C_1' \times C_2' \rightarrow B'$$

is finite and surjective: its composite with π is the finite surjective map obtained from F by the factor covers. Pullback of Equation (4.7) gives $F'^*(\pi^*D) = 2E$ on the smooth product $S' = C_1' \times C_2'$.

Let s_E be the canonical section of $\mathcal{O}_{S'}(E)$ and put

$$N = F'^*L \otimes \mathcal{O}_{S'}(-E), \quad u = F'^*(\pi^*d)/s_E^2 \in H^0(S', N^2).$$

The section u is everywhere nonzero. It defines an étale double cover

$$\tilde{S}' = \text{Spec}_{S'}(\mathcal{O}_{S'} \oplus N^{-1}) \rightarrow S'.$$

This is the normalization of $S' \times_{B'} X$ and has an everywhere-defined morphism to that fiber product. Explicitly, where E has local equation $f = 0$, the pulled-back equation of X is $z^2 = uf^2$. Its normalization is $y^2 = u$, and the morphism back is $z = fy$. This formula also applies at singular points and intersections of E and glues by the definition of N .

The étale double cover $\tilde{S}' \rightarrow S'$ corresponds to a character

$$\rho : \pi_1(C_1') \times \pi_1(C_2') \rightarrow \mathbb{Z}/2.$$

It is the sum of its two factor restrictions ρ_1, ρ_2 . Let $C_i \rightarrow C_i'$ be the connected finite étale cover corresponding to $\ker \rho_i$, using the identity cover if $\rho_i = 0$. These covers are algebraic and projective. The pullback of $\tilde{S}'$ to $C_1 \times C_2$ is trivial, so choosing either section and composing the preceding morphisms gives $g : C_1 \times C_2 \rightarrow X$. Its composite to B' is finite surjective, so its image has dimension two. The surface X is irreducible: its nonempty reduced branch divisor makes its function-field square class nontrivial. Hence g is dominant and generically finite.

The cohomological correspondence. The projection formula gives $g_*g^* = (\deg g) \text{id}$ on rational cohomology, so g^* is injective. Suppose U_X has no nonzero rational $(1, 1)$ -class. Then it has no nonzero Hodge map to $\mathbb{Q}(-1)$: dualizing such a map with a polarization would give a rational $(1, 1)$ -class in U_X . Thus the two outer projections of g^*U_X in the decomposition

$$\begin{aligned} H^2(C_1 \times C_2, \mathbb{Q}) &= (H^2(C_1) \otimes H^0(C_2)) \oplus (H^1(C_1) \otimes H^1(C_2)) \\ &\quad \oplus (H^0(C_1) \otimes H^2(C_2)) \end{aligned}$$

vanish. Choose $p_i \in C_i$ and the algebraic projectors

$$\pi_1^{C_i} = \Delta_{C_i} - [p_i \times C_i] - [C_i \times p_i].$$

With e from Equation (3.2), the correspondence

$$(\pi_1^{C_1} \boxtimes \pi_1^{C_2}) \circ {}^t\Gamma_g \circ e \in \mathrm{CH}^2(X \times C_1 \times C_2)_{\mathbb{Q}}$$

induces the asserted injection on the whole rational space U_X . Proposition 3.3 supplies the additional Hodge hypothesis at a very general parameter. $\square$

4.4. Successive generic choices. We call a set residual if its complement is a countable union of closed nowhere-dense subsets. In the connected coordinate domains below, every residual set is dense by Baire's theorem.

Lemma 4.5 (Ordered construction choices). *Every nonempty regular period chart admits a smaller normalized product chart $A \times Q$ of abelian data and affine section coordinates as in Section 3.1, mapping into it and lying in the simultaneous geometric open of Propositions 3.3 and 4.4. For any prescribed countable collection of proper closed analytic period loci in the original chart, the following choices are available in this order:*

- (i) *abelian data a in a residual subset of A ;*
- (ii) *for each such a , an actual section pencil in a residual subset of each prescribed nonempty admissible pencil open meeting Q ;*
- (iii) *after that pencil is fixed, all but countably many points of its intersection with Q and with a nonempty algebraic good open.*

The selected points avoid the prescribed period loci. The geometric requirements hold on the algebraic good open, and the exceptional points for proper marked rational Hodge-tensor conditions on its whole connected smooth part form a countable set. The pencils in (ii) have the form $d_t = d_0 + td_1$ with $\ell(d_0) = 1$ and $\ell(d_1) = 0$, and their leading section d_1 is smooth and avoids two-torsion. The final countable subset may include any further countable exceptions arising after the pencil is fixed; no uniform exceptional list across pencils is required.

Proof. Start with any nonempty regular period chart. The simultaneous geometric open of Propositions 3.3 and 4.4 is dense, so choose a datum $(a_*, [d_*])$ in its intersection with this chart. In the same section fiber, choose an independent section d_∞ whose divisor is smooth and avoids two-torsion. Trivialize the section bundle, choose $\ell(d_*) = 1$ and $\ell(d_\infty) = 0$, extend these choices in a local frame, and shrink the abelian base. Smoothness and avoidance of two-torsion are open conditions, so $\mathbf{P}(\ker \ell)$ contains a good section over every point of this base.

Use the normalization $\ell(d) = 1$ of Section 3.1 and choose a product polydisc $A \times Q$ within the original chart and the simultaneous geometric open. Its period map is a local isomorphism. These choices precede any actual pencil. Affine lines in each section chart are the pencils $d_t = d_0 + td_1$ on the original affine t -line, with $\ell(d_0) = 1$ and $\ell(d_1) = 0$. The good directions form a nonempty dense Zariski-open subset of $\mathbf{P}(\ker \ell)$. Since the direction map on lines meeting Q is a surjective submersion, pencils with a smooth leading section avoiding two-torsion form a dense open in every nonempty admissible pencil open. Every pencil meeting Q lies in the geometric open at all but finitely many parameters. Its period map is nonconstant, since the period differential is injective on section directions.

Pull back the prescribed analytic conditions to $H_m \subset A \times Q$. For each m , the set

$$A_m = \{a \in A : \{a\} \times Q \subset H_m\}$$

is closed: it is the intersection, over $q \in Q$, of the closed sets $\{a : (a, q) \in H_m\}$. It has empty interior, since otherwise H_m would contain an open product and hence the connected chart by

the identity theorem. The complement of $\bigcup_m A_m$ is therefore residual in A . For every a in this complement, each $H_{m,a}$ is a proper closed analytic subset of Q .

Fix such a . In the open space of affine lines meeting Q , put

$$\mathcal{L}_{m,a} = \{\Lambda : \Lambda \cap Q \subset H_{m,a}\},$$

where Λ denotes an affine line. This set is closed: every point of a limiting line inside Q is a limit of points on the approximating lines inside Q . It has empty interior, since an open set of lines meeting Q sweeps out an open subset of Q ; equivalently, evaluation from the incidence space of lines and points is a submersion. Excluding these countably many closed nowhere-dense sets leaves a residual set in every nonempty open of admissible pencils. For any remaining line, $\Lambda \cap Q$ is connected, and each condition restricts to a proper analytic subset of this one-dimensional domain, hence to a discrete and countable set. Their union is countable.

For the exceptional Hodge-tensor conditions on the connected smooth open of a whole pencil, enlarge the initial list to include every proper Hodge locus of a rational tensor in the marked local system. Continue markings on the universal cover. A tensor condition holding identically on any open would, by analytic continuation, hold on the selected pencil disc for one of the marked rational tensors already excluded there. Thus every such condition has discrete zeros on marked pencil charts. There are only countably many rational tensors, continuation choices, and charts, so only countably many exceptional pencil parameters.

Intersect the residual pencil choices with the dense open of good directions. The geometric bad parameters are finite, while the local analytic exceptions and the continued Hodge-tensor exceptions are countable. Any further pencil-specific countable set can be added after fixing that pencil. This proves the stated order and conclusions. $\square$

5. ALGEBRAIC K3 PENCILS WITH CONTROLLED COMPACTIFICATIONS

We keep the notation of Proposition 3.1. The integral lattice V has signature $(2, 14)$, whereas its rank-eight rational summand U has signature $(2, 6)$. It is useful to embed the larger lattice: the local vanishing cycles below are roots in V , although their projections to U need not be roots in an integral K3 lattice. We first realize the periods by polarized K3 surfaces with rational double points. Their canonical contractions retain enough descent data to work over the original parameter curve. A smooth K3 family is obtained on a smaller open of that curve, and its quadratic form on U is the form inherited from Y .

Proposition 5.1 (K3 realization of a section pencil). *Choose the abelian data and the pencil $d_t = d_0 + td_1$ in the generic sets of Lemma 4.5. There is a nonempty algebraic open $O \subset \mathbf{A}_t^1$ and a smooth projective family of K3 surfaces $S^0 \rightarrow O$ with a flat isometric embedding of rational Hodge variations*

$$U \hookrightarrow R^2(S^0/O)_*\mathbb{Q}.$$

The form on U here is its cup-product form inherited from Y . For very general t , this summand is the transcendental part of $H^2(S_t^0, \mathbb{Q})$. For every s in a nonempty Zariski-open subset of $\mathbf{A}_s^1$, every smooth projective model of the total family obtained by $t = s + u^2$ is uniruled. The K3 family over O is defined over the original t -line; an additional unspecified finite base change is not part of the conclusion.

5.1. The lattice extension. In this section M denotes the lattice enlargement, and $\ell(A)$ denotes the minimum number of generators of a finite abelian group A .

Lemma 5.2 (Enlargement of the lattice). *Let V be an even lattice of signature $(2, 14)$ with elementary 2-primary discriminant group. Suppose a group $G \subset O(V)$ acts trivially on that group. There are an even finite-index overlattice M of V , a primitive embedding $M \hookrightarrow \Lambda_{K3}$, and*

a primitive vector $h \in M^\perp$ with $(h, h) > 0$, such that the action of G extends to Λ_{K3} and is the identity on $M^\perp$.

Proof. Put $A_V = V^*/V$, and denote its discriminant quadratic and bilinear forms by $q : A_V \rightarrow \mathbb{Q}/2\mathbb{Z}$ and $b : A_V \times A_V \rightarrow \mathbb{Q}/\mathbb{Z}$. On $A_V \simeq \mathbf{F}_2^n$, the form $\beta = 2b$, valued in $\mathbb{Z}/2\mathbb{Z}$, is nondegenerate and symmetric. Let $E = \ker(x \mapsto \beta(x, x))$, so $\dim E \geq n - 1$. The restriction of β to E is alternating. The inverse image in E of a Lagrangian in its symplectic quotient by the radical is a totally isotropic subspace of dimension

$$\frac{\dim E + \dim \text{rad}(\beta|_E)}{2} \geq \left\lfloor \frac{n-1}{2} \right\rfloor.$$

On this subspace, $b(x, x) = 0$ implies that $q(x)$ is integral modulo $2\mathbb{Z}$; moreover $q(x+y) - q(x) - q(y) = 2b(x, y) = 0$ modulo $2\mathbb{Z}$. Thus q restricts to a linear functional with values in $\mathbf{F}_2$. If $n \geq 5$, the subspace has dimension at least two, and this functional has a nonzero isotropic vector.

Adjoin the corresponding isotropic element to the lattice. The new lattice is even and has discriminant form $H^\perp/H$, where H is the order-two subgroup generated by that vector. This reduces the discriminant length by two and leaves the rational quadratic space unchanged. Repetition gives an even overlattice M with $\ell(A_M) \leq 4$. Each enlargement is G -stable, since G acts trivially on the preceding discriminant group; its action on the new discriminant group is again trivial.

The sufficient primitive-embedding criterion of Nikulin [22, Chapter 14, Theorem 1.12] applies, since

$$3 > 2, \quad 19 > 14, \quad 22 - 16 = 6 \geq \ell(A_M) + 2.$$

Choose such an embedding. Its complement has signature $(1, 5)$, and therefore contains a primitive positive integral vector h . The complement is primitive in the unimodular K3 lattice, so h is also primitive in Λ_{K3} . Unimodular gluing identifies the discriminant forms of M and $M^\perp$ with opposite signs. Consequently an isometry of M trivial on A_M , together with the identity on $M^\perp$, extends uniquely to Λ_{K3} . These extensions respect composition.

In particular, a root $r \in V$, with $r^2 = -2$, remains an integral root throughout the construction. Its reflection is

$$s_r(x) = x + (x, r)r.$$

It is integral on every even overlattice under consideration and on Λ_{K3} , and is the identity on $M^\perp$. Whenever a monodromy operator on V is a product of these reflections, its extension is the same product of K3 root reflections. $\square$

5.2. Local monodromy of the section pencil. We analyze the normalized original-line pencil $d_t = d_0 + td_1$ already chosen in Proposition 5.1 through Lemma 4.5. Its abelian data (B, H, χ) are fixed. The leading member d_1 is smooth and avoids two-torsion. The corresponding variations V and U are those of Proposition 3.1. The next lemma distinguishes the finite root monodromies from the sign at infinity; the substitution $t = s + u^2$ will kill the latter.

Lemma 5.3 (Local monodromy of the section pencil). *At every finite singular parameter of this general pencil, the monodromy on V is a product of reflections in mutually orthogonal integral (-2) -roots in V . These roots are perpendicular to the limiting Hodge line. The periods extend to interior periods after killing these finite monodromies. At infinity the monodromy on V is $-\text{id}$.*

Proof. *The possible singular members.* The morphism defined by $|2H|$ factors through inversion and realizes the quotient as the classical quartic Kummer surface. A general line in the dual projective space meets its discriminant either at a simple Lefschetz contact in the smooth locus or at a hyperplane through one ordinary Kummer node. Here is the transversality assertion being used. The hyperplanes containing each Kummer node form a plane in the dual space. A general point of such a plane avoids every other node and is transverse to the quadratic tangent cone at

that node. For tangencies in the smooth locus, the conormal incidence has generically separable projection to the dual variety. If its image has dimension less than two, a general line avoids it. If it has dimension two, its general smooth point represents a unique contact, because that contact point is recovered as the tangent hyperplane to the dual variety. Separability makes the quadratic contact nondegenerate. The smooth-contact dual component cannot be a node plane, by biduality. The higher contacts, intersections of these discriminant components, and their singular loci have codimension at least two in the dual projective space. A general line avoids them and meets each remaining component transversely. Upstairs on B , the first event gives two ordinary nodes exchanged by inversion. At a two-torsion point in the second event, the invariant local equation has a nondegenerate quadratic term and its constant term vanishes to first order in t . Thus the branch has an ordinary node with transverse smoothing. The base points of a general pencil are smooth on every member, since the differentials of d_0 and d_1 there are independent. Pulling back along the isogeny to B' gives disjoint copies of these events. We examine their inversion quotients.

An exchanged pair. At an exchanged pair away from the fixed points, the branch curve on $K = \text{Km}(B')$ acquires one ordinary node. The local double surface is $z^2 = xy + t$. Its vanishing sphere has square -2 , and the deck involution δ negates it. It therefore gives an integral root of V and its Picard–Lefschetz reflection.

A positive-sign fixed point. At a fixed point with nonbranch exceptional curve on K , choose coordinates in which the upstairs model is

$$z^2 = x^2 + y^2 + t, \quad j(x, y, z) = (-x, -y, z).$$

These coordinates can be chosen equivariantly: the branch equation is even in (x, y) , and the holomorphic Morse lemma, applied to its nondegenerate even quadratic-coefficient matrix, preserves that parity. The parameter is replaced locally by its constant term, which is a coordinate because the discriminant is met transversely. On the chart $a = x^2$, $v = y/x$ of the resolved base quotient, the double surface has equation

$$z^2 = a(1 + v^2) + t.$$

At $t = 0$, its branch is the exceptional curve $a = 0$ together with a residual curve meeting it at $v = i, -i$. At each crossing the equation is $z^2 = ab + t$. There are consequently two disjoint ordinary node smoothings, and two orthogonal integral anti-invariant roots in V . The local monodromy is their product. The difference of the two lifts of the nonbranch exceptional curve, of square -4 , need not be fixed by this monodromy; it is not being used as one of the individual roots.

A negative-sign fixed point. At a fixed point with branch exceptional curve, the involution in the upstairs quadratic model is instead

$$j(x, y, z) = (-x, -y, -z).$$

It acts antipodally on the real vanishing sphere S^2 , and hence by -1 on its degree-two class. There is no vanishing class in the j -invariant quotient cohomology. To check the resolution contribution, for $t \neq 0$ the points $(0, 0, \pm\sqrt{t})$ are exchanged by j . Their quotient is one smooth point. On the chart represented near one chosen square root, the induced sheet involution is $(x, y) \mapsto (-x, -y)$. Blowing up this point therefore gives a $\mathbf{P}^1$ fixed pointwise by δ ; its cohomology class is δ -invariant. More explicitly, on the Kummer-resolution chart $a = x^2$, $v = y/x$, put $\eta = xz$. The finite double-cover model is

$$\eta^2 = a(a(1 + v^2) + t), \quad \delta(\eta) = -\eta.$$

For $t \neq 0$, the preimage of $a = 0$ is that ramification curve; its square is -1 , because the base exceptional curve has square -2 . Its class is the only added resolution class at this point. Thus this event contributes no negative eigenspace to V .

The full monodromy operator. All the displayed roots are integral classes on the nearby smooth Y . They are mutually orthogonal because their Milnor neighborhoods are disjoint, and the local

monodromy negates each of them. To see that there is no further monodromy, use the orthogonal decomposition of variations in Proposition 3.1,

$$V_{\mathbb{Q}} = U \oplus \langle a_1, \dots, a_8 \rangle_{\mathbb{Q}}, \quad U \simeq U_X.$$

On X , the complement of the disjoint node neighborhoods is locally topologically trivial, so Picard–Lefschetz gives the product of the corresponding commuting reflections. An exchanged pair has a one-dimensional j -invariant vanishing space. At a positive fixed point, j has degree $+1$ on the vanishing sphere and contributes one dimension to U_X . At a negative fixed point it has degree -1 and contributes none. On the difference lines $\mathbb{Q}a_i$, the two labels are exchanged by $\sqrt{t} \mapsto -\sqrt{t}$ exactly at the affected positive fixed points; all other difference lines are fixed. Consequently the full operator on $V_{\mathbb{Q}}$ is an involution whose negative eigenspace has one dimension per exchanged pair and two per positive fixed point. This is precisely the number of mutually orthogonal roots found above. Those roots span that negative eigenspace, proving that the entire operator is their product of reflections. The same argument includes all disjoint lifts of a single event through the isogeny.

The limiting periods. After a quadratic cover at each ordinary node, the upstairs surface family has a simultaneous local analytic resolution. In the A_1 case used here, write $xy = (z - v)(z + v)$ and blow up the non-Cartier ideal $(x, z - v)$. On the chart $z - v = xa$, the equations are $z = v + xa$ and $y = a(2v + xa)$, so (x, a, v) are smooth coordinates. On the other chart $x = (z - v)b$, they are $z = by - v$ and $x = b(by - 2v)$, with smooth coordinates (b, y, v) . Both projections to v are smooth and their central fibers resolve the node. The relative two-form $dx \wedge dz/x$ becomes $dx \wedge da$ on the first chart, while the equal expression $-dy \wedge dz/y$ becomes $db \wedge dy$ on the second. Thus the resolution is crepant. These are the local A_1 charts of the classical simultaneous-resolution constructions [7, 3]. They glue to the unchanged locus because the exceptional neighborhoods are disjoint.

Choose a nonzero $\theta_+ \in H^0(B', L)^{j=+1}$. Using an invariant volume form on B' , the Hodge line of U_X has frame

$$\frac{\theta_+ dx \wedge dy}{z}.$$

Crepantness extends it regularly after the local base change and resolution. It remains a nonzero global form on the resolved central surface: test at a point of B' outside the branch divisor and the zero divisor of θ_+ . At a minus-sign node its coefficient is odd and vanishes at that node, but this does not make the global Hodge-line frame vanish over the parameter.

The chosen simultaneous resolution need not carry the involutions. Their flat actions on cohomology nevertheless extend after the base change, preserve the limiting Hodge filtration by continuity, and retain the prescribed eigenspaces. The nonzero limiting holomorphic form on the projective resolved central surface has positive norm. Thus U_X , and hence U , has an interior limiting period. The section just used was single-valued before the local base change, so inertia fixes its limiting vector. In particular the negative eigenspace of local monodromy is perpendicular to that vector. The additional exceptional-difference classes remain algebraic, giving the same conclusion for the roots in V .

Infinity. Near infinity write $q_{\infty} = t^{-1}$. The equation is $z^2 = q_{\infty}^{-1}(d_1 + q_{\infty}d_0)$. The leading member d_1 is smooth and avoids the fixed points. After adjoining $\sqrt{q_{\infty}}$ and rescaling z , the family extends smoothly. The only twist around $q_{\infty} = 0$ is the change of sign of this square root. This is δ , which acts as $-\text{id}$ on V . $\square$

5.3. The period stack and polarized contraction. The monodromies just calculated describe integral period data; they do not yet give an algebraic K3 family on the original curve. To construct that family, we pass through polarized contractions, for which reflections in contracted roots act trivially. A *polarized ADE K3 surface* is a normal surface with rational double points

whose minimal resolution is a K3 surface, together with a primitive ample polarization. Here a polarization in a family is a relative Picard class, locally represented by a line bundle. An ADE K3 surface is the canonical polarized contraction of its resolution with the pulled-back big and nef class; this contraction is given by a sufficiently high power of the class. The relative-Picard convention removes scalar automorphisms of a representing line bundle from the moduli problem.

Fix a primitive positive vector $h \in \Lambda_{K3}$. Choose a connected component $\mathcal{D}_h$ of

$$\{[\sigma] \in \mathbf{P}(h^\perp \otimes \mathbb{C}) : (\sigma, \sigma) = 0, (\sigma, \bar{\sigma}) > 0\},$$

and let Γ_h be the isometries of Λ_{K3} fixing h and preserving this component. Unimodular gluing identifies Γ_h with the stable orthogonal group of $h^\perp$ preserving $\mathcal{D}_h$: restriction acts trivially on the discriminant group, and every such restricted isometry extends by the identity on $\mathbb{Z}h$. Thus $[\Gamma_h \backslash \mathcal{D}_h]$ is the integral period stack for primitive degree (h, h) polarizations. This identification is independent of the choice of the primitive vector of that square; the usual transitivity follows from the Eichler criterion for the K3 lattice, which contains two hyperbolic planes; see [22, Chapter 14, Corollary 1.10].

Lemma 5.4 (The contraction morphism from the period stack). *Let $\mathcal{P}_h = [\Gamma_h \backslash \mathcal{D}_h]$, and let $\mathcal{M}_h$ denote the moduli stack of primitively polarized ADE K3 surfaces of degree (h, h) . There is a morphism*

$$q : \mathcal{P}_h \longrightarrow \mathcal{M}_h$$

whose composition with the period map of quasi-polarized smooth K3 surfaces is their canonical polarized contraction. The pullback of the Hodge line of $\mathcal{M}_h$ is the tautological period line, including its Γ_h -linearization.

For $[\sigma] \in \mathcal{D}_h$, let W_σ be generated by the reflections in the roots

$$R_\sigma = \{r \in \Lambda_{K3} : r^2 = -2, (r, h) = 0, (r, \sigma) = 0\}.$$

The kernel of the homomorphism on stabilizers induced by q is exactly W_σ .

Proof. The existence and factorization of q are [16, Lemma 1.12]. We use the analytic form of that morphism. Its hypotheses are primitive ample polarization classes on ADE K3 surfaces, primitive big and nef classes on smooth K3 surfaces, and the full integral period stack over $\mathbb{C}$. The stable group and the representable, surjective étale period map of the quasi-polarized stack are described in [16, Sections 1.6 and 1.8; proof of Lemma 2.6 for surjectivity]. Taking one component of the period domain with its stabilizer gives the equivalent stack used here. We now prove the two stated compatibilities.

Let $\mathcal{K}_h$ be the smooth quasi-polarized K3 stack. Over a local étale period chart its universal smooth family contracts to the pullback of the ADE family. The latter is flat with Gorenstein fibers, so its relative dualizing sheaf is invertible. Off the exceptional curves, contraction identifies the relative two-forms. In the total smooth family over this full period chart the exceptional locus has codimension at least two: the curves occur only over root walls. The identification of the two line bundles therefore extends over the exceptional locus. On each fiber it is the usual crepant identification for resolution of rational double points. This gives the claimed comparison of Hodge lines after pullback to $\mathcal{K}_h$.

For descent, consider $\mathcal{K}_h \times_{\mathcal{P}_h} \mathcal{K}_h$. It is locally smooth over the period domain, and the ample locus is dense in every component. There each period arrow comes from an actual polarized isomorphism by Torelli, so the two induced Hodge-line identifications agree. Equality extends over this reduced relation space. Hence the comparison descends with its entire Γ_h -linearization, including on the root mirrors.

Fix a root $r \in h^\perp$ and consider its mirror. Its reflection s_r fixes the period vector, so its image under q acts trivially on the Hodge line. At a very general point of the mirror, the resolution has rational Picard space $\mathbb{Q}h \oplus \mathbb{Q}r$. The polarized automorphism induced by $q(s_r)$ lifts uniquely to

that resolution. It fixes h , and preservation of effectivity forces it to fix the effective choice of r . It is symplectic. The transcendental Hodge structure is simple, being polarizable of K3 type with no nonzero rational $(1, 1)$ -classes. A Hodge endomorphism fixing its $(2, 0)$ -line pointwise is the identity: the kernel of the difference from the identity is a nonzero Hodge substructure. The lifted automorphism thus acts identically on all second cohomology and is the identity by Torelli.

On the whole mirror, $q(s_r)$ and the identity are sections of a separated isomorphism space. Their equality on the dense locus just considered extends everywhere. Since root reflections generate W_σ , this group is killed even at points with several or persistent contracted roots. This agrees with the generic root-reflection kernel identified in [16, Lemma 3.6].

Conversely, an integral Hodge isometry fixing h takes the ample chamber of a chosen resolution to another chamber whose closure contains h . The Weyl group generated by the algebraic roots perpendicular to h acts transitively on these chambers: only walls through h separate chambers incident to h . After composition with an element of W_σ , strong Torelli realizes the isometry by an automorphism of the resolution. By the factorization defining q , its image is the induced automorphism of the contraction. If the image is the identity, its lift is the identity on the dense complement of the exceptional curves and hence everywhere. The original isometry then belongs to W_σ . $\square$

5.4. Descent and the relative canonical class. The use of the integral period stack in Lemma 5.4 is essential: the local system and its monodromy supply a map to that stack, whereas a map to the coarse period space need not supply a family. The following curve statement also keeps track of ramified descent and of the relative canonical bundle.

Lemma 5.5 (Realization over the original curve). *Let C be a smooth projective complex curve, let $D \subset C$ be finite, and put $C^\circ = C \setminus D$. Suppose the following data are given:*

- (i) *a local system $\mathcal{L}$ of lattices isometric to Λ_{K3} , with a primitive flat section h of positive square;*
- (ii) *a holomorphic line $\mathcal{F} \subset \mathcal{L} \otimes \mathcal{O}_{C^\circ}$ with*

$$(\mathcal{F}, \mathcal{F}) = 0, \quad (\sigma, \bar{\sigma}) > 0 \quad (0 \neq \sigma \in \mathcal{F}), \quad (\mathcal{F}, h) = 0;$$

- (iii) *finite local monodromy at every point of D , and an interior limit of the period line after a local cover killing that monodromy.*

Then there is a normal projective model $f : \mathcal{Z} \rightarrow C$ whose restriction to C° is a family of polarized ADE K3 surfaces realizing these integral period data modulo contracted roots.

Suppose, in addition, that the local monodromy at each $c \in D$, viewed on the limiting marked lattice, belongs to the Weyl group generated by

$$(*) \quad R_c = \{r \in \mathcal{L}_c : r^2 = -2, (r, h) = 0, (r, \mathcal{F}_c) = 0\}.$$

Then $\mathcal{Z} \rightarrow C$ can be chosen to be a flat family of polarized ADE K3 surfaces on all of C . Its extended Hodge line descends to a line bundle λ on C , and

$$(5.1) \quad \omega_{\mathcal{Z}/C} \simeq f^* \lambda, \quad K_{\mathcal{Z}} = f^*(K_C \otimes \lambda).$$

Under hypotheses (i)–(iii), independently of (), let $\mathcal{U} \subset \mathcal{L}_{\mathbb{Q}}$ be a nondegenerate rational sub-local system containing $\mathcal{F}$, and suppose every flat root which is of type $(1, 1)$ on an open set is perpendicular to $\mathcal{U}$. After shrinking a nonempty algebraic open in C° , there is a smooth projective K3 family $\mathcal{S}^0$ there and an identification of $\mathcal{U}$ with an isometric Hodge summand of $R^2 \mathcal{S}_*^0 \mathbb{Q}$. Here a flat root means a locally flat section; it need not be a globally invariant vector. No change of the base curve occurs in this conclusion.*

Proof. The family and its algebraization. The torsor of markings of $(\mathcal{L}, h)$, with a component chosen by $\mathcal{F}$, and the equivariant period map define a holomorphic map $C^\circ \rightarrow \mathcal{P}_h$. Its monodromy lies in Γ_h : it fixes h , has trivial restricted discriminant action by unimodular gluing, and preserves

the period component by continuity. Pullback along q in Lemma 5.4 gives an actual analytic family of polarized ADE K3 surfaces on C° , with its compatible descent data.

Choose a finite Galois cover $\tilde{C} \rightarrow C$ whose ramification indices at the points above D are divisible by the local monodromy orders. Such a cover exists: choose a common multiple N of these orders, use Riemann–Roch with sufficiently large poles away from D to prescribe a simple zero at each point of D , and normalize the cyclic cover obtained by adjoining an N -th root of that function. Additional branch points lie over points where the original family is already defined. If D is empty the identity cover suffices. The pulled-back local system and period extend across the punctures: in a flat trivialization the latter is the holomorphic interior extension in hypothesis (iii). They define a map $\tilde{C} \rightarrow \mathcal{P}_h$, and hence a proper analytic family $\tilde{\mathcal{Z}} \rightarrow \tilde{C}$ of polarized ADE K3 surfaces. The original monodromy data equip it, functorially through q , with an action of the Galois group. All identifications and cocycles are provided by this stack morphism; no choices of smooth simultaneous resolutions have to glue.

This compact analytic family is algebraic and normal. For normality, use flatness and the normal ADE fibers over the smooth base. To check the projectivity needed for GAGA, represent its relative polarizing class by line bundles on small open sets of the base. A common sufficiently divisible power is relatively ample. On an overlap, the ratio of two such representatives is trivial on each fiber. Its direct image is a line bundle, and evaluation is an isomorphism, by cohomology and base change and $H^1(\mathcal{O}) = 0$ on the fibers. Thus the ratio comes from the base. After local trivializations, the remaining gluing obstruction is a scalar cocycle in $H^2(\tilde{C}, \mathcal{O}_{\tilde{C}}^*)$. The exponential sequence makes this group zero, since

$$H^2(\tilde{C}, \mathcal{O}_{\tilde{C}}) = 0, \quad H^3(\tilde{C}, \mathbb{Z}) = 0.$$

The polarizing power consequently glues to a relatively ample line bundle. For a sufficiently high power its direct image is a vector bundle, and the family embeds into the associated projective bundle on $\tilde{C}$. That vector bundle and the closed embedded family are algebraic by GAGA [36, Theorems 2–3]. More explicitly, first algebraize the vector bundle on the projective curve, then algebraize the coherent ideal of the family in its projective bundle. The finite group action algebraizes by the same fully faithful comparison. Twist the relative ample bundle by a sufficiently positive base bundle to make it ample on the total space. The tensor product of its translates by the Galois group is ample and has its natural linearization; invariant sections of a sufficiently high power give a projective finite quotient. That quotient is a normal projective model $\mathcal{Z} \rightarrow C$. Over C° its construction is descent of the pullback of the initial family, also at any auxiliary branch points. This proves the first assertion, with no cover remaining in its conclusion.

Descent at a Weyl puncture. Assume (*). On a disc killing the monodromy, write the map to the original disc as $t = w^m$. The cyclic inertia group acts on the central polarized surface via the homomorphism on stabilizers induced by q . Its image is the identity by Lemma 5.4. Thus the group acts pointwise on the whole central surface as the identity, including its singular points.

Here is the corresponding scheme-theoretic descent calculation. Let A be a local completed ring of $\tilde{\mathcal{Z}}$ along that fiber. Flatness makes w a nonzerodivisor. If g generates inertia, then $g(w) = \zeta w$, with ζ primitive of order m , and g is the identity on $A/(w)$. Set $B = A^{(g)}$. Reynolds averaging makes $B \rightarrow A/(w)$ surjective. If $a = wb$ is invariant, then $g(b) = \zeta^{-1}b$. For $m > 1$, reduction modulo w forces b to be divisible by w . At each subsequent division the character changes by ζ^{-1} ; until the m -th division it is nontrivial and the reduction must again vanish. This gives, also trivially when $m = 1$,

$$B \cap wA = w^m B, \quad B/(t) \simeq A/(w).$$

There is also an exact base-change statement. Write $A_j = \{a : g(a) = \zeta^j a\}$, for $0 \leq j < m$. Successive division by w gives $A_j = w^j B$; character projection and the fact that w is a

nonzerodivisor then give

$$A = \bigoplus_{j=0}^{m-1} w^j B, \quad B \otimes_{\mathbb{C}[[t]]} \mathbb{C}[[w]] \xrightarrow{\sim} A, \quad t \mapsto w^m.$$

These assertions pass from completed rings to the local rings by faithful flatness of completion. Thus the quotient central fiber is exactly the original reduced ADE surface, and the upstairs family is its flat pullback. The quotient is flat over the parameter disc, since t is a nonzerodivisor in B , a torsion-free module over the discrete valuation ring of the disc. This proves the assertion about all fibers and rules out a multiple central fiber.

The underlying ADE family has now descended. To retain its primitive polarization, choose on a smaller invariant analytic w -disc a line bundle L representing its relative Picard class; the preceding scalar-cocycle argument applies on the disc as well. The same base-change argument used above shows that $g^*L \otimes L^{-1}$ comes from the disc and that automorphisms of L are base units; the base line is trivial after shrinking. For a g -semilinear lift T , write $T^m = c \text{ id}$, where commutation with T makes $c = c(t)$ an invariant base unit. Replacing T by aT replaces c by $c \prod_{i=0}^{m-1} g^i(a)$. Every unit in $\mathbb{C}\{t\}$ has an m -th root after shrinking, whose norm as an invariant upstairs unit is its m -th power. Taking such a root of c^{-1} therefore arranges $T^m = \text{id}$. On the proper connected reduced central fiber, T is one scalar $\chi \in \mu_m$. Replacing T by $\chi^{-1}T$ makes it the identity on $L/(w)L$, without changing the relative Picard class or the separately prescribed Hodge-line linearization.

At any central point, including an ADE point, work with the actual analytic local rings and let s be a local generator of L . Then $s_* := m^{-1} \sum_{i=0}^{m-1} T^i(s)$ is invariant and congruent to s modulo w , hence is still a generator. The invariant local modules are consequently free of rank one over the invariant local rings and pull back to L . Their invariant transition units glue them to an actual invertible sheaf on the analytic quotient. Under $B/(t) \simeq A/(w)$ its central restriction is exactly L_0 , so the primitive ample class, not a power, descends. These local classes agree with the original one off the punctures and glue on C ; the preceding scalar-cocycle argument glues their primitive representatives, which GAGA algebraizes on the projective quotient.

The Hodge line must descend with its prescribed character, separately from the scalar adjustment of the polarizing bundle. Its limiting inertia action is trivial: root reflections fix a vector perpendicular to their roots. Averaging a local frame which is nonzero at the center gives an invariant frame with that same central value, and hence a nowhere-vanishing frame on a smaller disc. Consequently the extended Hodge line descends to a line bundle λ on C . By Lemma 5.4, it is the Hodge line of the descended family, with precisely the linearization prescribed by $\mathcal{L}$. No character twist has been discarded.

Every ADE K3 fiber is Gorenstein with trivial dualizing sheaf. A flat family with these Gorenstein fibers over a smooth curve is a Gorenstein morphism, so $\omega_{Z/C}$ is invertible. Its direct image is a line bundle and formation commutes with base change, since every fiber has a one-dimensional space of canonical forms. Evaluation is an isomorphism on each fiber and therefore everywhere. The identification of this direct image with λ proves (5.1). For the subsequent intersection argument, its restriction to codimension one would already suffice.

A smooth family on an open of the original curve. This part uses the normal contraction family from the first assertion, not the additional Weyl condition. We resolve its generic fiber over $k = \mathbb{C}(C)$, then recover $\mathcal{U}$ by comparing local period lifts through the cohomology of that same contraction. The generic polarized ADE surface has a projective minimal resolution over k . One way to see the field assertion is to resolve its rational double points over a finite Galois extension. Every Galois translate is a minimal resolution of the same surface; the unique isomorphism over that surface supplies a descent datum and automatically satisfies its cocycle. A norm of a relatively ample bundle makes this descent projective. The descended resolution is defined over k itself.

Spreading the morphism and its projective embedding, and then shrinking an algebraic open of C° , gives a smooth projective K3 family $\mathcal{S}^0 \rightarrow C_0$ resolving the fixed contraction family. All fibers and exceptional configurations can be assumed to be of the generic type on this open.

Here are the details of the cohomological comparison on C_0 . For the minimal resolution $\pi : S \rightarrow Z$ of a complex ADE K3 surface, let $E_Z \subset H^2(S, \mathbb{Q})$ be spanned by its exceptional curves. The rational cohomology sequence for the resolution gives

$$\pi^* : H^2(Z, \mathbb{Q}) \hookrightarrow H^2(S, \mathbb{Q}), \quad \text{im}(\pi^*) = E_Z^\perp.$$

Indeed, the exceptional fibers are trees of rational curves, with $H^1 = 0$, and the restriction map in degree two records intersection with their components. It is onto on their span because the ADE intersection matrix is negative definite. This proves both the injectivity and the stated image. In a resolving family these pullbacks are maps of higher direct-image sheaves. Shrinking C_0 once more makes the sheaves local systems, by constructibility for the algebraic proper morphisms. They preserve Hodge structures and the cup-product forms on the indicated orthogonal complements.

Locally lift the marked period map through the étale map $\mathcal{K}_h \rightarrow \mathcal{P}_h$. Its K3 family has an identification of its contraction with the fixed family obtained from q . The exceptional root lattice is locally constant on C_0 ; all its roots are therefore persistent in the sense of the hypothesis, and are perpendicular to $\mathcal{U}$. The marked copy of $\mathcal{U}$ in this local lift consequently lies in the image of H^2 of the contraction. The displayed pullback description gives a flat, isometric map from it to $R^2\mathcal{S}_*^0\mathbb{Q}$. For two such local lifts, compare their minimal resolutions over the identity of their common contraction on each fiber. The resulting marking discrepancy has trivial image under q , so belongs to the Weyl group in Lemma 5.4. This group fixes $\mathcal{U}$ pointwise. Thus the maps just constructed agree on overlaps, including their flat structures, and glue over C_0 . Nondegeneracy of $\mathcal{U}$ makes its image an orthogonal Hodge summand. This proves the final assertion on the original curve.

For clarity about the shrinking in this argument, the mirrors of the h -perpendicular roots are locally finite in $\mathcal{D}_h$. On a compact set of periods, a root orthogonal to $\langle h, \Re\sigma, \Im\sigma \rangle$ has bounded norm in a fixed positive definite metric, since its square is -2 . Only finitely many lattice vectors can occur. A finite collection of such local period charts covers the compact finite cover used above. In each chart a nonpersistent mirror has only isolated intersections with the period curve. Local finiteness and compactness therefore leave only finitely many such intersection parameters. Removing their images, as well as the exceptional points from spreading, gives the open used here. The argument uses constant root type there; it requires no uniqueness assertion for simultaneous resolutions across jumps in the root configuration. $\square$

Corollary 5.6 (A canonical-degree criterion). *Under the Weyl condition of Lemma 5.5, suppose $\deg(K_C \otimes \lambda) < 0$. Every smooth projective model of the function field of $\mathcal{Z}$ is uniruled.*

Proof. Let $p : \tilde{\mathcal{Z}} \rightarrow \mathcal{Z}$ be a smooth projective resolution, and let A be sufficiently ample on $\mathcal{Z}$. Since $\mathcal{Z}$ is normal and Gorenstein, the difference $K_{\tilde{\mathcal{Z}}} - p^*K_{\mathcal{Z}}$ is supported on the exceptional divisors. Every such divisor has zero intersection with $(p^*A)^2$: its image has codimension at least two, and the projection formula applies. Geometrically, two general members of $|A|$ have a complete-intersection curve avoiding all these images. Equation (5.1) consequently gives

$$K_{\tilde{\mathcal{Z}}} \cdot (p^*A)^2 = \deg(K_C \otimes \lambda) (A|_{\mathcal{Z}_c})^2 < 0.$$

Here $\mathcal{Z}_c$ is any general fiber, and its displayed self-intersection is positive by ampleness. The divisor p^*A is nef, so every effective divisor, and by continuity every pseudo-effective divisor, has nonnegative intersection with $(p^*A)^2$. Thus $K_{\tilde{\mathcal{Z}}}$ is not pseudo-effective. The theorem of Boucksom–Demailly–Păun–Peternell [6, Corollary 0.3] applies to this smooth projective threefold and gives uniruledness. This is a birational property, proving the assertion for every smooth projective model. $\square$

5.5. Realization and the Hodge line.

Proof of Proposition 5.1. Realization over the t -line. By Proposition 3.1, the monodromy of V is trivial on its discriminant group. Apply Lemma 5.2, extending that monodromy to the K3 lattice and choosing its fixed positive class h . The lattice embedding preserves the form inherited from Y . Give this lattice the period line of V , which is also that of U . Over $\mathbb{Q}$ its orthogonal decomposition is

$$\Lambda_{K3, \mathbb{Q}} = U \oplus \langle a_1, \dots, a_8 \rangle_{\mathbb{Q}} \oplus M_{\mathbb{Q}}^{\perp}.$$

The last two summands, of dimensions eight and six, are of type $(1, 1)$, so the complement of U has dimension fourteen. For the choices specified in Lemma 4.5, and a very general parameter, U has no rational $(1, 1)$ -classes by Proposition 3.3. If a locally flat rational root remains algebraic on a nonempty open, its projection to U in the displayed orthogonal decomposition is a rational flat class of type $(1, 1)$ there. Analytic continuation keeps it of that type wherever its flat continuation is defined. At a very general point this projection is zero, so it is identically zero. Every persistent root is therefore perpendicular to U . Lemmas 5.3 and 5.5 now give, over a nonempty open of the original line, the asserted smooth projective K3 family with the specified U -variation. Its orthogonal complement has dimension fourteen and is entirely of type $(1, 1)$. Lefschetz's $(1, 1)$ -Theorem and the absence of rational $(1, 1)$ -classes in U show that its very general transcendental part is exactly U . This determines the rational Hodge summand and its pairing; the integral Néron–Severi lattice of the auxiliary K3 surface is not specified by this construction.

The quadratic substitution. Let Δ be the finite union of the critical values and $\mathbf{A}_t^1 \setminus O$. For every $s \notin \Delta$, make the substitution $t = s + u^2$. Its finite branch value is $t = s$, a point of the smooth open O . It is unramified over every finite critical value, so those monodromies remain the products in Lemma 5.3. At any other removed finite value the original period data extend, and the local monodromy is the identity. At infinity the ramification index is two and kills the square-root twist. Thus the full extended K3 lattice has trivial monodromy there: the square is the identity on M , and the action on $M^{\perp}$ was already the identity. Every remaining finite monodromy is a product of integral root reflections fixing h and the limiting period vector, by Lemma 5.2. All local monodromies therefore satisfy the Weyl condition in Lemma 5.5. Apply that lemma on $\mathbf{P}_u^1$ to obtain a flat projective ADE K3 family $f : \mathcal{Z} \rightarrow \mathbf{P}_u^1$ and its Hodge line λ .

The Hodge line and uniruledness of the ADE model. We first prove uniruledness for this model using its Hodge line, whose degree is exactly one. We then identify its function field with the square pullback of the fixed S^0 -family.

Indeed, the compactified upstairs double cover is defined over $B' \times \mathbf{P}_u^1$ using $L \boxtimes \mathcal{O}_{\mathbf{P}^1}(1)$. In homogeneous coordinates $[u_0 : u_1]$ its branch section is $(d_0 + sd_1)u_0^2 + d_1u_1^2$, with pullback to B' understood. On the affine chart $u = u_1/u_0$ its equation is

$$z^2 = d_0 + (s + u^2)d_1.$$

The double-cover formula for the relative dualizing sheaf, with $K_{B'} \simeq \mathcal{O}_{B'}$, gives the anti-sheet part of the direct image of relative canonical forms as

$$H^0(B', L) \otimes \mathcal{O}_{\mathbf{P}^1}(1).$$

Taking $j = +1$ picks out the fixed one-dimensional eigenspace of $H^0(B', L)$ from Proposition 3.1, hence gives $\mathcal{O}_{\mathbf{P}^1}(1)$. Choose a nonzero $\theta_+ \in H^0(B', L)^{j=+1}$. For completeness, with $v = 1/u$ and $w = z/u$ the equation becomes

$$w^2 = d_1 + (d_0 + sd_1)v^2,$$

which is smooth at infinity. Local relative forms transform as

$$\frac{\theta_+ dx \wedge dy}{z} = v \frac{\theta_+ dx \wedge dy}{w},$$

the transition for $\mathcal{O}(1)$. In particular the displayed frame on the finite chart has a simple zero, measured against the nonzero frame on the infinity chart, at $v = 0$. At every finite nodal fiber, Lemma 5.3 shows that the frame extends after the local resolving cover and is a nonzero global form on the central resolution. There is therefore no additional integral zero or pole at a finite parameter. The limiting inertia acts trivially on that form: its vanishing roots are perpendicular to the limiting period vector. The invariant-frame argument in Lemma 5.5 consequently introduces no fractional local weight either. The Hodge-line identification in Lemma 5.4 includes these exact characters, so the period realization transfers this extension to λ . We obtain $\lambda \simeq \mathcal{O}_{\mathbf{P}^1}(1)$, as line bundles with the stated local extensions.

The canonical bundle of this normal polarized contraction model is

$$f^*(\mathcal{O}_{\mathbf{P}^1}(-2) \otimes \lambda) = f^*\mathcal{O}_{\mathbf{P}^1}(-1).$$

Thus $\deg(K_{\mathbf{P}^1} \otimes \lambda) = -2 + 1 = -1$, and Corollary 5.6 shows that every smooth projective model of its function field is uniruled.

The algebraic square-pullback function field. We use a common finite cover for this construction and the fixed construction over the t -line. Put $K = \mathbb{C}(t)$, $K_s = \mathbb{C}(u)$ with $t = s + u^2$, and $\widetilde{K} = \mathbb{C}(\widetilde{C})$, where $\widetilde{C} \rightarrow \mathbf{P}_t^1$ is the finite Galois cover chosen in Lemma 5.5. Choose the fields in a common algebraic closure of K , put $\widetilde{K}_s = \widetilde{K}K_s$, and let $\widetilde{C}_s$ be the normalization of $\mathbf{P}_u^1$ in $\widetilde{K}_s$. This is the normalization of an irreducible component of $\widetilde{C} \times_{\mathbf{P}_t^1} \mathbf{P}_u^1$. Since $\widetilde{K}/K$ is Galois, so is $\widetilde{K}_s/K_s$; its group G_s is identified by restriction with a subgroup of $G = \text{Gal}(\widetilde{K}/K)$.

This common cover is admissible in the second application of Lemma 5.5. Indeed, at a point where the original cover has ramification index e , the substitution has index d , and the original local monodromy has order $m \mid e$, normalization of the fiber product gives index $e/\gcd(e, d)$. The pulled-back monodromy has order $m/\gcd(m, d)$, which divides that index. This also covers auxiliary branch points, where $m = 1$. Thus all pulled-back local monodromies are killed. We take the extended period map on $\widetilde{C}_s$ to be the pullback of the previous extended map, with the restricted equivariance data.

Write $\widetilde{Z}_t \rightarrow \widetilde{C}$ for the algebraized family in the first construction. Functoriality of the stack morphism q identifies the analytic family on $\widetilde{C}_s$, together with its G_s -action, with the analytification of the cartesian algebraic pullback

$$\widetilde{Z}_t \times_{\widetilde{C}} \widetilde{C}_s \longrightarrow \widetilde{C}_s.$$

This pullback is already flat and projective with ADE K3 fibers. If the new analytic family is algebraized separately as in the lemma, the fully faithful GAGA comparison used there, applied to the projective total spaces over $\widetilde{C}_s$, algebraizes the displayed analytic identification and its compatible group action. We may therefore use this literal algebraic pullback, with the induced G_s -action, in forming the second quotient.

Let Z_K be the generic fiber of the original contraction over K . Original descent identifies $\widetilde{Z}_{t, \widetilde{K}}$ with $Z_K \times_K \widetilde{K}$, including its G -descent datum. After extending scalars to $\widetilde{K}_s$ and restricting to G_s , this is exactly the descent datum of $Z_K \times_K K_s$. Uniqueness of effective finite Galois descent therefore identifies the generic fiber of the second quotient with $Z_K \times_K K_s$. The fixed smooth generic K3 surface S_K^0 is a resolution of Z_K ; its base change is still birational to $Z_K \times_K K_s$, since geometric integrality ensures that the base change retains the dense open on which the resolution is an isomorphism. Consequently the second model $\mathcal{Z}$ has the function field of the actual algebraic square pullback of the fixed S^0 -family. The uniruledness just proved therefore applies to every smooth projective model of that fixed pulled-back family, as required. $\square$

6. AN ALGEBRAIC COMPARISON BY TWO SQUARE SUBSTITUTIONS

We construct a correspondence between two surface fibers by passing through the third cohomology of uniruled threefolds. The first substitution, $t = s + u^2$, makes degree-three Hodge maps algebraic. The second, $s = b + v^2$, produces a reducible conic whose difference of components recovers degree two with the explicit factor -4 . Two points require care: the degree-three comparison must apply to the particular Leray class furnished by the conic, and the omitted boundary must have fixed Hodge structures before b varies. We formulate the argument for one fixed pencil, including this order of choices.

Theorem 6.1 (Two-substitution comparison). *Let $O \subset \mathbf{A}_t^1$ be a nonempty Zariski open set, and let*

$$f : \mathcal{S} \longrightarrow O, \quad g : \mathcal{X} \longrightarrow O$$

be smooth projective families of connected surfaces, with the fibers of f K3 surfaces. Suppose that the following conditions hold.

- (i) *A polarized rational subvariation $\mathcal{U} \subset R^2 f_* \mathbb{Q}$ of K3 type, with nondegenerate restricted cup product, is given. Its period map is nonconstant, and $U_t = \mathcal{U}_t$ is simple for all but countably many $t \in O$.*
- (ii) *There are commuting involutions j, δ of $\mathcal{X}$ over O such that*

$$e = \frac{1}{4}(1 + j)(1 - \delta)$$

annihilates $R^q g_ \mathbb{Q}$ for $q \neq 2$, together with an isomorphism of rational variations of Hodge structure*

$$\varphi : \mathcal{U} \xrightarrow{\sim} eR^2 g_* \mathbb{Q}.$$

- (iii) *For s in a nonempty Zariski open subset of $\mathbf{A}^1$, the total spaces of the two surface families pulled back by $t = s + u^2$ have smooth projective uniruled threefold models.*

Then there are a nonempty Zariski open set $G \subset O$ and a countable set $E \subset G$ such that, for every $b \in G \setminus E$, a correspondence

$$\Theta_b \in \mathrm{CH}^2(S_b \times X_b)_{\mathbb{Q}}$$

induces φ_b on U_b . The open set G and all auxiliary correspondences used before the choice of b can be fixed for the given pencil.

The isomorphism φ in the theorem is the specified map of variations; it need not preserve the two cup-product forms. A cycle on a product acts from its first factor to its second, with the convention of the Introduction. In particular, a codimension-three cylinder from a surface to a fourfold gives $H^2(-)(-1) \rightarrow H^4(-)$, and its transpose gives $H^4(-) \rightarrow H^2(-)(-1)$. We suppress Tate twists on individual vectors.

6.1. Third cohomology of uniruled threefolds.

Lemma 6.2. *If Z is a smooth projective uniruled threefold over $\mathbb{C}$, then $H^3(Z)$ is a direct summand of a finite sum of spaces $H^1(C_i)(-1)$, with both maps of the splitting induced by algebraic correspondences. Consequently, every rational Hodge map between the third cohomologies of two such threefolds is algebraic.*

Proof. Uniruledness gives a generically finite dominant rational map $T \times \mathbf{P}^1 \dashrightarrow Z$ for a smooth projective surface T . Indeed, choose a two-dimensional parameter space of rational curves dominating Z and, if necessary, make a generically finite change of its parameter surface to trivialize the generic rational curve. Principalize the base ideal of the rational map in the smooth product, using projective blowups with smooth centers [42, Theorem 1.0.1]. This gives a morphism

$a : \widetilde{T \times \mathbf{P}^1} \rightarrow Z$. The identity $a_* a^* = (\deg a) \text{id}$ gives an algebraic splitting of $H^3(Z)$ from the third cohomology of the source.

The product formula and the blowup formula give

$$H^3(T \times \mathbf{P}^1) = H^3(T) \oplus H^1(T)(-1),$$

and each blowup along a curve adds its $H^1(-)(-1)$; a point adds nothing in degree three. The product projections are algebraic, and the full blowup decompositions are given by algebraic projectors [26, Appendix A, equation (9) and Proposition A.1]; see also [19, Section 6.7] for the Chow blowup formalism. To see the sign in the only new summand, let $\pi : \tilde{Z} \rightarrow Z$ be a blowup along a smooth curve C , with exceptional divisor $j : E \hookrightarrow \tilde{Z}$ and $q : E \rightarrow C$. Its inclusion is $j_* q^*$ and its retraction is $-q_* j^*$, since

$$q_* j^* j_* q^* y = q_*(c_1(\mathcal{O}_E(-1)) \smile q^* y) = -y.$$

The other summand has inclusion π^* and projection π_* . These maps and the product projections provide the required algebraic splittings at every step of the resolution.

For completeness, the needed splittings for a surface use only divisor classes. Choose a smooth ample curve $i : C \hookrightarrow T$. Hard Lefschetz says that multiplication by $[C]$ is an isomorphism $H^1(T)(-1) \rightarrow H^3(T)$. Since this is $i_* i^*$, restriction $i^* : H^1(T) \rightarrow H^1(C)$ is injective and $i_* : H^1(C)(-1) \rightarrow H^3(T)$ is surjective. Polarizability gives a Hodge retraction r of i^* and a Hodge section s of i_* . Poincaré duality identifies their tensors with rational classes of type $(0,0)$ in, respectively,

$$H^1(C) \otimes H^1(T)(1) \quad \text{and} \quad H^1(T) \otimes H^1(C)(1).$$

The underlying untwisted classes have type $(1,1)$ on $C \times T$ and $T \times C$, so the Lefschetz $(1,1)$ theorem represents them by rational divisors. Their actions on the indicated degrees are r and s . Together with restriction and Gysin, they split both surface summands through $H^1(C)(-1)$. Taking products with the point and fundamental classes of $\mathbf{P}^1$ realizes these splittings in the product decomposition above. Composing with a^* and $(\deg a)^{-1} a_*$ proves the first claim.

Finally, a Hodge map between two finite sums of $H^1(C_i)(-1)$ is given by divisor correspondences on products of the curves. Composing with the splittings gives a cycle in $\text{CH}^3(Z \times Z')_{\mathbb{Q}}$ with exactly the prescribed action on H^3 . $\square$

6.2. Coefficient projections on affine curves. The degree-three comparison uses the canonical $H^1(R^2)$ term on the affine square-pullback curves. We first fix the coefficient projections on the original curve O , so that the same projections remain available after both substitutions. We work in the rational derived category of sheaves on a complex curve; Tate twists record the Hodge structures on the corresponding cohomology groups.

Lemma 6.3. *Let B be a smooth affine complex curve.*

- (i) *A bounded complex on B with finite-rank local-system cohomology only in even degrees has a unique splitting as the direct sum of its cohomology sheaves, shifted to their degrees, inducing the identity on each cohomology sheaf.*
- (ii) *If A is a bounded complex with local-system cohomology, then there is a natural exact sequence*

$$0 \longrightarrow H^1(B, \mathcal{H}^{n-1} A) \longrightarrow \mathbb{H}^n(B, A) \longrightarrow H^0(B, \mathcal{H}^n A) \longrightarrow 0.$$

For a morphism of such complexes, its action on the left-hand term is determined by its action on $\mathcal{H}^{n-1}$.

Proof. An affine complex curve has the homotopy type of a one-dimensional CW complex. For local systems L, M ,

$$\text{Ext}_B^r(L, M) = H^r(B, L^\vee \otimes M) = 0 \quad (r > 1).$$

For two distinct nonzero even degrees $i < j$, the obstruction to splitting the corresponding truncation triangle lies in an Ext^{j-i+1} group, and the ambiguity lies in an Ext^{j-i} group. Their degrees are at least three and two, respectively, so both vanish. Induction proves (i), including uniqueness with the specified action on cohomology. In (ii), the hypercohomology spectral sequence has only columns zero and one; its degeneration and functoriality give the assertion. $\square$

For the K3 family of Theorem 6.1, this gives a canonical splitting of rational complexes

$$(6.1) \quad Rf_*\mathbb{Q} \simeq \mathbb{Q} \oplus (R^2f_*\mathbb{Q})[-2] \oplus \mathbb{Q}(-2)[-4].$$

We make this splitting on the original affine curve O and pull it back under subsequent base changes. On the other side, the group-algebra idempotent has a single nonzero cohomology sheaf and therefore gives canonically

$$(6.2) \quad eRg_*\mathbb{Q} \simeq (eR^2g_*\mathbb{Q})[-2] \simeq \mathcal{U}[-2].$$

Use the restricted cup product to write $p : R^2f_*\mathbb{Q} \rightarrow \mathcal{U}$ for the orthogonal projection. Pullback of (6.1), followed by p , and (6.2), followed by φ^{-1} , define coefficient projections after any base change from O . These are morphisms of rational sheaf complexes. Their use on a surface base requires neither a new derived splitting there nor algebraicity of the projector p .

6.3. The geometric Hodge comparison in degree three. The next lemma gives the precise coefficient comparison that we need. It uses a Hodge class on a relative product to construct a Hodge map; algebraicity of that map will then follow from Lemma 6.2.

Lemma 6.4. *Let I be a smooth affine curve, let $h_S : E^S \rightarrow I$ be a smooth projective K3 family, and let $h_X : E^X \rightarrow I$ be a smooth projective surface family with a relative algebraic idempotent e acting on the direct-image complex whose action on $R^q h_{X*}\mathbb{Q}$ vanishes for $q \neq 2$. Suppose that*

$$a : R^2 h_{S*}\mathbb{Q} \longrightarrow eR^2 h_{X*}\mathbb{Q}$$

is a morphism of rational Hodge variations. Let Z^S, Z^X be smooth projective compactifications which extend the two maps to a smooth projective compactification of I and are unchanged over I . Write $\bar{I}$ for that compactified curve. There is a rational Hodge map $c : H^3(Z^S) \rightarrow H^3(Z^X)$ satisfying

$$e \text{ res}_X(cz) = H^1(I, a)(\text{res}_S z), \quad z \in H^3(Z^S),$$

where $H^3(E^S)$ and $eH^3(E^X)$ are identified canonically with $H^1(I, R^2 h_{S}\mathbb{Q})$ and $H^1(I, eR^2 h_{X*}\mathbb{Q})$.*

Proof. Put $R = E^S \times_I E^X$, and resolve its closure in $Z^S \times_{\bar{I}} Z^X$, preserving R . This gives a smooth projective fivefold $\bar{R}$ with proper maps $\pi_S : \bar{R} \rightarrow Z^S$ and $\pi_X : \bar{R} \rightarrow Z^X$. On each fiber, Poincaré duality represents a by the unique tensor in the $H^2(E_t^S) \otimes H^2(E_t^X)$ summand of $H^4(R_t)$ with that correspondence action. The tensors form a flat rational section α of $R^4(R/I)_*\mathbb{Q}$ and have type $(2, 2)$.

The global invariant cycle theorem [11, Theorem 4.1.1] identifies the image of $H^4(\bar{R}) \rightarrow H^4(R_t)$ with the monodromy invariants. Restriction is a morphism of polarizable pure Hodge structures. Its image therefore contains α_t , and semisimplicity gives a rational lift $\tilde{\alpha} \in H^4(\bar{R})$ of type $(2, 2)$. For example, split restriction onto its image as a Hodge map and apply the splitting to α_t . The restriction of this lift to every fiber is α , because both define flat sections. Define

$$c(z) = (\pi_X)_*(\tilde{\alpha} \smile \pi_{S*}^* z).$$

Cup product raises degree by four and the proper pushforward lowers degree by four, so this is a rational Hodge map from H^3 to H^3 .

Its restriction to I is the relative cohomological correspondence defined by $\tilde{\alpha}|_R$. Pullback, cup product, and integration along the smooth proper relative surface projection give a morphism $Rh_{S*}\mathbb{Q} \rightarrow Rh_{X*}\mathbb{Q}$; on degree-two cohomology sheaves it is a . For the source, $R^3 h_{S*}\mathbb{Q} = 0$, and

for the projected target, $eR^3h_{X*}\mathbb{Q} = 0$. Lemma 6.3(ii), with $n = 3$, therefore identifies the entire source and projected target H^3 with their canonical $H^1(R^2)$ terms. Functoriality shows that the projected open action of c is exactly $H^1(I, a)$, as asserted. The proof uses no claim that $\tilde{a}$ is algebraic. $\square$

6.4. Compatible projective models. The following construction will be used for both surface families. It fixes the first compactifications before the relative correspondence and the point b are chosen.

Lemma 6.5. *For a smooth projective surface family $\mathcal{Y} \rightarrow O$, its first square pullbacks admit models Z projective over $\mathbf{A}_s^1$ and a nonempty open set $G \subset O$ such that $Z_G \rightarrow G$ is smooth. These models may be chosen with the following additional property. For every $b \in G$ there is a smooth projective fourfold M_b , with morphisms to $\mathbf{P}_t^1$ and $\mathbf{P}_v^1$, whose inverse images of $V_0 \subset \mathbf{P}_v^1$ and $O \subset \mathbf{P}_t^1$ are, respectively,*

(i) $Z_G \times_G V_0$, where $V_0 \rightarrow G$ is $v \mapsto b + v^2$ and $V_0 = \{v \in \mathbf{A}^1 : b + v^2 \in G\}$;

(ii) $\mathcal{Y} \times_O Q_b$, where

$$(6.3) \quad Q_b = \{u^2 + v^2 = (t - b)w^2\} \subset O \times \mathbf{P}^2.$$

The two identifications agree on their intersection. In particular, both specified spaces are preserved as actual open subvarieties. Shrinking G before choosing b preserves these conclusions.

Proof. Choose a smooth projective compactification $\overline{\mathcal{Y}} \rightarrow \mathbf{P}_t^1$ whose inverse image over O is $\mathcal{Y}$. It is obtained by taking a projective closure of the family and resolving away from its smooth open. Take the component containing the given open. Its source is smooth and integral, and its map to the nonsingular curve is flat. The generic fiber is geometrically integral: a smooth geometrically connected surface is geometrically integral. Hence the base changes below have the dense unchanged open and the same function field as the indicated pullback. The map

$$C = \mathbf{P}_u^1 \times \mathbf{A}_s^1 \longrightarrow \mathbf{P}_t^1, \quad t = s + u^2,$$

is a morphism: in homogeneous coordinates on $\mathbf{P}_u^1$, with $u = U_0/U_1$, it is $[U_0 : U_1] \mapsto [U_0^2 + sU_1^2 : U_1^2]$. Set $W = \overline{\mathcal{Y}} \times_{\mathbf{P}_t^1} C$. Choose a projective resolution $Z \rightarrow W$ as a sequence of blowups, all supported over $t \notin O$. The flat base change is generically smooth and therefore reduced, with no component supported only over the boundary; its open over O is smooth. Resolution is performed on each component if needed. Strong projective resolution in characteristic zero, preserving the regular locus, permits this construction [42, Theorems 1.0.2–1.0.3]. The map $Z \rightarrow \mathbf{A}_s^1$ is projective, and is smooth after deleting finitely many values of s . Fix this entire construction, and take G inside its smooth locus and inside O .

Let B_b be the normalization of $\mathbf{P}_t^1 \times \mathbf{P}_v^1$ in the field

$$\mathbb{C}(t, v)(\sqrt{t - b - v^2}).$$

It is a projective surface finite over the product. Its finite- v open is $\mathbf{P}_u^1 \times \mathbf{A}_v^1$, with $t = b + u^2 + v^2$. Its open over O is Q_b : the latter is smooth and the map

$$Q_b \longrightarrow O \times \mathbf{P}_v^1, \quad (t, [u : v : w]) \longmapsto (t, [v : w])$$

has no base point: $v = w = 0$ would force $u = 0$. It is proper and has finite fibers, hence is finite of degree two, with the required function field. The same verification on the finite- v open uses $\mathbf{P}_u^1 \times \mathbf{A}_v^1$ and the morphism $[U_0 : U_1] \mapsto [U_0^2 + (b + v^2)U_1^2 : U_1^2]$. These assertions identify both opens by the universal property of normalization. Smoothness of Q_b follows in the affine chart $w = 1$ from the derivative in t , and at infinity from the derivatives in u, v .

Start with the projective fourfold $W_b = \overline{\mathcal{Y}} \times_{\mathbf{P}_t^1} B_b$. On its finite- v open, pull back the chosen blowup sequence $Z \rightarrow W$ by $s = b + v^2$. This map of affine lines is finite flat, so blowup commutes

with the base change. Each blowup extends to the whole current projective model without altering its open over O . Explicitly, if $j : U \hookrightarrow W_b^{(r)}$ is its finite- v open and $\mathcal{I} \subset \mathcal{O}_U$ is the next center ideal, extend it by

$$\mathcal{J} = \ker(\mathcal{O}_{W_b^{(r)}} \longrightarrow j_*(\mathcal{O}_U/\mathcal{I})).$$

This is a coherent ideal: the schemes are noetherian, the open immersion is quasi-compact, and the kernel is a quasi-coherent submodule of $\mathcal{O}_{W_b^{(r)}}$. It restricts to $\mathcal{I}$ and is the unit ideal off the closed inverse image of $\mathbf{P}_t^1 \setminus O$. Thus its blowup has the asserted properties.

After extending the finite sequence, both opens in the statement are present and smooth. A final projective resolution which is an isomorphism on the regular locus preserves their union. It gives M_b and the two morphisms to the parameter lines. The construction uses one fixed set of discarded s -values, independent of b . $\square$

6.5. Proof of the comparison theorem.

Proof of Theorem 6.1. We retain the notation $\mathcal{U}$, p , and φ above.

The fixed part. We first prove the vanishing that will make each cylinder class restrict to zero on the projective threefold fibers:

$$(6.4) \quad H^0(O, \mathcal{U}) = 0.$$

By the fixed-part theorem, the invariant part of the geometric polarized variation is a constant Hodge subvariation; see [11, Theorem 4.1.1 and Corollary 4.1.2]. A nonzero such part would equal U_t at a simple fiber, and hence everywhere as a local system. The fixed-part theorem would then make its period constant, contrary to the hypothesis.

For $s \in O$, put $I_s = \{u \in \mathbf{A}^1 : s + u^2 \in O\}$. The pullback of $\mathcal{U}$ to I_s has the same monodromy image as $\mathcal{U}$ on O . To see this, remove the branch value s temporarily. A loop about s changes the sheet of the double cover and has trivial monodromy on $\mathcal{U}$; it can therefore be used to adjust the parity of any loop without changing its monodromy. Consequently,

$$(6.5) \quad H^0(I_s, \mathcal{U}) = 0 \quad (s \in O).$$

A relative correspondence in degree three. Apply Lemma 6.5 to both families, obtaining models $Z^S \rightarrow \mathbf{A}_s^1$ and $Z^X \rightarrow \mathbf{A}_s^1$. Shrink a common nonempty open $G \subset O$ so that both families $\zeta^k : Z_G^k \rightarrow G$ are smooth projective and their fibers are uniruled. Write E_s^S and E_s^X for their unchanged opens over I_s . The Leray spectral sequence over the affine curve I_s and the degree-two projections give maps

$$\begin{aligned} r_s^S : H^3(Z_s^S) &\longrightarrow H^3(E_s^S) = H^1(I_s, R^2 f_* \mathbb{Q}) \xrightarrow{p} H^1(I_s, \mathcal{U}), \\ r_s^X : H^3(Z_s^X) &\longrightarrow eH^3(E_s^X) = H^1(I_s, eR^2 g_* \mathbb{Q}) \xrightarrow{\varphi^{-1}} H^1(I_s, \mathcal{U}). \end{aligned}$$

All coefficients in these expressions have been pulled back to I_s .

Here is the Hodge-theoretic content of this comparison. Give $H^1(I_s, \mathcal{U})$ the mixed Hodge structure transported from $eH^3(E_s^X)$ by the second identification above. The weight-three part

$$T_s = W_3 H^1(I_s, \mathcal{U})$$

is pure and polarizable. Restriction from a smooth projective compactification has image equal to the lowest weight [11, Corollary 3.2.17]; the algebraic projector on the open commutes with the weight filtration. Thus r_s^X is a surjective Hodge map onto T_s . This argument uses e only on E_s^X .

Apply Lemma 6.4 with $a = \varphi p$, after pullback to I_s . It gives a rational Hodge map

$$c_s : H^3(Z_s^S) \longrightarrow H^3(Z_s^X), \quad r_s^X c_s = r_s^S.$$

In particular, r_s^S is also a Hodge map into this same pure space T_s . This proves the coefficient compatibility at exactly the strength required, using geometric invariant cycles and the canonical Leray term over a curve. Lemma 6.2 now realizes c_s by a cycle in $\mathrm{CH}^3(Z_s^S \times Z_s^X)_{\mathbb{Q}}$.

We next spread the cycles just obtained. Write $\Delta = \mathbf{A}^1 \setminus O$. The punctures of I_s are the roots of $u^2 = c - s$ for $c \in \Delta$, together with infinity in its compactification. Because $s \in G \subset O$, they remain distinct. Locally in s their motion gives a topologically trivial punctured curve with its pulled-back coefficient system. Consequently $H^1(I_s, \mathcal{U})$ forms a local system over G , and r^S, r^X are morphisms of local systems.

The projective family $Z_G^S \times_G Z_G^X$ has relative dimension six. Use the relative Hilbert schemes for polynomials of degree three to parameterize its codimension-three subschemes, and take all finite products together with all rational coefficient lists [21, Theorem 3.2]. This is a countable collection of finite-type parameter spaces and includes a representative of every rational codimension-three cycle. Stratify them into connected smooth locally closed varieties. The degree-six fundamental cycle class of a universal flat subscheme is locally constant in the cohomology of the ambient fibers. Indeed, over a smooth stratum the ambient total space is smooth, so the universal structure sheaf has a finite locally free resolution. Flatness over the stratum makes derived restriction to any fiber equal to ordinary restriction. The degree-six part of its Chern character restricts to the fundamental cycle class, with generic multiplicities. This follows from Grothendieck–Riemann–Roch, dévissage, and the compatibility of algebraic and topological Chern classes [19, Theorem 15.2, Example 15.2.16(a), and Proposition 19.1.2]. Since these characteristic classes come from a class on the total space, their fiber restrictions are flat. Lemma 8.2 proves the corresponding assertion for the cycle parameter spaces used in specialization. Thus on each stratum the equality

$$(6.6) \quad r^X C_* = r^S \quad \text{on } R^3 \zeta_*^S \mathbb{Q}$$

holds everywhere or nowhere. Successful strata cover G pointwise, since a cycle has been constructed for every $s \in G$. A nondominating finite-type stratum has finite image in the irreducible curve G . If all successful strata were nondominating, their images would be countable, whereas $G(\mathbb{C})$ is uncountable.

Choose a successful dominating stratum. A closed point of its generic fiber gives a generically finite multisection; normalize its closure and shrink G so that it is a finite *étale* cover $\nu : G' \rightarrow G$, of degree d . The universal rational cycle restricts to a cycle C' over G' . Its trace

$$(6.7) \quad C = \frac{1}{d} \nu_* C' \in \mathrm{CH}^3(Z_G^S \times_G Z_G^X)_{\mathbb{Q}}$$

satisfies (6.6): over every s , its action is the average of the d successful actions over $\nu^{-1}(s)$. Generic flatness permits a final shrinking so that the chosen cycle representatives are flat over G . From this point onward, fix G , both models, and this relative cycle C .

The second substitution and its cylinders. Choose $b \in G$, subject to the countable exclusions specified below, and set $s = b + v^2$. Lemma 6.5 supplies smooth projective fourfolds M^S, M^X with the two required opens. Let

$$V_0 = \{v \in \mathbf{A}^1 : b + v^2 \in G\}, \quad Y^k = \mathcal{Y}^k \times_O Q_b \quad (k = S, X),$$

where $\mathcal{Y}^S = \mathcal{S}$ and $\mathcal{Y}^X = \mathcal{X}$. In the chart $w = 1$, put

$$Q_b^\circ = \{(u, v) \in \mathbf{A}^2 : v \in V_0, b + u^2 + v^2 \in O\}.$$

The substitutions are $t = s + u^2$ and $s = b + v^2$. The two projections $\pi_t : Q_b \rightarrow O$ and $\pi_v : Q_b \rightarrow \mathbf{P}_v^1$ are given by t and $[v : w]$. For each $k = S, X$, the preserved opens form the square

$$\begin{array}{ccc} \mathcal{Y}^k \times_O Q_b^\circ & \hookrightarrow & Z_G^k \times_G V_0 \\ \downarrow & & \downarrow \\ Y^k = \mathcal{Y}^k \times_O Q_b & \hookrightarrow & M^k. \end{array}$$

All arrows are open immersions, and the upper-left open is the intersection of the two opens in M^k . The relative product over V_0 of the two upper-right opens carries the pullback of the relative cycle C . The lower-left open contains the complete cylinders $F_b^k \times L_\pm$ defined below, and the upper-left open is the coefficient-comparison locus. The choice order is summarized by

Fixed before b	Depending on b
Original pencil, G , Z_G^S, Z_G^X , relative cycle C	Q_b, V_0, M^k and the cycles $\widehat{C}, D_k$ constructed below

Pull (6.7) back to V_0 and close its cycle representatives in $M^S \times_{\mathbf{P}_v^1} M^X$. Viewed in the absolute product, the result is a codimension-four cycle $\widehat{C}$, hence acts degree-preservingly. Its support is over the diagonal of the v -line. Restriction of its action to V_0 is consequently the original relative action: restricting the target restricts the source as well, and proper pushforward commutes with this open restriction.

The fiber of $Q_b \rightarrow O$ at b is the union of the two lines

$$L_+ = \{u = iv\}, \quad L_- = \{u = -iv\}.$$

Write $F_b^S = S_b$ and $F_b^X = X_b$ for the original surface fibers. The inclusions $i_{k,\pm} : F_b^k \times L_\pm \hookrightarrow M^k$, together with the projections $p_{k,\pm} : F_b^k \times L_\pm \rightarrow F_b^k$, define cylinder correspondences from F_b^k to M^k . Let D_k be their difference, with the plus component first. Thus

$$D_k \in \mathrm{CH}^3(F_b^k \times M^k)_{\mathbb{Q}}, \quad d_k = (i_{k,+})_* p_{k,+}^* - (i_{k,-})_* p_{k,-}^*,$$

and its cohomological action is

$$d_k : H^2(F_b^k)(-1) \longrightarrow H^4(M^k).$$

The full lines, including their points at infinity, are contained in the preserved open Y^k . Since these cylinders are projective, they are closed in M^k .

In the chart $w = 1$, the two lines meet at $(u, v) = (0, 0)$ with independent tangent directions. Their intersection multiplicity is therefore one. Their sum is the divisor of the base parameter $t - b$, whose restriction to either compact line has degree zero. Thus, on the smooth surface Q_b ,

$$L_+ \cdot L_- = 1, \quad L_+^2 = L_-^2 = -1.$$

The normal bundles of the cylinder divisors are pulled back from these normal bundles on Q_b . The self-intersection and excess-intersection formulas [19, Theorem 6.3, Corollary 6.3, and Theorem 19.2] therefore give

$$(6.8) \quad {}^t d_k d_k = -4 \mathrm{id} \quad \text{on } H^2(F_b^k)(-1).$$

More explicitly, each self-term is the identity on $H^2(F_b^k)$ multiplied by $\deg N_{L_\pm/Q_b} = -1$, and each cross-term is the identity multiplied by the intersection multiplicity one. The difference convention gives $(-1) + (-1) - 1 - 1 = -4$. Every operation on the surface factor is the identity, so the formula holds on all of $H^2(F_b^k)$.

Coefficient classes of the cylinders. The relative comparison acts on coefficient cohomology over the affine conic chart. We must therefore identify the coefficient images of the two cylinder classes. For either complete component $L_\pm \subset Q_b$, its inverse image in Y^k is $F_b^k \times L_\pm$. We shall show that the coefficient projection carries its total-space Gysin class to the Gysin class of the same coefficient on Q_b . We verify the compatibility on a general surface base before restricting to the comparison chart.

Let B_2 be a smooth surface with a map $\tau : B_2 \rightarrow O$, and let $\sigma : D \hookrightarrow B_2$ be a smooth divisor whose map to O is constant, with value b . Write $\mathcal{S}_{B_2} = \mathcal{S} \times_O B_2$ and similarly for $\mathcal{X}_{B_2}$. The inverse image of D is $S_b \times D$, respectively $X_b \times D$. For $x \in U_b$, its pullback to this inverse image

has a Gysin class in $H^4(\mathcal{S}_{B_2})$, with domain understood as $U_b(-1)$. Under the coefficient projection its image is

$$(6.9) \quad \sigma_*(x) \in H^2(B_2, \tau^*\mathcal{U}).$$

The same formula holds for the e -part on $\mathcal{X}_{B_2}$, with coefficient $\varphi_b(x)$.

To verify the formula, make the coefficient Thom construction explicit. The complex orientation of the normal line of σ gives a morphism

$$u_\sigma : \sigma_*\mathbb{Q}_D[-2] \longrightarrow \mathbb{Q}_{B_2}.$$

On a normal disc, the relative cochain complex of the disc and its punctured disc has one cohomology group, generated in degree two by this orientation. Tensoring this local calculation with a bounded locally constant rational complex K gives the natural purity identification $\sigma^!K \simeq \sigma^*K[-2]$. More precisely, the globally defined map

$$\mu_K : \sigma^*K \otimes \sigma^!\mathbb{Q}_{B_2} \longrightarrow \sigma^!K$$

is adjoint to the projection-formula/counit composite

$$\sigma_*(\sigma^*K \otimes \sigma^!\mathbb{Q}_{B_2}) \simeq K \otimes \sigma_*\sigma^!\mathbb{Q}_{B_2} \longrightarrow K.$$

The normal-disc calculation shows that μ_K is an isomorphism for such K , and the complex orientation identifies $\sigma^!\mathbb{Q}_{B_2} \simeq \mathbb{Q}_D[-2]$. This gives the stated natural purity isomorphism globally. Equivalently, the coefficient Gysin morphism is

$$\sigma_*\sigma^*K[-2] \simeq (\sigma_*\mathbb{Q}_D[-2]) \otimes K \xrightarrow{u_\sigma \otimes \text{id}} K.$$

The first isomorphism is the projection formula, and this construction is natural in K . In the rational sheaf category the displayed shifts are only $[-2]$; the Tate (-1) records the Hodge degree of the complex normal orientation on cohomology.

For completeness, let $h : Y \rightarrow B_2$ be either smooth proper surface family, let $Y_D = Y \times_{B_2} D$, and write $j : Y_D \hookrightarrow Y$ and $h_D : Y_D \rightarrow D$. The normal line of j is $h_D^*N_{D/B_2}$ with its pulled complex orientation. The normal-disc construction therefore identifies the total-space Thom morphism with the pullback of u_σ . Under the natural proper base-change identification

$$Rh_*j_*\mathbb{Q}_{Y_D}[-2] \simeq \sigma_*Rh_{D*}\mathbb{Q}_{Y_D}[-2] \simeq \sigma_*\sigma^*Rh_*\mathbb{Q}_Y[-2],$$

the projection formula consequently identifies $Rh_*(u_j)$ with the coefficient Gysin morphism for $K = Rh_*\mathbb{Q}_Y$. The formal base-change and projection-formula statements used here are natural for these bounded rational complexes [35, Theorems 5.10 and 6.3].

Finally $D \rightarrow O$ factors through b . Proper base change identifies the restricted splitting with the canonical splitting of $R\Gamma(S_b, \mathbb{Q})$, respectively $eR\Gamma(X_b, \mathbb{Q})$, pulled back from a point. The class pulled back from x is therefore its degree-two product coefficient. Naturality of the coefficient Gysin map makes it commute with the splitting pulled from (6.1), its degree-two projection, and the idempotent (6.2). This proves (6.9). The construction applies to each divisor separately, including when two such divisors intersect.

Comparison on the affine conic chart. For $x \in U_b$, put $a = \widehat{C}_*d_S(x)$. Our immediate aim is to prove $ea = d_X(\varphi_b x)$ on $\mathcal{X} \times_O \mathbb{Q}_b^\circ$, using only the degree-three action of C . On $\mathbb{Q}_b^\circ$, write $q : \mathbb{Q}_b^\circ \rightarrow V_0$ for projection to v and $\tau(u, v) = b + u^2 + v^2$. Put $c = d_S(x)|_{Z_C^S \times_G V_0}$. The restriction of c to every threefold fiber over v is zero. Each cylinder meets that fiber transversally: the map $L_\pm \rightarrow \mathbf{P}_v^1$ is an isomorphism. Its restriction is therefore the ordinary surface Gysin map at $u = \pm iv$, with no excess class, including at $v = 0$. Pair either surface Gysin image of x with an arbitrary class in the second cohomology of the threefold. This testing class restricts to a monodromy-invariant class on surface fibers over I_{b+v^2} . Its orthogonal projection to $\mathcal{U}$ is zero by (6.5), so the pairing is zero.

Poincaré duality on the projective threefold proves the claimed vanishing of each term. At $v = 0$ the two surface inclusions also coincide.

Write $\rho_k : Z_G^k \times_G V_0 \rightarrow V_0$ and $A_k = R\rho_{k*}\mathbb{Q}$. Lemma 6.3(ii) places c in the canonical Leray subspace

$$(6.10) \quad H^1(V_0, \mathcal{H}^3 A_S) \subset \mathbb{H}^4(V_0, A_S).$$

Set $K = Rq_*\tau^*\mathcal{U}[-2]$. Its cohomology sheaves are local systems, by the same noncollision of punctures used over G , now pulled back to V_0 . Restriction to the open surface family over Q_b° , followed by the coefficient projections (6.1)–(6.2), gives morphisms of rational complexes

$$\beta_S : A_S \longrightarrow K, \quad \beta_X : A_X \longrightarrow K.$$

The relative correspondence gives $\gamma : A_S \rightarrow A_X$: on the relative product, take cup product with its class of degree six and integrate along the proper relative threefold projection. Its action on R^3 is the action prescribed by C . Equation (6.6) says exactly that

$$\mathcal{H}^3(\beta_X \gamma) = \mathcal{H}^3(\beta_S).$$

Functoriality in Lemma 6.3(ii), applied to the subspace (6.10), now yields

$$(6.11) \quad (\beta_X \gamma)_* c = (\beta_S)_* c \quad \text{in } \mathbb{H}^4(V_0, K) = H^2(Q_b^\circ, \tau^*\mathcal{U}).$$

The reason degree-three agreement suffices is precisely the membership (6.10): the left term of the functorial exact sequence in Lemma 6.3(ii) only sees $\mathcal{H}^3$. No equality on the $H^0(\mathcal{H}^4)$ quotient is used. The u -fibers and the v -base have cohomological dimension one, so the last group is also canonically

$$H^1(V_0, R^1 q_* \tau^* \mathcal{U}).$$

The restrictions of the cylinders to Q_b° are the two smooth section divisors $\sigma_\pm(v) = (\pm iv, v)$. Formula (6.9) shows that the right side of (6.11) is

$$\sigma_{+*}(x) - \sigma_{-*}(x).$$

The same formula on the X side identifies this with the coefficient class of $d_X(\varphi_b x)$. Although the sections meet at $v = 0$, purity has been applied to each regular embedding separately. By (6.2), the coefficient projection identifies the entire e -part of the fourth cohomology over Q_b° with $H^2(Q_b^\circ, \tau^*\mathcal{U})$. Hence we have proved

$$(6.12) \quad e a = d_X(\varphi_b x) \quad \text{on } \mathcal{X} \times_O Q_b^\circ.$$

Excluding the boundary error. We must prove (6.12) on all of Y^X , since the complete cylinders also contain their points at infinity. Let $F = \mathbf{A}^1 \setminus G$, a fixed finite set. The complement of $\mathcal{X} \times_O Q_b^\circ$ in Y^X is a disjoint union of smooth divisors:

- (i) the two infinity sections $[u : v : w] = [\pm i : 1 : 0]$, each carrying the fixed family $\mathcal{X} \rightarrow O$;
- (ii) for each $c_0 \in F$ and each of the two roots a_0 of $a_0^2 = c_0 - b$, the divisor $v = a_0$, carrying the fixed open threefold

$$E_{c_0}^X = \{(x, u) : g(x) = c_0 + u^2 \in O\}.$$

The second description follows by substituting $v = a_0$ in the affine conic equation. The projective equation of a finite divisor is $v = a_0 w$; a point with $w = 0$ on it would also have $v = u = 0$, which is impossible. Thus it has no point on either infinity section. Different finite values of v give disjoint divisors, and all the roots listed above are distinct since $b \in G$. Consequently the entire boundary is a disjoint union, with the fixed isomorphism types just listed. Their smoothness follows from smoothness of g over O , including when c_0 is not a good value for a compactified threefold.

Denote their union by D . Topological localization and the complex-oriented Thom isomorphism for each smooth closed divisor give the underlying exact sequence

$$(6.13) \quad H^2(D)(-1) \longrightarrow H^4(Y^X) \longrightarrow H^4(\mathcal{X} \times_O Q_b^\circ).$$

This is exact in mixed Hodge structures as well. The closed immersion $D \hookrightarrow Y^X$ is proper even though both varieties may be nonproper. For smooth separated complex algebraic varieties, the adjunction and duality constructions of restriction and proper Gysin lift to mixed Hodge modules; the smooth duality normalization gives exactly the codimension-one shift $[-2](-1)$ [33, Sections 1.2–1.4]. Their cohomology mixed Hodge structures agree with Deligne’s [34, Theorems 0.1–0.2 and the following paragraph]. Thus the two arrows in (6.13) are morphisms for the canonical mixed Hodge structures. Their underlying sequence is exact by the Thom calculation, hence it is exact in that abelian category. The first term’s weight-four part is isomorphic, independently of b , to $H_F(-1)$, where the fixed polarizable pure Hodge structure H_F is

$$H_F = (W_2 H^2(\mathcal{X}))^{\oplus 2} \oplus \bigoplus_{c_0 \in F} (W_2 H^2(E_{c_0}^X))^{\oplus 2}.$$

Here these spaces are polarizable because lowest-weight cohomology of a smooth variety is an image of the cohomology of a smooth projective compactification.

For all but countably many $b \in G$,

$$(6.14) \quad \text{Hom}_{\text{HS}}(U_b, H_F) = 0.$$

To check this precisely, exclude the countably many nonsimple fibers and trivialize $\mathcal{U}$ on a countable cover by simply connected discs. A nonzero map from a remaining U_b is injective. There are countably many rational linear injections into the fixed rational vector space H_F . For each such injection, the condition of being a Hodge map determines the entire Hodge filtration on U_b uniquely as the inverse image of the filtration on H_F . The nonconstant period map meets this one fixed filtration in a discrete set on a disc. Taking the indicated countable unions proves (6.14).

Consider the error map

$$U_b(-1) \longrightarrow H^4(Y^X), \quad x \longmapsto e a - d_X(\varphi_b x).$$

Both terms are morphisms of Hodge structures: the cylinder and $\widehat{C}$ act algebraically on the projective models, restriction is a mixed-Hodge morphism, e is algebraic on Y^X , and φ_b is Hodge by hypothesis. Equation (6.12) places the error in the kernel of the last arrow of (6.13). Its source is pure of weight four. Strictness for the weight filtration [11, Theorem 2.3.5(iii)] therefore puts its image in the Gysin image of $W_4(H^2(D)(-1)) \simeq H_F(-1)$. This image is a quotient of a polarizable pure Hodge structure. Semisimplicity splits that quotient, lifting the error map to $U_b(-1) \rightarrow H_F(-1)$; by (6.14) the lift is zero. We have proved

$$(6.15) \quad e \widehat{C}_* d_S(x) = d_X(\varphi_b x) \quad \text{in } H^4(Y^X), \quad x \in U_b.$$

Returning to the surfaces. The transpose cylinder can be evaluated on a class of Y^X by restriction to $X_b \times L_{\pm}$ and integration along $L_{\pm}$. These are complete projective cylinders inside Y^X , so this operation agrees with its action on classes restricted from M^X . Moreover, $e_b {}^t d_X = {}^t d_X e$, since j, δ act only on the surface coordinate. Define the algebraic correspondence

$$(6.16) \quad \Theta_b = -\frac{1}{4} e_b \circ {}^t d_X \circ \widehat{C} \circ D_S \in \text{CH}^2(S_b \times X_b)_{\mathbb{Q}}.$$

Applying the transpose to (6.15) and using (6.8) gives, for every $x \in U_b$,

$$(\Theta_b)_* x = -\frac{1}{4} {}^t d_X e \widehat{C}_* d_S(x) = -\frac{1}{4} {}^t d_X d_X(\varphi_b x) = \varphi_b x.$$

In this calculation e acts after restriction to Y^X . Thus the action is exactly the specified map on the whole of U_b . The codimension is $3 + 4 + 3 - 4 - 4 = 2$; composing with $e_b \in \text{CH}^2(X_b \times X_b)_{\mathbb{Q}}$ preserves it. Every exceptional choice of b used after fixing G belongs to the countable set excluded in (6.14), together with the nonsimple fibers. This proves the theorem with the asserted order of choices. $\square$

6.6. Application to the abelian-cover pencil.

Corollary 6.6. *Fix abelian-cover data and a section pencil for which Theorems 3.1, 3.3 and 5.1 apply, with nonconstant U -period along the pencil and simple U_t outside a countable set. On a nonempty open of that pencil, for all but countably many b , there is an algebraic correspondence*

$$\Theta_b \in \mathrm{CH}^2(S_b^0 \times X_b)_{\mathbb{Q}}$$

whose action on U_b is the given identification with $H^2(X_b)^{j=+, \delta=-}$.

Proof. The variation, the projector, and the identification are supplied by Theorems 3.1 and 5.1. Theorem 5.1 gives the uniruled threefold models on the K3 side. For X , after $t = s + u^2$ the generic fiber over B' is the conic

$$z^2 - \pi^* d_1 u^2 = \pi^*(d_0 + s d_1)$$

over $\mathbb{C}(B')$. A finite extension splits this conic, so a generically finite surface cover of B' yields a dominant rational map from a product of a surface and $\mathbf{P}^1$. Every smooth projective model of the threefold is therefore uniruled. Theorem 6.1 applies. It applies separately to every fixed pencil with these properties; its countable set of excluded b is chosen only after the pencil, the degree-three correspondence, and the good s -open have been fixed. $\square$

7. RECOVERY OF THE PRESCRIBED KUGA–SATAKE TENSOR

The comparison of the preceding section and the domination by curves give an algebraic injection from the auxiliary K3 cohomology into a tensor product of two curve cohomologies. This does not yet realize the standard tensor required by Theorem 1.1. We bridge the difference by proving that its compositions with rational Hodge maps on the two curve factors, together with their interchanges, span every Hodge map into the tensor square of the full even Clifford module. This permits us to select the prescribed right factor and polarization. The calculation applies at the construction periods selected in Lemma 7.1, where the Hodge group is the full special orthogonal group. At the end of the section we transfer the curve source by a rational Hodge isometry; Section 8 will use that transfer to obtain seeds with $\mathrm{NS} = P$ and extend one fixed tensor to every ample P -polarized target.

7.1. An algebraic embedding into a tensor of curve cohomologies.

Lemma 7.1. *Choose the abelian data and the pencil as in Lemma 4.5, and then choose b in the common good open outside the countable exceptional sets of that lemma and Corollary 6.6. Then there are smooth projective curves C_1, C_2 and a rational algebraic correspondence*

$$Z \in \mathrm{CH}^2(S_b^0 \times C_1 \times C_2)_{\mathbb{Q}}$$

whose action restricts to an injection

$$f : U_b \longrightarrow H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q}).$$

Here $U_b = T_{S_b^0}$ and $\mathrm{Hg}(U_b) = \mathrm{SO}(U_b)$.

Proof. Let $\Theta_b \in \mathrm{CH}^2(S_b^0 \times X_b)_{\mathbb{Q}}$ be the correspondence of Corollary 6.6; its restriction to U_b is the specified identification $\varphi_b : U_b \rightarrow U_{X_b}$. At these periods U_{X_b} has no nonzero rational $(1, 1)$ -class by Proposition 3.3. Proposition 4.4 therefore supplies curves C_1, C_2 , a morphism $g : C_1 \times C_2 \rightarrow X_b$, and the algebraic correspondence

$$Z_X = (\pi_1^{C_1} \boxtimes \pi_1^{C_2}) \circ {}^t \Gamma_g \circ e \in \mathrm{CH}^2(X_b \times C_1 \times C_2)_{\mathbb{Q}},$$

where the curve projectors are those constructed there. Its action on U_{X_b} is the injection g^* , with image entirely in $H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q})$. Set

$$Z = Z_X \circ \Theta_b.$$

Its action on U_b is $f = g^* \varphi_b$ and is injective. The composition passes through the surface X_b , so its codimension is $2 + 2 - 2 = 2$. The equality $U_b = T_{S_b^0}$ and the full Hodge group follow from Propositions 3.3 and 5.1 at these selected periods. $\square$

The comparison just constructed fits into

$$(U_b = T_{S_b^0}, q_Y) \xrightarrow{\varphi_b = (\Theta_b)_*} (U_{X_b}, q_X) \xrightarrow{g^*} H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q}).$$

Here q_X is the cup-product form on X_b , and q_Y is the form inherited from Y in Proposition 3.1; the latter is exactly the K3 cup-product form by Proposition 5.1. The resolved pullback satisfies $\varphi_b^* q_X = 2q_Y$. Thus the first map is an algebraic Hodge identification with a factor-two change of form, and the second is the algebraic injection supplied by the curves. The rational isometry used below concerns q_Y , not the rescaled form from X_b .

7.2. The Clifford map and its tensor form. We recall the tensor of Section 1 for an eight-dimensional quadratic space (U, q) . With the Clifford convention $v^2 = q(v, v)$, the full even module is

$$W = C^+(U, q), \quad \dim_{\mathbb{Q}} W = 2^7 = 128.$$

If e_1, e_2 form an oriented orthonormal basis of the positive period plane, with $U^{2,0} = \mathbb{C}(e_1 + ie_2)$, put $J = e_1 e_2$. Then $J^2 = -1$, and the Kuga–Satake Hodge circle acts on $W_{\mathbb{R}}$ by left multiplication by $\exp(\theta J)$. Conjugation by the same element acts on $U^{2,0}$ by $\exp(2i\theta)$, as required by the weight-two Hodge structure.

Fix the prescribed rational vector $w \in U$ with $q(w, w) \neq 0$. The standard embedding on the even module is

$$(7.1) \quad j_w : U(1) \longrightarrow \text{End}(W), \quad j_w(v)(a) = vaw \quad (a \in C^+(U, q)).$$

The output is even, and $w^{-1} = w/q(w, w)$ shows, by evaluation at $a = 1$, that j_w is injective. For $g \in \text{Spin}(U, q)$ and left multiplication L_g one has

$$L_g j_w(v) L_g^{-1}(a) = (g v g^{-1}) a w = j_w(g v g^{-1})(a).$$

Thus j_w is a rational Hodge map for every period; the right factor w is fixed and need not have any particular Hodge type.

Let E be the prescribed rational polarization of $W = H^1(A, \mathbb{Q})$. With $\text{End}(W) = W \otimes W^{\vee}$, its identification $I_E = (E^b(1))^{-1} : W^{\vee} \xrightarrow{\sim} W(1)$ converts j_w into

$$(7.2) \quad \iota_{w,E} : U \longrightarrow W \otimes W \subset H^2(A \times A, \mathbb{Q}).$$

Explicitly, $(\text{id} \otimes I_E) j_w$ maps $U(1)$ into $(W \otimes W)(1)$, and twisting by $\mathbb{Q}(-1)$ gives Equation (7.2). This is the specified untwisted tensor. Its recovery will use the full rational space of Hodge maps to $W \otimes W$.

7.3. A rational spanning lemma for the two half-spin factors. The Clifford tensor and the even-module half-spin decomposition are classical; see [37, Proposition 6.3 and Example 6.6]. The spanning argument below also uses rational Hodge-Hom descent and both tensor orientations. The following statement is about effective weight-one Hodge structures: their only types are $(1, 0)$ and $(0, 1)$.

Lemma 7.2. *Let (U, q) be an eight-dimensional polarized rational Hodge structure of K3 type, of signature $(2, 6)$, and suppose $\text{Hg}(U) = \text{SO}(U, q)$. Let W_1, W_2 be polarized effective weight-one Hodge structures and let $W = C^+(U, q)$ have its Kuga–Satake Hodge structure. For every nonzero Hodge map*

$$f : U \longrightarrow W_1 \otimes W_2,$$

put

$$\mathcal{H} = \mathrm{Hom}_{\mathrm{HS}}(W_1, W) \otimes_{\mathbb{Q}} \mathrm{Hom}_{\mathrm{HS}}(W_2, W).$$

If $\tau : W \otimes W \rightarrow W \otimes W$ is the ordinary tensor flip, the rational linear map

$$\Phi : \mathcal{H} \oplus \mathcal{H} \longrightarrow \mathrm{Hom}_{\mathrm{HS}}(U, W \otimes W),$$

defined by

$$\begin{cases} \Phi(a \otimes b, 0) = (a \otimes b) \circ f, \\ \Phi(0, a \otimes b) = \tau \circ (a \otimes b) \circ f \end{cases}$$

is surjective.

If $U \subset H^2(M, \mathbb{Q})$ for a smooth projective surface M , $W_i = H^1(C_i, \mathbb{Q})$, $W = H^1(A, \mathbb{Q})$, and f is the restriction of a codimension-two algebraic correspondence from M to $C_1 \times C_2$, then every map in the target of Φ is induced by a cycle in $\mathrm{CH}^2(M \times A \times A)_{\mathbb{Q}}$.

Proof. We prove surjectivity of the rational map Φ by spanning both complex half-spin tensor orientations. We first isolate the rational orthogonal Lie algebra in the common Hodge group. Weight-one effectivity will restrict the nontrivial constituents of W_1, W_2 to half-spins and identify their rational intertwiners with Hodge maps to W . This allows the tensor-product calculation to be made over $\mathbb{C}$ and its surjectivity to be descended to $\mathbb{Q}$.

The rational Hodge group. Let

$$G = \mathrm{Hg}(W_1 \oplus W_2 \oplus W).$$

The Hodge group is the connected rational algebraic group generated by the Hodge circle. Any rational invariant subspace of the polarized pure weight-one structure $W_1 \oplus W_2 \oplus W$ has a Hodge complement, obtained from its polarization. The complement is again G -invariant. Its faithful representation is therefore completely reducible, so G is reductive.

The embedding in Equation (7.1) realizes $U(1)$ in $\mathrm{End}(W)$; Tate twists do not change the Hodge-circle action. This gives a representation of G on U preserving q . Since G is connected, its image lies in $\mathrm{SO}(U, q)$. It contains the Hodge circle of U and therefore $\mathrm{Hg}(U) = \mathrm{SO}(U, q)$, so the map is surjective. Write $\mathfrak{g} = \mathrm{Lie}(G)$. The derived Lie algebra of $\mathfrak{g}$ is a direct sum of rational simple ideals. Since $\mathfrak{so}(U, q)$ is absolutely simple, exactly one of those ideals maps nontrivially to it: two such images would be commuting copies of the same nonabelian simple algebra. The nonzero map is an isomorphism, and the remaining ideals and the center form its kernel. We obtain rational Lie ideals

$$(7.3) \quad \mathfrak{g} = \mathfrak{s} \oplus \mathfrak{k}, \quad \mathfrak{s} \xrightarrow{\sim} \mathfrak{so}(U, q), \quad \mathfrak{k} = \ker(\mathfrak{g} \rightarrow \mathfrak{so}(U, q)).$$

The ideal $\mathfrak{s}$ integrates to a connected rational normal semisimple subgroup. Finite central kernels are harmless here; we do not identify a spin representation with a representation of $\mathrm{SO}(U, q)$.

The allowed weight-one constituents. Normalize the complexified infinitesimal Hodge-circle generator D so that its eigenvalues on weight one are $1, -1$. Its image on U has eigenvalues $2, 0, \dots, 0, -2$. Write $D = D_{\mathfrak{s}} + D_{\mathfrak{k}}$ according to Equation (7.3). Each $\mathfrak{s}_{\mathbb{C}}$ -isotypic summand of $W_{i, \mathbb{C}}$ has the form $R_{\lambda} \otimes M$, with R_{λ} irreducible. The commuting operator $D_{\mathfrak{k}}$ acts only on M . The two projections of the semisimple element D are semisimple, so their eigenvalues add on simultaneous eigenspaces.

Identify the root system of $\mathfrak{s}_{\mathbb{C}}$ with D_4 , with roots $\pm e_i \pm e_j$, using a Cartan subalgebra in which $D_{\mathfrak{s}}$ pairs with weights as $2e_1$. A dominant highest weight has

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq |\lambda_4|,$$

with coordinates either all integral or all half-integral. Its Weyl orbit contains weights giving the eigenvalues $2\lambda_1$ and $-2\lambda_1$ for $D_{\mathfrak{s}}$. Translation by any eigenvalue of the commuting factor cannot reduce this spread. Effectivity of weight one therefore gives $4\lambda_1 \leq 2$. The only possibilities

are the trivial representation and the two half-spin representations S_+, S_- , with highest weights $\frac{1}{2}(e_1 + e_2 + e_3 + e_4)$ and $\frac{1}{2}(e_1 + e_2 + e_3 - e_4)$. Their weights are all sign choices $\frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)$ of the respective parities. Each half-spin therefore has both $D_{\mathfrak{s}}$ -eigenvalues $1, -1$. If μ is an eigenvalue of $D_{\mathfrak{k}}$ on its multiplicity space, both $\mu + 1$ and $\mu - 1$ must belong to $\{1, -1\}$; hence $\mu = 0$. Semisimplicity now shows that $D_{\mathfrak{k}}$ is zero on that multiplicity space.

The rational decompositions

$$W_i = W_i^{\mathfrak{s}} \oplus W_i^{\text{nt}}, \quad W_i^{\text{nt}} = \mathfrak{s}W_i$$

are G -stable. On W_i^{nt} the entire Hodge-circle generator consequently acts through $\mathfrak{s}$. The same assertion holds on W . Its Hodge group lies in the rational spin group acting by left multiplication on $C^+(U, q)$ and maps onto $\text{SO}(U, q)$. The spin covering has finite kernel, so the Lie algebra of this image on W maps isomorphically to $\mathfrak{so}(U, q)$; in particular $\mathfrak{k}$ acts trivially on W .

For completeness, write $U_{\mathbb{C}} = F \oplus F^{\vee}$ with F maximal isotropic of dimension four. Exterior multiplication and contraction give the Clifford action on $\bigwedge^{\bullet} F$. These operators generate its matrix units, and the even algebra preserves parity. Thus, with $S_+ = \bigwedge^{\text{even}} F$ and $S_- = \bigwedge^{\text{odd}} F$, one has

$$C^+(U)_{\mathbb{C}} \simeq \text{End}(S_+) \oplus \text{End}(S_-).$$

Since $\dim S_+ = \dim S_- = 8$, left multiplication gives eight copies of each half-spin. In particular W has no $\mathfrak{s}$ -invariant summand.

Rational Hodge maps and complex multiplicities. A rational $\mathfrak{s}$ -intertwiner from W_i^{nt} to W commutes with D and hence with the connected Hodge circle. Extended by zero on $W_i^{\mathfrak{s}}$, it is therefore a Hodge map. Conversely every rational Hodge map commutes with G and hence with $\mathfrak{s}$. We have proved

$$(7.4) \quad \begin{aligned} \text{Hom}_{\text{HS}}(W_i, W) &= \text{Hom}_{\mathfrak{s}}(W_i^{\text{nt}}, W), \\ \text{Hom}_{\text{HS}}(U, W \otimes W) &= \text{Hom}_{\mathfrak{s}}(U, W \otimes W). \end{aligned}$$

In the second equality, both Hodge structures have weight two and their Hodge-circle actions factor through $\mathfrak{s}$, so the same commutation argument applies. For any two rational $\mathfrak{s}$ -modules V_1, V_2 the intertwiner space is the kernel of the rational linear map

$$\text{Hom}_{\mathbb{Q}}(V_1, V_2) \longrightarrow \text{Hom}_{\mathbb{Q}}(\mathfrak{s} \otimes V_1, V_2), \quad a \longmapsto ((x, v) \mapsto a(xv) - xa(v)).$$

Extension of scalars is exact. Hence $\text{Hom}_{\mathfrak{s}}(V_1, V_2) \otimes_{\mathbb{Q}} \mathbb{C} = \text{Hom}_{\mathfrak{s}_{\mathbb{C}}}(V_{1, \mathbb{C}}, V_{2, \mathbb{C}})$. In particular, after complexifying the rational Hodge Hom spaces in Equation (7.4), all complex maps between the half-spin multiplicity spaces are available. This assertion does not require either half-spin type to be defined separately over $\mathbb{Q}$.

Spanning both tensor orientations. Write

$$\begin{aligned} W_{i, \mathbb{C}}^{\text{nt}} &= (S_+ \otimes M_{i, +}) \oplus (S_- \otimes M_{i, -}), \\ W_{\mathbb{C}} &= (S_+ \otimes N_+) \oplus (S_- \otimes N_-), \quad \dim N_+ = \dim N_- = 8. \end{aligned}$$

Both half-spins are self-dual: the Weyl group contains simultaneous negation of all four coordinates, preserving the two weight parities. Clifford antisymmetrization identifies $C(U)_{\mathbb{C}}$ with $\bigwedge^{\bullet} U_{\mathbb{C}}$ as a module for conjugation. The even part consists of the diagonal endomorphism blocks of $S_+ \oplus S_-$, and the odd part of the two off-diagonal blocks. Choose a normalized volume element ω with $\omega^2 = 1$. Left multiplication by ω exchanges exterior degrees k and $8 - k$ and distinguishes the two blocks of each parity. Each block therefore contains one copy of each paired degree; degree four

splits into its two 35-dimensional eigenspaces. This gives

$$(7.5) \quad \begin{aligned} S_+ \otimes S_- &\simeq U_{\mathbb{C}} \oplus \bigwedge^3 U_{\mathbb{C}}, \\ S_{\pm} \otimes S_{\pm} &\simeq \mathbb{C} \oplus \bigwedge^2 U_{\mathbb{C}} \oplus \bigwedge_{\pm}^4 U_{\mathbb{C}}. \end{aligned}$$

The choice of the signs on $\bigwedge_{\pm}^4 U_{\mathbb{C}}$ follows that of $S_{\pm}$. To check the needed multiplicity, $\bigwedge^3 U_{\mathbb{C}}$ has no standard summand. Indeed, an equivariant map from $U_{\mathbb{C}}$ to this exterior power would send a unit vector u to an invariant of its stabilizer $\mathrm{SO}(u^{\perp}) \simeq \mathrm{SO}_7$. Neither $\bigwedge^3 u^{\perp}$ nor $u \wedge \bigwedge^2 u^{\perp}$ has such an invariant: determinant-one sign changes in an orthonormal basis kill every exterior basis coefficient. Thus the map vanishes on the orbit of u , which spans $U_{\mathbb{C}}$. The standard summand in each opposite-sign product has multiplicity one. There is none in a same-sign product, since $-\mathrm{id} \in \mathrm{SO}(U_{\mathbb{C}})$ acts trivially on all even exterior powers and by -1 on $U_{\mathbb{C}}$.

Choosing a nonzero equivariant map from $U_{\mathbb{C}}$ into each opposite-sign product identifies the complex target space with

$$\mathrm{Hom}_{\mathfrak{s}_{\mathbb{C}}}(U_{\mathbb{C}}, W_{\mathbb{C}} \otimes W_{\mathbb{C}}) \simeq (N_+ \otimes N_-) \oplus (N_- \otimes N_+).$$

The ordinary flip interchanges these two summands. In particular, Equation (7.4) and base change give

$$\dim_{\mathbb{Q}} \mathrm{Hom}_{\mathrm{HS}}(U, W \otimes W) = 8^2 + 8^2 = 128.$$

It remains to generate both multiplicity spaces from the one map f .

A tensor involving a trivial $\mathfrak{s}$ -constituent of either W_i is a sum of trivial or half-spin representations, so it also contains no standard summand. Therefore $f_{\mathbb{C}}$ is described by two multiplicity tensors

$$t_{+-} \in M_{1,+} \otimes M_{2,-}, \quad t_{-+} \in M_{1,-} \otimes M_{2,+},$$

at least one of which is nonzero.

Suppose, for example, that $t_{+-} \neq 0$. Choose linear functionals ℓ_1, ℓ_2 with $(\ell_1 \otimes \ell_2)(t_{+-}) \neq 0$. For arbitrary $n_+ \in N_+$ and $n_- \in N_-$, the rank-one maps $m \mapsto \ell_1(m)n_+$ and $m \mapsto \ell_2(m)n_-$ send this component to a nonzero scalar multiple of $n_+ \otimes n_-$. Taking the maps zero on the other signs isolates the component. Such decomposable tensors span $N_+ \otimes N_-$, and the independent second copy of $\mathcal{H}$ in Φ supplies the other orientation by τ . The case $t_{-+} \neq 0$ is identical. Hence $\Phi_{\mathbb{C}}$ is surjective. Its rational cokernel has zero complexification and is therefore zero, proving surjectivity of Φ . This descent accounts for arbitrary rational forms and Schur multiplicities; neither half-spin type need be defined separately over $\mathbb{Q}$.

Algebraic realization. Let $a : H^1(C_i, \mathbb{Q}) \rightarrow H^1(A, \mathbb{Q})$ be a rational Hodge map. Poincaré duality on C_i identifies it with a unique rational class α_a in the middle Künneth summand $H^1(C_i, \mathbb{Q}) \otimes H^1(A, \mathbb{Q}) \subset H^2(C_i \times A, \mathbb{Q})$, normalized by

$$(p_A)_*(\alpha_a \smile p_{C_i}^* x) = a(x).$$

The Hodge condition says precisely that α_a has type $(1, 1)$. After clearing denominators, the exponential sequence realizes it as the first Chern class of a holomorphic line bundle: the obstruction is its zero image in $H^2(\mathcal{O}_{C_i \times A})$. Projective GAGA makes this line bundle algebraic. Dividing by the same denominator gives a divisor $D_a \in \mathrm{CH}^1(C_i \times A)_{\mathbb{Q}}$ with $\mathrm{cl}(D_a) = \alpha_a$. Thus its cohomology class has exactly the required middle component.

For maps $a : H^1(C_1) \rightarrow W$ and $b : H^1(C_2) \rightarrow W$, the exterior product $D_a \boxtimes D_b$, with its factors ordered as $(C_1 \times C_2) \times (A \times A)$, has codimension two and acts on $H^1(C_1) \otimes H^1(C_2)$ by $a \otimes b$. If Z is the given cycle from a surface, its composition with this product has codimension $2 + 2 - 2 = 2$ and has image in $W \otimes W$.

Let $s : A \times A \rightarrow A \times A$ interchange the factors. For $u, v \in H^1(A, \mathbb{Q})$,

$$s_*(p_1^*u \smile p_2^*v) = p_2^*u \smile p_1^*v = -p_1^*v \smile p_2^*u.$$

Consequently $-\Gamma_s$ induces the ordinary tensor flip τ on this summand. Given a target map h , choose a rational preimage under Φ and write it as a finite sum

$$h = \sum_{\nu} c_{\nu}(a_{\nu} \otimes b_{\nu})f + \sum_{\mu} d_{\mu}\tau(a'_{\mu} \otimes b'_{\mu})f.$$

It is realized exactly by the rational cycle

$$\sum_{\nu} c_{\nu}(D_{a_{\nu}} \boxtimes D_{b_{\nu}}) \circ Z - \sum_{\mu} d_{\mu}\Gamma_s \circ (D_{a'_{\mu}} \boxtimes D_{b'_{\mu}}) \circ Z.$$

Composition with Γ_s , whose codimension is $\dim(A \times A)$, also preserves codimension two. On the specified source subspace the resulting action equals h in the full $H^2(A \times A, \mathbb{Q})$. $\square$

7.4. Transfer by a rational Hodge isometry.

Proposition 7.3. *Let b be a construction point of Lemma 7.1, and let S be a projective K3 surface equipped with a rational Hodge isometry $r_T : T_S \xrightarrow{\sim} U_b = T_{S_b^0}$ for the cup-product forms. For every full even-Clifford model A of T_S , every rational nonisotropic right factor w , and every rational polarization E , there is a cycle*

$$\Gamma_S \in \text{CH}^2(S \times A \times A)_{\mathbb{Q}}, \quad (\Gamma_S)_*x = \iota_{w,E}(x) \quad (x \in T_S),$$

where the equality holds in $H^2(A \times A, \mathbb{Q})$.

Proof. The assumed r_T uses the form on U_b inherited from Y , which the auxiliary K3 realizes isometrically. The factor-two pullback to X_b enters only the curve correspondence of Lemma 7.1 and is not part of this isometry.

Choose markings of the full second cohomology spaces by $\Lambda_{K3, \mathbb{Q}}$. They turn r_T into an isometry between two nondegenerate rational subspaces of this one quadratic space. Witt's Extension Theorem [30, Corollary 7.2.8] extends it to an isometry of $\Lambda_{K3, \mathbb{Q}}$. Returning through the markings gives

$$r : H^2(S, \mathbb{Q}) \xrightarrow{\sim} H^2(S_b^0, \mathbb{Q}).$$

The isometry r maps the orthogonal complement of T_S to the orthogonal complement of $T_{S_b^0}$. These are exactly $\text{NS}(S)_{\mathbb{Q}}$ and $\text{NS}(S_b^0)_{\mathbb{Q}}$, both of type $(1, 1)$. Since its restriction r_T is Hodge, r is therefore a rational Hodge isometry on the full second cohomology. Buskin's Theorem [9, Theorem 1.1], applied to these smooth projective complex K3 surfaces, gives a cycle

$$R \in \text{CH}^2(S \times S_b^0)_{\mathbb{Q}}, \quad R_*|_{H^2(S, \mathbb{Q})} = r.$$

The hypotheses here are precisely full rational Hodge isometry and projectivity; the integral Néron–Severi lattice of S_b^0 is not required to equal P .

Compose R with the cycle Z of Lemma 7.1. The resulting codimension-two correspondence induces the injection

$$f_S : T_S \xrightarrow{r_T} U_b \xrightarrow{f} H^1(C_1, \mathbb{Q}) \otimes H^1(C_2, \mathbb{Q}).$$

At these periods $\text{Hg}(T_S) = \text{SO}(T_S)$. Apply Lemma 7.2 to f_S and to the given full even-Clifford model $W = H^1(A, \mathbb{Q})$ of dimension 128. The prescribed map $\iota_{w,E}$ of Equation (7.2) belongs to the target of Φ . A rational preimage under Φ , realized by the divisor products and signed interchange in that lemma, supplies Γ_S with exactly the asserted action. In particular the equality is on all of T_S and has no extra component in the other Künneth summands of $H^2(A \times A, \mathbb{Q})$.

The choices of abelian data, pencil, and final parameter retain the successive order of Lemma 4.5 and Corollary 6.6. We have used the full Hodge group only at these construction periods.

Proposition 8.1 will realize one fixed standard tensor throughout a target period ball, including Picard jumps. $\square$

Remark 7.4. Buskin's theorem does not require integrality or preservation of an ample chamber. In its cyclic reduction, order one means an integral isometry. After a possible sign change and finitely many reflections in algebraic (-2) -classes, it maps a Kähler chamber to a Kähler chamber, so strong Torelli realizes it by an isomorphism; its graph and the sign and reflection correspondences are algebraic. In the final rational-reflection chain, transport an initial integral ample class through the partial isometries. On each intermediate period-realized K3 it is a rational $(1, 1)$ -class of positive square. Clearing denominators gives an integral such class, so the K3 projectivity criterion makes every intermediate surface projective, as required for the cyclic steps.

8. ALGEBRAIC CYCLES AT EVERY PERIOD

Proposition 7.3 constructs the prescribed correspondence at the selected construction periods of Lemma 7.1. To reach a Picard-jump target, we keep the perpendicular of P fixed, rather than varying the source with the transcendental space.

Proposition 8.1. *Let S have an ample primitive P -polarization and put $U_P = P_{\mathbb{Q}}^{\perp} \subset H^2(S, \mathbb{Q})$. There are a full even-Clifford realization $W_U = C^+(U_P, q) = H^1(A_U, \mathbb{Q})$, a rational polarization E_0 , a nonisotropic rational $w_0 \in U_P$, and a cycle*

$$\Gamma_U \in \text{CH}^2(S \times A_U^2)_{\mathbb{Q}}$$

whose full degree-two action on every $x \in U_P$ is exactly $\iota_{w_0, E_0}(x)$. This assertion concerns the whole fixed rank-eight space, including its rational $(1, 1)$ classes at a Picard jump.

The parameter-space lemma below records equality of a cycle's action on all of this fixed source. Its proper images will supply actual cycles at every target, without specializing the curves used to construct a seed.

8.1. Proper parameter spaces of successful cycles.

Lemma 8.2. *Let $f : \mathcal{Z} \rightarrow B$ be a smooth proper holomorphic family of pure relative dimension n over a connected coordinate ball, with a closed relative embedding $\mathcal{Z} \subset \mathbf{P}^N \times B$. Fix an integer $0 \leq r \leq n$, a rational local system $\mathcal{V}$ on B , a morphism of local systems*

$$L : R^{2r} f_* \mathbb{Q} \longrightarrow \mathcal{V},$$

and a flat section τ of $\mathcal{V}$. The set of $b \in B$ admitting a rational codimension- r algebraic cycle Z_b such that

$$(8.1) \quad L_b(\text{cl}(Z_b)) = \tau_b$$

is a countable union of closed analytic subsets of B . Each subset in this union is the image of a proper analytic parameter space carrying a fixed rational combination of universal flat subschemes; every point of that space supplies a cycle satisfying (8.1).

If the successful set contains a nonempty open subset of B , it equals B . The same conclusion holds if, for every countable union of proper closed analytic subsets of B , there is a successful point outside that union.

Proof. Embedded parameters over an analytic base. Let R be the Hilbert polynomial of the embedded fibres of f ; it is constant because the family is flat. For any fixed Hilbert polynomial Q , write $K_Q = \text{Hilb}^Q(\mathbf{P}^N)$. This is a projective scheme with a flat universal subscheme, by the projective Hilbert-scheme theorem [21, Theorem 3.2]. We use the incidence space

$$H_Q = \{(b, z) \in B \times K_Q^{\text{an}} : \mathcal{U}_z \subset \mathcal{Z}_b\}.$$

It is closed analytic, with its natural incidence structure. To see this directly, let $\mathcal{J}$ be the ideal sheaf of $\mathcal{Z}$ in $\mathbf{P}^N \times B$. The bounded regularity for the fixed polynomials R and Q supplies an integer k such that $\mathcal{J}(k)$ is relatively generated, its direct image is locally free with base change, and the universal subscheme for Q has the corresponding vanishing and base-change properties. On $B \times K_Q^{\text{an}}$, restriction defines a holomorphic vector-bundle map from sections of $\mathcal{J}(k)$ to sections of $\mathcal{O}_{\mathcal{U}}(k)$. Its zero scheme is exactly H_Q : vanishing of these generating sections is equivalent to containment of subschemes. Since K_Q^{an} is compact, $H_Q \rightarrow B$ is proper. Its universal subscheme is flat over H_Q , being the pullback of the universal Hilbert family. No algebraicity assumption on B or on f is required.

The ambient Betti class on singular parameters. For polynomials of degree $n - r$, associate to each universal fibre its $(n - r)$ -dimensional fundamental cycle, with the generic lengths of its structure sheaf as multiplicities. We claim that its class in the ambient $H^{2r}(\mathcal{Z}_b, \mathbb{Q})$ is locally constant on H_Q , including at singular or nonreduced Hilbert parameters.

Here are details that justify this assertion in the analytic setting. The same high-degree ideal construction gives a holomorphic classifying map $B \rightarrow \text{Hilb}^R(\mathbf{P}^N)^{\text{an}}$: the bundle of degree- k ideals defines the map to the Hilbert scheme's Grassmannian realization. Its image is in the open locus K_R^{sm} of smooth, pure n -dimensional fibres. Over this algebraic locus let $\mathfrak{Z} \rightarrow K_R^{\text{sm}}$ be the universal smooth family, and let $\mathfrak{H}_Q$ be its relative Hilbert scheme. Thus H_Q and its universal pair are analytic pullbacks of this algebraic universal pair.

On $\mathfrak{Z} \times_{K_R^{\text{sm}}} \mathfrak{H}_Q$, the universal structure sheaf $\mathcal{F}$ is flat over $\mathfrak{H}_Q$ and is a perfect complex on the ambient space. For this last assertion, take successive locally free surjections onto $\mathcal{F}$ and their kernels on the quasi-projective ambient space. All the kernels remain flat over $\mathfrak{H}_Q$. The fibres of the ambient family are regular of dimension n , so after at most n syzygies the fibrewise kernels are locally free. The local flatness criterion then makes this kernel locally free on the ambient space. This gives a bounded vector-bundle resolution even when $\mathfrak{H}_Q$ is singular. Flatness over the parameter space also shows that the resolution restricts to a resolution of the ordinary structure sheaf on every ambient fibre.

The topological Chern character of this complex, pulled back to the analytic universal pair, consequently satisfies

$$(8.2) \quad \text{ch}_r(\mathcal{F})|_{\mathcal{Z}_b} = \text{cl}([U_z]_{n-r}) \quad \text{in } H^{2r}(\mathcal{Z}_b, \mathbb{Q}).$$

This is the leading Chern-character identity for a coherent sheaf supported in codimension at least r . It follows from Grothendieck–Riemann–Roch (GRR) and dévissage, followed by the comparison of algebraic and topological Chern classes [19, Theorem 15.2, Example 15.2.16(a), and Proposition 19.1.2]. For the precise form needed here, remove a subset of codimension at least $r + 1$: dévissage then reduces to the generic lengths on smooth codimension- r supports, where the regular-embedding case of GRR gives their fundamental classes as the leading terms. The removed subset contributes nothing in degree $2r$, since its cohomology with supports vanishes below degree $2r + 2$ [19, Section 19.1, equation (3) and Lemma 19.1.1]. This also explains why lower-dimensional embedded components cause no problem.

A class on the total space of a pulled-back smooth proper family restricts to a flat section of its fibre cohomology. Indeed, locally trivialize the original differentiable family over B and pull that trivialization back to the parameter space; inclusions of fibres along a path are homotopic. Complex analytic spaces are locally path connected, and their nilpotents do not change their topology. Equation (8.2) proves the claimed constancy on all connected components, without assuming that the universal subschemes themselves are smooth.

Successful components and their images. Take every finite list $Q_1, \dots, Q_s$ of polynomials of degree $n - r$ and every list $c_1, \dots, c_s \in \mathbb{Q}$, and form

$$H_Q = H_{Q_1} \times_B \cdots \times_B H_{Q_s}, \quad \mathcal{Z}_{c,z} = \sum_{i=1}^s c_i [\mathcal{U}_{i,z}]_{n-r}.$$

The empty list represents the zero cycle. These spaces are proper over B , and the displayed universal combinations have constant ambient Betti class on each connected component. Because B is simply connected, both L and τ are constant in local-system trivializations. Hence (8.1) holds either everywhere or nowhere on each component.

Retain exactly the successful components. There are countably many polynomial and coefficient lists and countably many components: all the parameter spaces are second countable and locally connected, so their connected components are open and closed. Each retained component is a closed analytic subspace, proper over B . Its image is closed analytic by Remmert's proper mapping theorem; for a primary statement, see [20, §7.1, Satz 1], applied to the reduction of the component. Reduction changes neither its points nor its image. For a textbook statement of the proper mapping theorem, see [15, Chapter II, Section 8.2, Theorem 8.8].

Every rational algebraic cycle on a fibre is a finite rational sum of integral subvarieties and therefore occurs in this list. Conversely, every point of any retained image has an actual universal fibre combination above it. Their supports in the projective fibre are algebraic by Chow's theorem, so their fundamental cycles are algebraic. We have proved that the successful set is exactly the stated countable union, with the asserted cycle-theoretic meaning at every image point.

A proper closed analytic subset of a connected ball is nowhere dense. If the union contains a nonempty open set, Baire's theorem forces one image to have interior; the analytic identity principle makes that image all of B . If instead the union escapes every countable union of proper closed analytic subsets, the same conclusion follows: otherwise the successful images would themselves be such a union. $\square$

8.2. Realization in an ample period ball. We now apply the proper-parameter lemma directly in a small period ball through the desired target. A rational change of period coordinates places the generic geometric construction in that ball; it does not change the algebraic pencil used in the construction.

Lemma 8.3. *Let (V, q) be a rational quadratic space of signature $(2, n)$. The group $O(V)(\mathbb{Q})$ is dense in $O(V_{\mathbb{R}})$, and the rational orbit of every point of its type-IV period domain meets every nonempty open subset of that domain.*

Proof. Cartan–Dieudonné expresses each real orthogonal transformation as a finite product of reflections in nonisotropic real vectors. The reflection

$$r_v(x) = x - \frac{2q(x, v)}{q(v, v)}v$$

depends continuously on v on the nonisotropic open and is rational for rational v . Approximating the finitely many vectors proves density. The full real orthogonal group acts transitively on the oriented positive two-planes, which are the period-domain points; it also interchanges the two components. Density proves the orbit assertion. We use the standard reflection theorem in [30, Theorem 7.2.4]. $\square$

Proof of Proposition 8.1. The fixed local family and tensor. Choose a marking of the target and an integral ample class $h \in P$. The local K3 period theorem gives a six-dimensional marked deformation ball B through S on which P remains algebraic, identified with an open subset of the period domain of U_P . The class h lifts locally to a line bundle; shrink B so it is relatively ample and a power embeds the family $\mathcal{S} \rightarrow B$ projectively. Only P , not the target's whole Picard

group, is kept algebraic. Thus U_P is a fixed nondegenerate rational Hodge subvariation even at jumps. These local assertions follow from local Torelli and the deformation theory of polarized K3 surfaces [22].

Fix any rational $w_0 \in U_P$ with $q(w_0, w_0) \neq 0$ and put $W_U = C^+(U_P, q)$. There is one rational alternating form polarizing the left Clifford Hodge structures on the chosen period component. Here is a construction that also fixes its characters. Let x^* be reversal and $B_C(x, y) = \text{scal}(x^*y)$. Choose two orthogonal positive rational vectors with oriented product β in that component and set, up to an overall sign,

$$E_0(x, y) = B_C(x, y\beta).$$

Reversal sends β to $-\beta$, and cyclic invariance of scalar part proves alternation. Left spin multiplication preserves E_0 because $g^*g = 1$. At the chosen plane write $\beta = cJ$, $c > 0$. For an even orthogonal monomial containing a of the two positive generators, $B_C(x, x)$ has sign $(-1)^a$ and $Jx\beta = (-1)^{a+1}cx$. Thus $E_0(x, Jx)$ is definite after one overall choice of sign. Transitivity of the connected real spin group on the component proves positivity throughout it. This is the standard Kuga–Satake polarization [24].

The $(1, 0)$ space of W_U is $\ker L_\sigma$ for a period vector σ , and varies holomorphically. A rational symplectic basis for E_0 and its self-dual integral lattice give, over a smaller B , a polarized holomorphic abelian family $\mathcal{A} \rightarrow B$ with this fixed rational H^1 . The usual torus construction over Siegel space and a relative theta bundle give a relative projective embedding after shrinking. The tensor $\tau = \iota_{w_0, E_0}$ is a fixed rational linear map $U_P \rightarrow W_U \otimes W_U$ and a Hodge map at every period, by spin equivariance of j_{w_0} .

A geometric period chart in the target ball. Choose a geometric datum in the simultaneous regular-period and domination open of Sections 3 and 4. On a local marking fix a rational isometry $\theta : U \simeq U_P$ for the form inherited from the quotient surface Y ; this is possible by Proposition 3.1 and Lemma 1.3. Lemma 8.3 gives a rational $g \in O(U_P)(\mathbb{Q})$ moving its period into B . Fix g and θ , and shrink the original normalized geometric product chart so that its period map followed by $g\theta$ is a submersion into B . The change $g\theta$ acts only on rational period coordinates. Pencils subsequently selected still have the form $d_0 + td_1$ in the normalized algebraic section chart, with directions in the original good open at infinity.

Proper images for the full action. Apply Lemma 8.2 to $\mathcal{Z} = \mathcal{S} \times_B \mathcal{A}^2$, of relative dimension 130, with $r = 2$ and

$$\mathcal{V} = \text{Hom}_{\mathbb{Q}}(U_P, R^2(\mathcal{A}^2/B)_*\mathbb{Q}), \quad L_b(\gamma)(x) = (p_{AA})_*(\gamma \smile p_S^*x).$$

The success condition is equality to τ on all of U_P in the full target $H^2(A_b^2)$, not only after a projection or on the holomorphic line. It is a flat local-system condition. The successful set is therefore a countable union of closed analytic images Y_i of proper cycle-parameter spaces, with actual successful cycles over every point of each image.

A seed outside every proposed proper image. Suppose every Y_i is proper. In the regular construction period chart, pull back the Y_i by the fixed map $g\theta$ and also pull back the extra-Picard hyperplanes in B . These are countably many proper closed analytic loci: the period submersion is open, and an extra integral class has a nonzero U_P projection because $\Lambda_{K3} \cap P_{\mathbb{Q}} = P$. Its period equation is therefore nontrivial. Include also the proper marked Hodge-tensor conditions used in Lemma 7.1.

Apply Lemma 4.5 to this prescribed list, using the admissible pencil open of the construction. It selects abelian data, then an actual admissible section pencil with good leading section. Once that pencil, its first models, and its relative correspondence are fixed, intersect its algebraic good open with that of Corollary 6.6 and add the corollary's countable exceptional parameters. The intersection is still a nonempty algebraic open of the pencil. The last choice in the lemma supplies a point in this open outside those exceptions and every prescribed period locus. In particular, the

full special orthogonal Hodge group and the curve source of Lemma 7.1 hold at this construction point.

The target fiber at its moved period has $\text{NS} = P$, so its full U_P is transcendental. The auxiliary K3 has transcendental space U . The rational period isometry $g\theta$ identifies these transcendental Hodge structures with their cup forms. Proposition 7.3 therefore supplies on the target fiber a cycle realizing exactly the fixed tensor τ . That period lies in a successful image, contrary to its choice. Thus some image is all of B , and its fiber over the original target supplies Γ_U .

The rank-eight source and its full action have remained fixed throughout the ball. Neither a global moduli cycle nor specialization of a smooth curve presentation is required. $\square$

9. EXACT RESTRICTION AND RATIONAL TRANSFER

We first convert the fixed-source result to the standard tensor of the actual transcendental space at an ample P -polarized surface. A separate rational period transfer will then reach every surface in the quadratic locus of Theorem 1.2.

9.1. Restriction at Picard jumps. At a Picard jump the inclusion $T_S \subset U_P$ induces an inclusion of full even-Clifford modules. The following identity compresses the ambient tensor to this submodule with its prescribed polarization; it does not require the ambient polarization to restrict to that choice.

Lemma 9.1. *Let $i : W_T \hookrightarrow W_U$ be an inclusion of polarizable rational weight-one Hodge structures, let $p : W_U \rightarrow W_T$ be a Hodge retraction, and let E_U, E_T be rational polarizations. Let H be a rational Hodge structure of weight zero. For a Hodge map $j : H \rightarrow \text{End}(W_U)$, write $\iota_{j,E_U} : H(-1) \rightarrow W_U \otimes W_U$ for its tensor form. Set*

$$b = (E_T^\flat)^{-1} i^\vee(-1) E_U^\flat : W_U \longrightarrow W_T.$$

Then

$$(9.1) \quad (p \otimes b) \iota_{j,E_U} = \iota_{pji,E_T}.$$

If the weight-one structures are abelian H^1 spaces, both maps on the tensor slots are induced by rational algebraic correspondences.

Proof. In the ordering $\text{End}(W_U) = W_U \otimes W_U^\vee$, tensor conversion is $\text{id} \otimes (E_U^\flat)^{-1}$ after the common Tate twist. Hence the left side of (9.1) is

$$(p \otimes (E_T^\flat)^{-1} i^\vee(-1)) j,$$

which is exactly the tensor conversion of pji . All maps are rational Hodge maps. For completeness, if $f : H^1(B, \mathbb{Q}) \rightarrow H^1(A, \mathbb{Q})$ is any such map, its dual on homology commutes with complex structures. For some positive integer n , $nf^\vee$ is integral and induces a holomorphic homomorphism $\phi : A \rightarrow B$. Projectivity and GAGA make ϕ algebraic, and $f = \phi^*/n$. Thus $[\Gamma_\phi^t]/n \in \text{CH}^{\dim B}(B \times A)_\mathbb{Q}$ induces f on H^1 . No equality of dimensions or isogeny assumption is needed. $\square$

Proposition 9.2. *Let S be an ample P -polarized K3 surface. For every prescribed nonisotropic $w \in T_S$, rational polarization E and transported even-Clifford realization of $C^+(T_S, q)$, its standard tensor $\iota_{w,E}$ has a rational codimension-two algebraic realization.*

Proof. Write $U = U_P$ and $M = U \cap \text{NS}(S)_\mathbb{Q}$. Since P contains an ample class, Hodge index makes M negative definite and

$$U = T_S \perp M.$$

The source projection to T_S is algebraic: choose divisor bases for $\text{NS}(S)_\mathbb{Q}$ and use their inverse intersection matrix together with the diagonal. Proposition 8.1 gives one exact tensor ι_{w_0, E_0} on all of U .

For the desired $w \in T_S$, put $r = w_0^{-1}w \in C^+(U, q)$. It is invertible. Right multiplication R_r commutes with the left Hodge complex structure, so is a rational Hodge automorphism, and associativity gives

$$j_w = R_r j_{w_0}, \quad \iota_{w, E_0} = (R_r \otimes \text{id}) \iota_{w_0, E_0}.$$

This is a change of the first, output tensor slot.

The natural Clifford inclusion $i : W_T = C^+(T_S, q) \hookrightarrow W_U = C^+(U, q)$ is injective by PBW and is Hodge: the left period bivector lies in $C^+(T_S, q)$ on both spaces. Polarizability gives a rational Hodge retraction p . Apply Lemma 9.1 with the desired E on W_T . For $v, w \in T_S$, $j_w(v)$ preserves W_T and restricts there to the standard operator $a \mapsto vaw$. Therefore

$$(pR_r \otimes (E^b)^{-1} i^\vee (-1) E_0^b) \iota_{w_0, E_0} = \iota_{w, E}^{T_S}.$$

The order matters: R_r itself need not preserve W_T when $w_0 \notin T_S$, but the resulting operator j_w does.

Both tensor-slot maps are algebraic on abelian H^1 by the lemma. After clearing denominators, the single product homomorphism on the two abelian factors and its normalized transpose graph realize their tensor product, with no unrecorded permutation sign. The normalized transpose graph has codimension $2 \dim A_U = \dim A_U^2$, so postcomposition preserves codimension two. Precompose with the algebraic source projector and postcompose with algebraic degree-one projectors on the abelian factors. The full degree-two action is then precisely the displayed tensor. The same argument with the prescribed rational Hodge identification transports it to any stipulated isogenous model. $\square$

9.2. Transfer to the rational quadratic locus.

Lemma 9.3. *Let S be a projective K3 surface with a rational isometric embedding $\iota : T_S \hookrightarrow V_P$. There is an ample P -polarized K3 surface R and a full rational Hodge isometry*

$$\Phi : H^2(S, \mathbb{Q}) \xrightarrow{\sim} H^2(R, \mathbb{Q}), \quad \Phi(T_S) = T_R.$$

Both directions of this isometry are induced by algebraic correspondences.

Proof. Use the primitive embedding in Lemma 1.3 and an exact- P surface supplied there. Choose an integral ample $h \in P$ and a small marked P -polarized period patch Ω about that surface, shrinking so that h stays ample. This is an actual open image in the period domain of V_P , and includes its Picard-jump fibers.

The line $z = \iota(H^{2,0}(S))$ is a point of this period domain. Lemma 8.3 gives $g \in O(V_P)(\mathbb{Q})$ with $gz \in \Omega$. Choose the fiber R with period exactly gz . Then $g\iota : T_S \rightarrow H^2(R, \mathbb{Q})$ is an isometry taking the holomorphic line to that of R . The two full second-cohomology spaces have the rational K3 form. Witt's extension theorem extends this map to a full rational isometry Φ . It preserves the holomorphic line, its conjugate by rationality, and their orthogonal complement, which is the $(1, 1)$ space. Hence Φ is Hodge. Lefschetz $(1, 1)$ gives $\Phi(\text{NS}(S)_\mathbb{Q}) = \text{NS}(R)_\mathbb{Q}$ and therefore $\Phi(T_S) = T_R$. Buskin's theorem [9, Theorem 1.1] algebraizes Φ and its inverse. It requires neither integrality nor preservation of an ample class. Density was used only to enter the patch; the comparison uses the exact realized period. $\square$

Proposition 9.4. *Under the isometric-embedding hypothesis of Theorem 1.2, the exact tensor $\iota_{w, E}$ on T_S is algebraic for every stipulated w, E , and transported isogenous even-Clifford model.*

Proof. Choose R, Φ by Lemma 9.3 and set $\phi = \Phi|_{T_S} : T_S \simeq T_R$. The isometry induces a unital Clifford algebra isomorphism

$$c_\phi : C^+(T_S, q) \xrightarrow{\sim} C^+(T_R, q), \quad c_\phi(vaw) = \phi(v)c_\phi(a)\phi(w).$$

It is a weight-one Hodge isomorphism because it takes the oriented period bivector defining left multiplication to the corresponding bivector for R . For the broader functoriality under Hodge

similarities, see [40, Proposition 3.1 and Lemma 3.5]; the multiplier here is one, with the prescribed right factor transported as in the displayed identity. Proposition 9.2 gives the standard tensor on T_R for $\phi(w)$ and the polarization transported from E . Precompose its cycle with the algebraic correspondence for Φ and transport both abelian slots by c_ϕ^{-1} . These rational Hodge maps on H^1 are algebraic by Lemma 9.1. The displayed Clifford identity proves equality with the specified tensor, not merely with an abstract Hodge embedding. Source and degree-one target projectors isolate its full action. All operations are degree-preserving on their intermediate factors, so the resulting cycle has codimension two. The same algebraic transport treats the stipulated isogenous model. $\square$

Proof of Theorem 1.2. Proposition 9.4 proves the exact standard-tensor assertion under the stated rational isometric-embedding hypothesis, with every prescribed right vector, polarization, and transported model. Choose any one such tensor and apply Theorem 1.1. This gives rational HC and GHC in every degree of every self-power. For an ample P -polarized surface, the actual transcendental space is a nondegenerate subspace of its fixed perpendicular $U_P \simeq V_P$ by Lemma 1.3, so this case is included as asserted. $\square$

9.3. Hilbert schemes and moduli spaces.

Corollary 9.5. *Let S be a smooth projective complex K3 surface admitting a rational isometric embedding $(T_S, q) \hookrightarrow V_P$. For every $n, r \geq 1$, rational HC and rational GHC hold in every degree of $(S^{[n]})^r$, where $S^{[n]}$ is the Hilbert scheme of length- n subschemes of S .*

More generally, let $\alpha \in \text{Br}(S)$ and let M be either a smooth projective moduli space of Gieseker-stable α -twisted sheaves on S , or a smooth projective moduli space of σ -stable objects in $D^b(S, \alpha)$ for a generic Bridgeland stability condition σ . Then rational HC and rational GHC hold in every degree of M^r for every $r \geq 1$. If M is hyperkähler, the same conclusions hold for every smooth projective hyperkähler variety birational to M and all its self-powers.

Proof. The assertion for zero-dimensional M is immediate. Write $h(-)$ for the Chow motive with rational coefficients. The decomposition of de Cataldo–Migliorini [10, Theorem 6.2.1] expresses $h(S^{[n]})$ as a sum of Tate twists of summands of motives of powers of S ; the summands corresponding to symmetric products are cut out by rational averaging projectors. For either stated moduli space, Büllers [8, Theorem 0.1] likewise makes $h(M)$ a direct summand of a finite sum of Tate twists of motives $h(S^k)$. Tensoring these split correspondences gives analogous splittings for every self-power, again through Tate twists of powers of S . Thus these varieties are motivated by S in the sense of Arapura. Theorem 1.2 supplies HC and GHC for every power of S , so Arapura’s transfer lemma [2, Lemma 4.2], which retains geometric coniveau with its Tate shifts, gives both conjectures for every power of each indicated variety. Finally, Rieß’s specialization correspondence for birational smooth projective hyperkähler varieties [32, Theorem 3.2 and Section 3.3] gives an isomorphism of Chow motives, as formulated in [8, Proposition 1.4]. This proves the last assertion. $\square$

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